THE SHARP QUANTITATIVE FABER–KRAHN INEQUALITY VIA THE LEVEL-SET MOVING-BALL ROUTE

ABSTRACT. We prove the sharp quantitative Faber–Krahn inequality: there is a dimensional constant $c_{\mathrm{FK}}(n) > 0$ such that every open set $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ satisfies

$$\left(\frac{|\Omega|}{\omega_n}\right)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \ \geq\ c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2,$$

where λ_1 is the first Dirichlet eigenvalue, B^* the ball of the same volume, and $\mathcal{A}$ the Fraenkel asymmetry; the exponent 2 is sharp. The proof is organised around the torsion function and the unscaled moving-ball distance $q(\rho) = \inf_z |E_\rho \Delta B_\rho(z)|$ on its superlevel sets E_ρ. On reduced boundaries and for almost every level we establish an exact level-set deficit identity expressing the Saint–Venant deficit as a weighted integral of a Cauchy–Schwarz defect D_H and an isoperimetric defect D_I; nestedness gives a perimeter-free Lipschitz bound for q; a finite-perimeter affine first variation for q, a positive kinetic estimate, and an endpoint trace yield bounded sharp Saint–Venant stability $\mathcal{A}(\Omega)^2 \leq C(n,R)\,\delta_T(\Omega)$ for $\Omega \subset B_R$. A constructive De Giorgi truncation removes the confinement, and the Kohler–Jobin rearrangement transfers the Saint–Venant estimate to the eigenvalue. The Fusco–Julin sharp quantitative isoperimetric inequality (Theorem 2.2), used to extract a same-centre control of volume asymmetry and boundary-normal oscillation, is itself reproved internally in §3 by a regularising selection that uses only the ε-regularity for almost-minimisers of perimeter, so the manuscript is self-contained modulo standard tools for sets of finite perimeter and the classical rearrangement results.

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1. Introduction

1.1. **History.** For an open set $\Omega \subset \mathbb{R}^n$ $(n \geq 2)$ with $0 < |\Omega| < \infty$, let $\lambda_1(\Omega)$ denote its first Dirichlet eigenvalue and B^* the ball of the same volume. The *Faber–Krahn inequality*

$$\lambda_1(\Omega) \geq \lambda_1(B^*), \qquad \text{equality if and only if } \Omega \text{ is a ball,} \tag{1}$$

goes back to a 1877 conjecture of Lord Rayleigh [21] and was proved independently by Faber [19] and Krahn [20] in the 1920s, by Schwarz symmetrisation of the Rayleigh quotient (23). It is the spectral analogue of the classical isoperimetric inequality and is the prototype for an entire family of spectral shape inequalities; for the modern theory we refer to the monograph of Henrot [22], the survey of Brasco–De Philippis [5], and the references therein.

The qualitative statement (1) is silent on the rate at which an almost-extremal domain must resemble a ball. Quantitative versions were first obtained by Hansen–Nadirashvili [23], Melas [24], Bhattacharya–Weitsman [25], and Fusco–Maggi–Pratelli [26], each producing an inequality of the form

$$\delta_{\mathrm{FK}}(\Omega) := \left(\tfrac{|\Omega|}{\omega_n}\right)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c(n)\,\mathcal{A}(\Omega)^p$$

for some exponent $p > 2$ ($p = 4$ in [26]), where $\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} |\Omega \Delta (x_0 + B^*)|/|B^*|$ is the *Fraenkel asymmetry.* Smooth ellipsoidal perturbations of the ball show that no exponent $p < 2$ is possible [4, 5], so the sharp question asks for $p = 2$. The sharp form was proved by Brasco–De Philippis–Velichkov [3] in the form

$$\delta_{\mathrm{FK}}(\Omega) \geq c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2, \tag{2}$$

using a Cicalese–Leonardi-style selection principle [28] adapted to a penalised Bernoulli functional, with the Alt–Caffarelli free-boundary regularity theory [29] providing the smoothing step. Their argument proceeds by a sequential compactness/contradiction scheme, and the resulting constant $c_{\mathrm{FK}}(n)$ is not explicit. This was pointed out by Brasco and De Philippis in their survey [5, Open Problem 2, p. 22], where the question of giving a sharp quantitative Faber–Krahn inequality with a *computable* dimensional constant – equivalently, with no compactness selection – is formally posed and listed as open; the same issue is reiterated in Allen–Kriventsov–Neumayer [35, §1], who also gave a related linear-implies-nonlinear stability result via the Alt–Caffarelli–Friedman monotonicity formula, with a constant that is likewise not explicit. Prior to the present work, an explicit constant had only been obtained in the *non-sharp* regime, with exponent 4 on the asymmetry, in dimension two by Melas [24] and in general dimension by Brasco–De Philippis [5, Thm. 2.10]. The present manuscript answers Open Problem 2 of [5] in the affirmative. For related Stekloff- and spectrum-type stability results see [36, 37].

The development on the perimeter side runs in parallel. The qualitative isoperimetric inequality, with its full equality cases, was settled by De Giorgi [13] in the class of sets of finite perimeter. Quantitative stability was first established by Fuglede [27] for nearly-spherical sets, then in full generality by Fusco–Maggi–Pratelli [7], and refined via mass transport by Figalli–Maggi–Pratelli [8] – both with the *volume* asymmetry $\mathcal{A}$ on the right-hand side. Fusco and Julin [6] subsequently obtained the stronger *normal-oscillation* form: there is a centre z_E about which *both* the volume asymmetry and the squared deviation of the unit normal from the radial direction are controlled by a single perimeter deficit. Their proof is a regularising selection in the class of additive perimeter almost-minimisers, closed by the ε-regularity for quasi-minimal hypersurfaces of Almgren [31] and Tamanini [30], followed by a nearly-spherical second variation in the spirit of Fuglede and of Acerbi–Fusco–Morini [33]; see also the survey [32] for a unified account.

The Fusco–Julin same-centre control is, as we shall see in §2.2, the input that makes the level-set route to (2) go through.

1.2. Main result.

Theorem 1.1 (Sharp quantitative Faber–Krahn)**.** *There is a constant* $c_{\mathrm{FK}}(n) > 0$*, depending only on* n*, such that every open set* $\Omega \subset \mathbb{R}^n$ *with* $0 < |\Omega| < \infty$ *satisfies*

$$\delta_{\mathrm{FK}}(\Omega) := \Big(\frac{|\Omega|}{\omega_n}\Big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \;\geq\; c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2, \tag{3}$$

where $\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} |\Omega \Delta (x_0 + B^*)|/|B^*|$ *is the Fraenkel asymmetry. The exponent* 2 *is sharp.*

This is the theorem of Brasco–De Philippis–Velichkov [3]; the novelty here is in the proof. The constant $c_{\mathrm{FK}}(n)$ is *computable*, in the sense of [5, Open Problem 2]: it is exhibited as an explicit finite composition of dimensional constants, with no compactness selection, no sequential contradiction, and no Bernoulli free-boundary input. Theorem 1.1 thus answers in the affirmative the open problem of producing a sharp quantitative Faber–Krahn inequality with a computable constant, as raised in [5, p. 22] and reiterated in [35, §1]. In particular the Fusco–Julin strong-form inequality, used as an internal ingredient (Theorem 2.2), is reproved *from scratch* in §3, so the entire manuscript is self-contained modulo the classical theory of sets of finite perimeter [1, 15] and the Talenti–Pólya–Szegő rearrangement results.

1.3. Strategy of proof.

Reaching the eigenvalue through the torsion function. The eigenvalue λ_1 is reached only at the very end. The backbone is the torsion function $u = u_\Omega \in H^1_0(\Omega)$ of $-\Delta u = 1$ and its Saint–Venant deficit

$$\delta_T(\Omega) := E(\Omega) - E(B^*), \qquad E(\Omega) = -\tfrac{1}{2}\int_\Omega u\,dx.$$

The classical Saint–Venant inequality $\delta_T(\Omega) \geq 0$ is the torsion analogue of (1) and, by the Kohler–Jobin [9, 10, 11, 4] rearrangement, comparison of the Faber–Krahn deficit with the Saint–Venant deficit is essentially free; we reproduce a fully elementary, scale-tracked version of this transfer in §2.4. The whole problem is therefore to establish the *sharp Saint–Venant stability inequality*

$$\delta_T(\Omega) \;\geq\; c_{\mathrm{SV}}(n)\,\mathcal{A}(\Omega)^2, \tag{4}$$

which is the heart of the paper.

Why we do not follow the route via free-boundary regularity. One *theoretically plausible* quantification of the route of Brasco–De Philippis–Velichkov to (4) would replace their selection principle by a single-set, penalised selection map $\Omega \mapsto \widetilde{\Omega}$ preserving the deficit-to-asymmetry ratio without sequential compactness, then close as in [3]. The hard inputs of such a route would still be those of [3]: a Bernoulli-type penalised functional whose minimisers have an Alt–Caffarelli flatness improvement, a bootstrap from $C^{1,\gamma}$ to a *small* C^{2,γ_0} graph on which the nearly-spherical second variation can be invoked, and the second variation itself. Quantifying every constant along that chain demands the full Alt–Caffarelli theory [29] – existence and regularity for one-phase Bernoulli free boundaries – both for the penalised-minimiser construction (density and non-degeneracy lemmas of [3]) and for the flatness improvement underlying the $C^{1,\gamma}$ entry. The hidden cost is then the $C^{1,\gamma} \to$ small C^{2,γ_0} bootstrap, where the deficit only controls a half-derivative of the boundary; closing the gap requires uniform Schauder estimates for the hodograph problem and a careful interpolation against a high-order fixed bound.

We chose not to pursue that route, although it would be theoretically viable, because its quantification rests on machinery – the Alt–Caffarelli existence and regularity theory and a quantitative Bernoulli bootstrap – whose effective form is precisely the part one would need to redo by hand. Our route is, by contrast, *more instructive and relies on a more reasonable input*: it builds directly on the geometry of the torsion superlevels and on the only sharp quantitative result for which a fully constructive proof is available without recourse to Bernoulli free boundaries, namely the Fusco–Julin strong form of the quantitative isoperimetric inequality. At the level of regularity theory we therefore use only the De Giorgi–Almgren–Tamanini ε-regularity for almost-minimal sets of finite perimeter [31, 30, 15], not the Alt–Caffarelli theory for one-phase free boundaries; this is the intrinsic reason for the particular architecture of our proof, and the reason we believe it is worth recording.

The level-set viewpoint and the central distance. We work directly with the superlevel sets of the torsion function. Write $t(\rho)$ for the unique level such that $E_\rho := \{u > t(\rho)\}$ has volume $\omega_n \rho^n$; the family $\{E_\rho\}_{0<\rho<1}$ is nested and decreasing in ρ. The geometric object that does the work is the *unscaled moving-ball distance*

$$q(\rho) := \inf_{z\in\mathbb{R}^n} |E_\rho \Delta B_\rho(z)|. \tag{5}$$

Because $\{E_\rho\}$ is nested, q is uniformly Lipschitz in ρ (Lemma 4.25), with a constant depending only on the dimension. This Lipschitz bound is what eventually lets the bad-density levels be charged by *Lebesgue measure*, hence by the deficit only indirectly, rather than by the deficit pointwise.

The exact level-set deficit identity. The starting point of the analysis is an exact identity that decomposes the Saint–Venant deficit into a non-negative sum of two reduced-boundary defects. For a.e. level $t \in (0, \|u\|_\infty)$ set

$$\alpha(t) := \int_{\partial^*\{u>t\}} \frac{d\mathcal{H}^{n-1}}{|\nabla u|}, \qquad \beta(t) := \int_{\partial^*\{u>t\}} |\nabla u|\, d\mathcal{H}^{n-1},$$

and define the Cauchy–Schwarz defect $D_H(t) := \alpha(t)\beta(t) - P(E_t)^2$ and the squared-perimeter isoperimetric defect $D_I(t) := P(E_t)^2 - c_n m(t)^{2-2/n}$, with $c_n = n^2\omega_n^{2/n}$ and $m(t) = |E_t|$. Cauchy–Schwarz on $\partial^* E_t$ applied to $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$ and the De Giorgi isoperimetric inequality [13] give $D_H, D_I \geq 0$. Then

$$2\,\delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{c_n\, m(t)^{1-2/n}}\, dt \tag{6}$$

(Theorem 4.9). Identity (6) is an exact equality, valid on the De Giorgi reduced boundary of any open Ω, with no smoothness, no graph parametrisation, and no pointwise lower bound on $|\nabla u|$. In particular, levels at which $|\nabla u|$ is far from constant on $\partial^* E_t$ pay through D_H, and levels at which E_t is far from a ball pay through D_I; a small deficit therefore forces both defects to be small on most levels.

The Fusco–Julin input and same-centre control. The Fusco–Julin strong form (Theorem 2.2) provides, for each finite-perimeter set E with $|E| = |B_\rho|$, a *single* centre z_E controlling simultaneously the volume asymmetry and the boundary-normal oscillation:

$$\frac{|E\Delta B_\rho(z_E)|^2}{\rho^{2n}} + \frac{1}{\rho^{n-1}}\int_{\partial^* E} |\nu_E - e_{z_E}|^2\, d\mathcal{H}^{n-1} \le C_{\mathrm{FJ}}(n)\, \frac{P(E) - P(B_\rho)}{\rho^{n-1}}.$$

This same-centre feature is essential, and is not given by the weaker Fusco–Maggi–Pratelli theorem alone (which controls only the volume asymmetry, possibly about a different centre). Combined with an oriented radial trace identity (Lemma 2.4), it converts a small D_I on a level into homothetic-trace control of E_ρ about the corresponding optimal centre (Theorem 2.5); a Cauchy–Schwarz computation on the coarea derivative converts a small D_H into a small velocity defect (Lemma 2.6).

We have made it a point that the Fusco–Julin theorem itself is proved internally, in §3, in a sharp truly quantitative form with a finite, explicit dimensional constant. The argument splits in two stages: §3.2 establishes an elementary Fuglede-type nearly-spherical inequality with a displayed constant, and §3.3 runs a regularising selection that reduces an arbitrary finite-perimeter set to the nearly-spherical regime, using only the ε-regularity for additive almost-minimisers of perimeter. Crucially – this is the point about machinery emphasised above – no Alt–Caffarelli theory, no one-phase Bernoulli regularity, and no contradiction argument is used; the input is the classical De Giorgi–Almgren–Tamanini regularity for almost-minimal hypersurfaces.

The affine first variation and the positive kinetic estimate. The next step is to convert the same-centre good-level control into a quantitative first variation of q. The key technical tool is an *affine moving-ball first variation*: at almost every level a finite-perimeter version of the formula $D^+q(\rho) \le -(\text{something positive}) + (\text{error})$ holds, with the error controlled by the level-set defects $D_H + D_I$ and the *something positive* – the *positive kinetic estimate* of Proposition 4.28 – bounded below by a multiple of $q(\rho)$ itself. After bookkeeping for the bad-density levels (which are charged by Lebesgue measure via the Lipschitz bound on q) and a moving-ball endpoint trace (Lemma 4.26), this closes the bounded Saint–Venant stability inequality $\mathcal{A}(\Omega)^2 \le C(n, R)\delta_T(\Omega)$ for sets confined to a fixed ball B_R (Theorem 4.29).

Bounded reduction and the Kohler–Jobin transfer. The confinement $\Omega \subset B_R$ is removed by a constructive De Giorgi L^∞-truncation argument in the spirit of Brasco–De Philippis–Velichkov [3, §5] and Figalli–Maggi–Pratelli [8]: at the cost of slightly increasing the deficit and slightly decreasing the asymmetry, one replaces Ω by a bounded surrogate of dimensional diameter, applies Theorem 4.29, and transfers the resulting estimate back. The complementary large-deficit regime is trivial because $\mathcal{A} < 2$. This yields the global sharp Saint–Venant inequality $\delta_T(\Omega) \ge c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$ (Theorem 4.12).

Finally, the Kohler–Jobin rearrangement is proved in §2.4 from scratch, in the modern formulation of Brasco [4] specialised to the linear case, producing the scale-invariant spectral-shape inequality $T(\Omega)\lambda_1(\Omega)^{(n+2)/2} \ge T_n L_n^{(n+2)/2}$; combined with the global Saint–Venant inequality and an elementary one-variable estimate (Lemma 4.31), it transfers (4) into (3) and completes the proof.

Outline of the paper. Section 2 gathers standing notation and the technical inputs used in the proof: the constructive Fraenkel and Fusco–Julin same-centre packages (§2.2), the coarea differentiation framework on shell-admissible radii (§2.3), and the Kohler–Jobin inequality (§2.4). Section 3 carries out the internal, computable proof of the Fusco–Julin strong form (Theorem 2.2). Section 4 then assembles the main theorem in five movements: the exact level-set deficit identity and its bad-density consequences (§4.1); the bounded reduction (§4.2); the affine moving-ball first variation (§4.3); the endpoint trace (§4.4); the positive kinetic estimate (§4.5); and the closing Kohler–Jobin transfer (§4.6).

2. Setting and technical inputs

This section fixes the standing notation, then prepares the four technical ingredients fed into the main proof: the constructive Fraenkel and Fusco–Julin same-centre packages on level sets (§2.2); the coarea differentiation framework and the shell-admissible radii on which the first variation will be carried out (§2.3); and the Kohler–Jobin spectral-shape inequality, reproved from scratch (§2.4).

2.1. **Standing notation and scope.** Fix $n \geq 2$, $R \geq 1$, and $0 < \rho_* < 1$. Let $\Omega \subset B_R$ be bounded and open with $|\Omega| = \omega_n$, and let $u = u_\Omega \in H_0^1(\Omega)$ be the torsion function,

$$-\Delta u = 1 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega \quad (H_0^1 \text{ sense}).$$

We write the torsional rigidity $T(\Omega) := \int_\Omega u\,dx$, the Saint–Venant energy $E(\Omega) := -\frac{1}{2}T(\Omega)$, and the Saint–Venant deficit

$$\delta_T(\Omega) := E(\Omega) - E(B_1) = -\frac{1}{2}\int_\Omega u + \frac{1}{2}\int_{B_1} u_{B_1} \geq 0.$$

We write $\lambda_1(\Omega)$ for the first Dirichlet eigenvalue, B^* for the ball with $|B^*| = |\Omega|$, $L_n := \lambda_1(B_1)$, and $T_n := T(B_1) = \omega_n/(n(n+2))$; these enter only in the bounded reduction (§4.2) and the Kohler–Jobin transfer (§2.4).

Let $t(\rho)$ be the volume-radius parametrisation of the superlevels:

$$E_\rho := \{u > t(\rho)\}, \qquad |E_\rho| = \omega_n \rho^n, \qquad 0 < \rho < 1.$$

Set

$$w(\rho) := -t'(\rho), \qquad d\mu(\rho) = w(\rho)\,d\rho.$$

For a.e. ρ, E_ρ has finite perimeter, $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$, and the De Giorgi outer normal satisfies

$$\nu_\rho = -\frac{\nabla u}{|\nabla u|} \quad \mathcal{H}^{n-1}\text{-a.e. on } \partial^* E_\rho$$

(these level conventions are established in Lemma 4.1). On those levels define

$$V_\rho := \frac{w(\rho)}{|\nabla u|}, \qquad H_{z,\rho}(x) := \frac{(x-z)\cdot \nu_\rho(x)}{\rho}, \qquad \bar{V}_\rho := \frac{P(B_\rho)}{P(E_\rho)}.$$

The coarea relation of §4.1.1 (Lemma 4.2, with $w(\rho)\,\alpha(t(\rho)) = P(B_\rho)$) gives

$$\int_{\partial^* E_\rho} V_\rho\,d\mathcal{H}^{n-1} = P(B_\rho), \qquad \bar{V}_\rho = \frac{1}{P(E_\rho)}\int_{\partial^* E_\rho} V_\rho\,d\mathcal{H}^{n-1}. \tag{7}$$

The two level-set defects are

$$D_H(t(\rho)) := \left(\int_{\partial^* E_\rho} \frac{1}{|\nabla u|}\,d\mathcal{H}^{n-1}\right)\left(\int_{\partial^* E_\rho} |\nabla u|\,d\mathcal{H}^{n-1}\right) - P(E_\rho)^2,$$

$$D_I(t(\rho)) := P(E_\rho)^2 - P(B_\rho)^2.$$

2.2. Quantitative isoperimetry and same-centre trace control.

Theorem 2.1 (Constructive Fraenkel quantitative isoperimetry)**.** *For every $n \geq 2$ there is a computable constant $C_{\mathrm{iso}}(n)$ with the following property. If $E \subset \mathbb{R}^n$ is a set of finite perimeter and $|E| = \omega_n \rho^n$, then there is a centre $z_E \in \mathbb{R}^n$ such that*

$$|E \Delta B_\rho(z_E)|^2 \leq C_{\mathrm{iso}}(n) \rho^{n+1} \big(P(E) - P(B_\rho)\big). \tag{8}$$

Consequently, if $\rho \in [\rho_*, 1]$, then

$$|E \Delta B_\rho(z_E)|^2 \leq C(n, \rho_*) \big(P(E)^2 - P(B_\rho)^2\big). \tag{9}$$

Proof. This is the sharp quantitative isoperimetric inequality in Fraenkel form, proved constructively by the symmetrisation and mass-transport methods of Fusco–Maggi–Pratelli [7]. In the normalisation $|E| = |B_\rho|$, their estimate gives

$$\left(\frac{|E \Delta B_\rho(z_E)|}{|B_\rho|} \right)^2 \leq C(n) \frac{P(E) - P(B_\rho)}{P(B_\rho)}.$$

Since $|B_\rho| = \omega_n \rho^n$ and $P(B_\rho) = n \omega_n \rho^{n-1}$, this is (8). The passage to (9) follows from

$$P(E) - P(B_\rho) \leq \frac{P(E)^2 - P(B_\rho)^2}{P(B_\rho)} \leq C(n, \rho_*) \big(P(E)^2 - P(B_\rho)^2\big).$$

□

Theorem 2.2 (Fusco–Julin strong quantitative isoperimetry)**.** *For every $n \geq 2$ there is a computable constant $C_{\mathrm{FJ}}(n)$ with the following property. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter and let $r_E > 0$ be defined by $|E| = \omega_n r_E^n$. Then there is a centre $z_E \in \mathbb{R}^n$ such that*

$$\left[\frac{|E \Delta B_{r_E}(z_E)|}{r_E^n} + \left(\frac{1}{r_E^{n-1}} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1}(x) \right)^{1/2} \right]^2 \leq C_{\mathrm{FJ}}(n) \, \frac{P(E) - P(B_{r_E})}{r_E^{n-1}}. \tag{10}$$

Proof. Proved as Corollary 3.30 in §3, with a dimensional constant $C_{\mathrm{FJ}}(n)$ defined there as a finite composition of explicit dimensional constants (no contradiction, no compactness selection). In §3 the result is stated in the normalisation $|E| = |B_\rho|$ for the combined Fusco–Julin asymmetry $\mathcal{A}_{\mathrm{FJ}}$; since $\nu_{B_{r_E}(z)}(\pi_{z,r_E}(x)) = (x - z)/|x - z|$, the minimising centre is one centre for both the symmetric-difference term and the boundary-normal oscillation term, so the displayed form (10) above is the same statement with r_E in place of ρ. The rescaling between the two normalisations is the standard one and is the same as in the proof of Corollary 2.3 below. □

Corollary 2.3 (Unnormalised same-centre consequences)**.** *Fix $0 < \rho_* < 1$. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter with $|E| = \omega_n \rho^n$, where $\rho \in [\rho_*, 1]$. Then there is a centre $z \in \mathbb{R}^n$ such that*

$$|E \Delta B_\rho(z)|^2 \leq C(n) \big(P(E) - P(B_\rho)\big), \tag{11}$$

and

$$\int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1} \leq C(n) \big(P(E) - P(B_\rho)\big). \tag{12}$$

Consequently, since $\rho \geq \rho_*$,

$$|E \Delta B_\rho(z)|^2 + \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1} \leq C(n, \rho_*) \big(P(E)^2 - P(B_\rho)^2\big). \tag{13}$$

Proof. Apply Theorem 2.2 with $r_E = \rho$. Each summand in the left side of (10) is nonnegative, so

$$\frac{|E\Delta B_\rho(z)|^2}{\rho^{2n}} + \frac{1}{\rho^{n-1}} \int_{\partial^* E} \left| \nu_E - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1} \le C(n) \frac{P(E) - P(B_\rho)}{\rho^{n-1}}.$$

Since $\rho \le 1$, this gives (11) and (12). Finally,

$$P(E) - P(B_\rho) = \frac{P(E)^2 - P(B_\rho)^2}{P(E) + P(B_\rho)} \le \frac{P(E)^2 - P(B_\rho)^2}{P(B_\rho)} \le C(n, \rho_*)\big(P(E)^2 - P(B_\rho)^2\big).$$

□

Lemma 2.4 (Oriented radial trace)**.** *Fix $R \ge 1$ and $0 < \rho_* < 1$. Let $E \subset B_R$ be a set of finite perimeter with $|E| = \omega_n \rho^n$, where $\rho \in [\rho_*, 1]$, and let $z \in \mathbb{R}^n$ satisfy $|z| \le R + 1$. Set*

$$e_z(x) := \frac{x-z}{|x-z|}, \qquad W_z(x) := \left| \frac{|x-z|}{\rho} - 1 \right|.$$

Then

$$\int_{\partial^* E} W_z \, d\mathcal{H}^{n-1} \le \frac{n}{\rho_*} |E\Delta B_\rho(z)| + C(R, \rho_*) \int_{\partial^* E} |\nu_E - e_z|^2 \, d\mathcal{H}^{n-1}. \tag{14}$$

Proof. Translate the notation temporarily so that $z = 0$, and write $B = B_\rho$, $r = |x|$, $e = x/r$. The point $x = 0$ is $\mathcal{H}^{n-1}$-negligible on $\partial^* E$, so the value of e there is irrelevant. Define the bounded radial field

$$X(x) := \left| \frac{r}{\rho} - 1 \right| e.$$

For $0 < r < \rho$,

$$\operatorname{div} X = \frac{n-1}{r} - \frac{n}{\rho},$$

while for $r > \rho$,

$$\operatorname{div} X = \frac{n}{\rho} - \frac{n-1}{r}.$$

The field is singular only at the centre and has a Lipschitz radial coefficient away from the centre. To justify Gauss–Green, replace X by $\chi_\varepsilon(r)X$, where $\chi_\varepsilon = 0$ on B_ε and $\chi_\varepsilon = 1$ outside $B_{2\varepsilon}$, smooth the radial profile, apply the classical finite-perimeter Gauss–Green formula [15], and let the smoothing and then $\varepsilon \downarrow 0$. The extra interior term $\int_E \nabla \chi_\varepsilon \cdot X \, dx$ is supported on the annulus $\{\varepsilon < |x| < 2\varepsilon\}$, where $|\nabla \chi_\varepsilon| \lesssim \varepsilon^{-1}$ and $|X|$ is bounded; since that annulus has volume $O(\varepsilon^n)$, this term is $O(\varepsilon^{n-1})$ and vanishes as $\varepsilon \downarrow 0$. The boundary error tends to zero because $\mathcal{H}^{n-1}(\partial^* E \cap B_{2\varepsilon}) \to \mathcal{H}^{n-1}(\partial^* E \cap \{0\}) = 0$, and the volume terms converge by dominated convergence, since r^{-1} is locally integrable in $\mathbb{R}^n$ for $n \ge 2$. Thus

$$\int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} = \int_E \operatorname{div} X \, dx.$$

The same identity for B gives zero, because $X = 0$ on ∂B. Subtracting it yields

$$\int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} = \int_{E \setminus B} \operatorname{div} X \, dx - \int_{B \setminus E} \operatorname{div} X \, dx.$$

On $E \setminus B$, $\operatorname{div} X \le n/\rho$. On $B \setminus E$, $-\operatorname{div} X \le n/\rho$. Hence

$$\int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} \le \frac{n}{\rho} |E\Delta B|. \tag{15}$$

On $\partial^* E$,

$$W_z = X \cdot \nu_E + W_z(1 - e_z \cdot \nu_E) = X \cdot \nu_E + \frac{W_z}{2} |\nu_E - e_z|^2.$$

Since $E \subset B_R$ and $|z| \le R+1$, every reduced-boundary point satisfies $|x-z| \le 2R+1$, and therefore

$$W_z(x) \le 1 + \frac{2R+1}{\rho_*}.$$

Combining this bound with (15) and using $\rho \ge \rho_*$ proves (14). □

Theorem 2.5 (Same-centre good-level geometric package)**.** *There exist constants*

$$\eta_I = \eta_I(n, R, \rho_*) > 0, \qquad C_{\mathrm{sc}} = C_{\mathrm{sc}}(n, R, \rho_*),$$

such that the following holds. Define

$$G_I := \{\rho \in [\rho_*, \rho_\delta] : D_I(t(\rho)) \le \eta_I\}.$$

For a.e. $\rho \in G_I$ there is a centre z_ρ^{sc} such that

$$|E_\rho \Delta B_\rho(z_\rho^{\mathrm{sc}})|^2 + \left(\int_{\partial^* E_\rho} |H_{z_\rho^{\mathrm{sc}},\rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1}\right)^2 \le C_{\mathrm{sc}} \, D_I(t(\rho)). \tag{16}$$

Proof. Let $\rho \in G_I$ be a finite-perimeter level. Apply Corollary 2.3 to E_ρ, and let $z = z_\rho^{\mathrm{sc}}$ be the centre obtained there. Since

$$D_I(t(\rho)) = P(E_\rho)^2 - P(B_\rho)^2,$$

(13) gives

$$|E_\rho \Delta B_\rho(z)|^2 + \int_{\partial^* E_\rho} |\nu_\rho - e_z|^2 \, d\mathcal{H}^{n-1} \le C(n, \rho_*) D_I(t(\rho)). \tag{17}$$

Choose η_I so small that the first term in (17) implies

$$|E_\rho \Delta B_\rho(z)| < |B_\rho|.$$

Then $E_\rho \cap B_\rho(z)$ has positive measure. Since $E_\rho \subset B_R$, this forces $B_\rho(z) \cap B_R \ne \emptyset$, and therefore

$$|z| \le R + \rho \le R + 1. \tag{18}$$

The radial trace lemma (Lemma 2.4) gives

$$\int_{\partial^* E_\rho} \left| \frac{|x-z|}{\rho} - 1 \right| d\mathcal{H}^{n-1} \le C(n, R, \rho_*) \big(D_I(t(\rho))^{1/2} + D_I(t(\rho)) \big).$$

Reducing $\eta_I \le 1$, this is bounded by

$$C(n, R, \rho_*) D_I(t(\rho))^{1/2}. \tag{19}$$

For $H_{z,\rho} = (|x-z|/\rho)(e_z \cdot \nu_\rho)$,

$$|H_{z,\rho} - 1| \le \left| \frac{|x-z|}{\rho} - 1 \right| + |1 - e_z \cdot \nu_\rho| = \left| \frac{|x-z|}{\rho} - 1 \right| + \frac{1}{2} |\nu_\rho - e_z|^2.$$

Together with (17) and (19), this gives

$$\int_{\partial^* E_\rho} |H_{z,\rho} - 1| \, d\mathcal{H}^{n-1} \le C(n, R, \rho_*) D_I(t(\rho))^{1/2}. \tag{20}$$

Finally,

$$\bar{V}_\rho = \frac{P(B_\rho)}{P(E_\rho)}, \qquad P(E_\rho)|1 - \bar{V}_\rho| = P(E_\rho) - P(B_\rho) \le C(n, \rho_*) D_I(t(\rho)).$$

Combining this with (20) and again using $\eta_I \le 1$,

$$\int_{\partial^* E_\rho} |H_{z,\rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \le C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Squaring and adding the first estimate in (17) proves (16). □

Lemma 2.6 (Velocity defect controlled by D_H)**.** *For a.e.* $\rho \in [\rho_*, \rho_\delta]$,

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \leq D_H(t(\rho)). \tag{21}$$

Proof. Write $P = P(E_\rho)$, $P^* = P(B_\rho)$, $\alpha = \int_{\partial^* E_\rho} |\nabla u|^{-1} \, d\mathcal{H}^{n-1}$, and $\beta = \int_{\partial^* E_\rho} |\nabla u| \, d\mathcal{H}^{n-1}$. The coarea parametrisation gives $w\alpha = P^*$, and the flux identity gives $\beta = |E_\rho| = \omega_n \rho^n$. With $f := V_\rho = w/|\nabla u|$ and $\bar{f} := \bar{V}_\rho = P^*/P$,

$$\int_{\partial^* E_\rho} f \, d\mathcal{H}^{n-1} = P^*.$$

Moreover

$$\begin{aligned} \int_{\partial^* E_\rho} \frac{(f - \bar{f})^2}{f} \, d\mathcal{H}^{n-1} &= \int f \, d\mathcal{H}^{n-1} - 2\bar{f} P + \bar{f}^2 \int \frac{1}{f} \, d\mathcal{H}^{n-1} \\ &= -P^* + \frac{(P^*)^2}{P^2} \frac{\beta}{w} = -P^* + \frac{P^*}{P^2} \alpha\beta \\ &= \frac{P^*}{P^2} \big(\alpha\beta - P^2\big) = \frac{P^*}{P^2} D_H(t(\rho)). \end{aligned}$$

By Cauchy–Schwarz,

$$\left(\int |f - \bar{f}| \, d\mathcal{H}^{n-1} \right)^2 \leq \left(\int \frac{(f - \bar{f})^2}{f} \, d\mathcal{H}^{n-1} \right) \left(\int f \, d\mathcal{H}^{n-1} \right) = \left(\frac{P^*}{P} \right)^2 D_H(t(\rho)).$$

The isoperimetric inequality gives $P^* \leq P$, hence (21). □

Lemma 2.7 (Superlevel asymmetry transfer)**.** *Let* $E \subset \Omega$, $|\Omega| = \omega_n$, *and* $|E| = \omega_n \rho^n$ *with* $\rho \in (0, 1]$. *Then*

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E) + \frac{2}{\omega_n} |\Omega \setminus E|.$$

Here $\mathcal{A}(E)$ *is the Fraenkel asymmetry of* E *relative to balls of volume* $|E|$.

Proof. Let $s = |E|$, $L = |\Omega| = \omega_n$, and $\eta = L - s = |\Omega \setminus E|$. Choose z such that, for an arbitrary $\varepsilon > 0$,

$$|E \Delta B_\rho(z)| \leq s(\mathcal{A}(E) + \varepsilon).$$

Since $B_\rho(z) \subset B_1(z)$,

$$\begin{aligned} |\Omega \Delta B_1(z)| &\leq |\Omega \Delta E| + |E \Delta B_\rho(z)| + |B_\rho(z) \Delta B_1(z)| \\ &= \eta + |E \Delta B_\rho(z)| + \eta. \end{aligned}$$

Divide by L, use $s \leq L$, and let $\varepsilon \downarrow 0$. □

2.3. **Coarea differentiation and shell-admissible radii.** The first variation is proved on a fixed full-measure set of good-density radii. The point of the definition below is that all coarea differentiations are registered in advance by a countable family independent of the radius being tested.

Lemma 2.8 (Two-sided variable-thickness coarea differentiation)**.** *Let* $\xi \leq \zeta$ *be bounded Borel functions on* $\mathbb{R}^n$. *Then for Lebesgue-a.e. positive level* $s \in (0, \|u\|_\infty)$, *with exceptional null set depending only on the pair* (ξ, ζ),

$$\lim_{h \downarrow 0} \frac{1}{h} \big| \{ x : h\,\xi(x) < u(x) - s < h\,\zeta(x) \} \big| = \int_{\partial^* \{u > s\}} \frac{\zeta - \xi}{|\nabla u|} \, d\mathcal{H}^{n-1}. \tag{22}$$

The statement is used only at levels for which the right side is finite and $\mathcal{H}^{n-1}(\partial^*\{u > s\} \cap \{|\nabla u| = 0\}) = 0$.

Proof. By Lemma 4.1(a), $|\{x \in \Omega : |\nabla u(x)| = 0\}| = 0$, so the critical set may be discarded in the volume computations below.

Suppose first that $\xi = \sum_j p_j \mathbf{1}_{A_j}$ and $\zeta = \sum_j q_j \mathbf{1}_{A_j}$ are simple over a common disjoint Borel partition, with $p_j \le q_j$. By the BV coarea formula applied to $\mathbf{1}_{A_j}|\nabla u|^{-1}\mathbf{1}_{\{|\nabla u|>0\}}$,

$$\big|\{hp_j < u - s < hq_j\} \cap A_j\big| = \int_{s+hp_j}^{s+hq_j} \int_{\partial^*\{u>\tau\}} \frac{\mathbf{1}_{A_j}}{|\nabla u|}\, d\mathcal{H}^{n-1}\, d\tau.$$

The interval of integration shrinks to s and has length $h(q_j - p_j)$; hence at a Lebesgue point of $\tau \mapsto \int_{\partial^*\{u>\tau\}} \mathbf{1}_{A_j}|\nabla u|^{-1}\, d\mathcal{H}^{n-1}$, division by h followed by $h \downarrow 0$ gives $(q_j - p_j)\int_{\partial^*\{u>s\}} \mathbf{1}_{A_j}|\nabla u|^{-1}\, d\mathcal{H}^{n-1}$. Summing over j proves (22) for simple pairs.

For bounded $\xi \le \zeta$, fix $\eta > 0$ and choose simple pairs that sandwich the slab. Outer: simple $\sigma \le \xi \le \zeta \le \sigma'$ with $\sigma' - \sigma \le (\zeta - \xi) + 2\eta$; then $\{h\xi < u - s < h\zeta\} \subseteq \{h\sigma < u - s < h\sigma'\}$, so the lim sup of the left side of (22) is at most

$$\int_{\partial^*\{u>s\}} \frac{\sigma' - \sigma}{|\nabla u|}\, d\mathcal{H}^{n-1} \le \int_{\partial^*\{u>s\}} \frac{\zeta - \xi}{|\nabla u|}\, d\mathcal{H}^{n-1} + 2\eta \int_{\partial^*\{u>s\}} \frac{1}{|\nabla u|}\, d\mathcal{H}^{n-1}.$$

Inner: simple $\sigma_* \ge \xi$, $\sigma'_* \le \zeta$ with $\sigma_* \le \sigma'_*$ and $\sigma'_* - \sigma_* \ge (\zeta - \xi - 2\eta)_+$; the reverse inclusion bounds the lim inf from below by

$$\int_{\partial^*\{u>s\}} \frac{(\zeta - \xi - 2\eta)_+}{|\nabla u|}\, d\mathcal{H}^{n-1}.$$

Letting $\eta \downarrow 0$ gives (22). The exceptional levels are fixed as follows: choose the outer and inner simple sandwiches for the dyadic sequence $\eta = 2^{-m}$, $m \ge 1$, by a fixed dyadic quantisation of the bounded functions ξ, ζ. The union of the exceptional null sets for those countably many simple pairs is still null and depends only on (ξ, ζ). □

Shell-admissible radii. Let $\mathcal{U}$ be the countable family of finite unions of rational balls whose closures are compactly contained in Ω. This family is cofinal: for every compact $K \Subset \Omega$ there is $U \in \mathcal{U}$ with $K \subset U \Subset \Omega$. For

$$c, p, q, \lambda, r \in \mathbb{Q}, \qquad -1 \le c \le 0, \quad p \le q, \quad 1 \le \lambda \le \rho_*^{-1}, \quad r > 0, \qquad z, a \in \mathbb{Q}^n,$$

and $U \in \mathcal{U}$, register the bounded Borel pairs

$$(p\mathbf{1}_U, q\mathbf{1}_U)$$

and

$$\big(\mathbf{1}_U\big(\min(c, \nabla u \cdot (\lambda(x - z) + a)) - r\big),\ \mathbf{1}_U\big(\max(c, \nabla u \cdot (\lambda(x - z) + a)) + r\big)\big).$$

This is a fixed countable family, independent of the radius being tested. We call $\rho \in G_\tau \cap (0,1)$ *shell-admissible* if:

(i) E_ρ has finite perimeter, $\partial^* E_\rho \subset \Omega$ up to $\mathcal{H}^{n-1}$-null sets, and $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$;
(ii) t is differentiable at ρ, and

$$w(\rho) \int_{\partial^* E_\rho} \frac{1}{|\nabla u|}\, d\mathcal{H}^{n-1} = P(B_\rho);$$

(iii) $s = t(\rho)$ is a level where Lemma 2.8 holds for every registered pair above.

Almost every $\rho \in G_\tau \cap (0,1)$ is shell-admissible. Indeed, each registered pair has an exceptional null set of levels s. If N is such a null set, then the one-dimensional area formula for the monotone absolutely continuous map t gives, on G_τ,

$$|\{\rho : t(\rho) \in N\}| \le \tau_0^{-1} \int_{\{t(\rho)\in N\}} w(\rho)\, d\rho \le \tau_0^{-1}|N| = 0.$$

The remaining conditions hold for a.e. radius by BV coarea and the finite-perimeter level-set package. The critical-boundary condition follows from the preceding interior critical-set nullity and the coarea formula:

$$\int_0^{\|u\|_\infty} \mathcal{H}^{n-1}(\partial^*\{u > s\} \cap \{|\nabla u| = 0\})\, ds = \int_{\{|\nabla u|=0\}} |\nabla u|\, dx = 0.$$

Lemma 2.9 (Local BV deformation tail)**.** *Let $E \subset B_R$ be a set of finite perimeter, let $\eta \in C^1_c(\mathbb{R}^n)$ satisfy $0 \le \eta \le 1$, and let $W \in C^1$ on a neighbourhood of a ball containing $E \cup \operatorname{spt}\eta$. Set $T_h(x) = x + hW(x)$. Then*

$$\limsup_{h\to 0} \frac{1}{|h|} \int_{\mathbb{R}^n} \eta(x)\big|\mathbf{1}_{T_h(E)}(x) - \mathbf{1}_E(x)\big|\, dx \le \int_{\partial^* E} \eta\, |W|\, d\mathcal{H}^{n-1}.$$

Proof. For $|h|$ small, T_h is a C^1 diffeomorphism on the fixed ball. Changing variables $x = T_h y$, it is enough, up to a multiplicative $1 + o(1)$ factor, to estimate

$$\int \eta(T_h y)\big|\mathbf{1}_E(T_h y) - \mathbf{1}_E(y)\big|\, dy.$$

Let $u_\varepsilon = \mathbf{1}_E * \varphi_\varepsilon$. For fixed h, the corresponding integrals with u_ε converge to the displayed integral because $u_\varepsilon \to \mathbf{1}_E$ in L^1 and composition with T_h has bounded Jacobian. For smooth u_ε,

$$|u_\varepsilon(T_h y) - u_\varepsilon(y)| \le |h| \int_0^1 |\nabla u_\varepsilon(y + \theta h W(y))|\, |W(y)|\, d\theta.$$

After integrating, change variables $x = y + \theta h W(y)$. These maps have Jacobians $1 + o(1)$ uniformly in θ. Thus, for fixed h,

$$\frac{1}{|h|} \int \eta(T_h y)|u_\varepsilon(T_h y) - u_\varepsilon(y)|\, dy \le (1 + o_h(1)) \int \psi_h\, d|Du_\varepsilon|,$$

where the continuous compactly supported weights ψ_h are uniformly bounded and satisfy $\psi_h \to \eta|W|$ uniformly as $h \to 0$. Letting $\varepsilon \downarrow 0$ at fixed h, strict BV_{loc} convergence of the standard mollifications and Reshetnyak continuity for continuous weights [1, 15] give the same bound with $d|D\mathbf{1}_E|$. Taking $\limsup_{h\to 0}$ then gives

$$\limsup_{h\to 0} \frac{1}{|h|} \int \eta(x)\big|\mathbf{1}_{T_h(E)}(x) - \mathbf{1}_E(x)\big|\, dx \le \int \eta|W|\, d|D\mathbf{1}_E|.$$

Since $|D\mathbf{1}_E| = \mathcal{H}^{n-1}\llcorner\partial^* E$, this is the claim. ☐

2.4. **The Kohler–Jobin inequality.** This section is logically self-contained. In Part 1 we reconstruct from scratch, by the Kohler–Jobin rearrangement technique, the spectral shape inequality

$$T(\Omega)\, \lambda_1(\Omega)^{(n+2)/2} \;\ge\; T(B)\, \lambda_1(B)^{(n+2)/2} \qquad (B \text{ any ball}),$$

with equality only for balls. In Part 2 we feed this inequality, together with the global sharp Saint–Venant stability estimate proved elsewhere in the manuscript, into a short and fully explicit computation that yields sharp quantitative Faber–Krahn stability, $\delta_{\mathrm{FK}}(\Omega) \ge c_{\mathrm{FK}}(n)\, \mathcal{A}(\Omega)^2$.

The technique, due to Kohler–Jobin [9, 10, 11], is a spherical rearrangement that, unlike Schwarz symmetrisation, preserves not the *volume* of the superlevel sets but a *torsional* functional of them. We follow the modern exposition of Brasco [4, §§3–5, Thm. 1.1], specialised throughout to the linear case (the Laplacian, $p = 2$, and the Dirichlet energy quotient).

The section is written for a reader meeting the argument for the first time: each displayed inference is accompanied by a short justification naming the identity, inequality, or theorem invoked and noting why its hypotheses are met, and the central computations – the radial-gradient expansion in Lemma 2.15, the closed form of $\mathrm{T_{mod}}$, the convexity chain in Lemma 2.16, the Rayleigh assembly, and the one-variable inequality of Part 2 – are carried out in full, with every constant tracked.

2.4.1. *Notation and scaling conventions.* Throughout, $n \geq 2$ and $\Omega \subset \mathbb{R}^n$ is open with finite measure. We use:

- $u_\Omega \in H_0^1(\Omega)$, the *torsion function*, the unique weak solution of $-\Delta u_\Omega = 1$ in Ω, $u_\Omega = 0$ on $\partial\Omega$;
- $T(\Omega) = \int_\Omega u_\Omega \, dx$, the *torsional rigidity*, and $E(\Omega) = -T(\Omega)/2$;
- $\lambda_1(\Omega)$, the first Dirichlet eigenvalue,

$$\lambda_1(\Omega) = \min_{0 \neq v \in H_0^1(\Omega)} \frac{\int_\Omega |\nabla v|^2 \, dx}{\int_\Omega v^2 \, dx}, \tag{23}$$

  attained by a first eigenfunction $u > 0$ solving $-\Delta u = \lambda_1(\Omega) u$;
- B^*, the ball centred at 0 with $|B^*| = |\Omega|$; B_r the open ball of radius r centred at 0; B_1 the unit ball; $\omega_n = |B_1|$;
- $L_n = \lambda_1(B_1) = j_{n/2-1,1}^2$ and $T_n = T(B_1) = \omega_n / \big(n(n+2)\big)$;
- $\mathcal{A}(\Omega) = \inf_{x_0 \in \mathbb{R}^n} |\Omega \triangle (x_0 + B^*)| \,/\, |B^*| \in [0, 2)$, the Fraenkel asymmetry;
- $\delta_{\mathrm{FK}}(\Omega) = (|\Omega| / \omega_n)^{2/n} \big(\lambda_1(\Omega) - \lambda_1(B^*)\big)$, the scale-invariant Faber–Krahn deficit.

Justification of the listed identities. We verify the two facts about balls that the conventions assert, since the whole argument is calibrated against them.

The torsion function of a ball. On B_r put $v(x) = (r^2 - |x|^2)/(2n)$. Then $v \in C^\infty(\overline{B_r})$, $v = 0$ on $\{|x| = r\} = \partial B_r$, and, writing $|x|^2 = \sum_i x_i^2$,

$$\Delta v = \frac{1}{2n} \Delta\big(r^2 - |x|^2\big) = \frac{1}{2n} \left(-\Delta |x|^2\right) = \frac{1}{2n} (-2n) = -1,$$

because $\Delta |x|^2 = \sum_i \partial_{x_i x_i}^2 \big(\sum_j x_j^2\big) = \sum_i 2 = 2n$. Thus $-\Delta v = 1$ in B_r with $v = 0$ on ∂B_r, so by uniqueness of the weak solution $u_{B_r} = v$. This justifies the formula $u_{B_r}(x) = (r^2 - |x|^2)/(2n)$.

The torsional rigidity of a ball. Integrating u_{B_r} in polar coordinates (writing $\sigma_{n-1} = \mathcal{H}^{n-1}(\partial B_1) = n\omega_n$ for the surface area of the unit sphere, the standard relation between the volume and surface measure of the unit ball),

$$T(B_r) = \int_{B_r} \frac{r^2 - |x|^2}{2n} \, dx = \frac{1}{2n} \int_0^r \frac{(r^2 - \rho^2)}{1} \sigma_{n-1} \rho^{n-1} \, d\rho = \frac{\sigma_{n-1}}{2n} \Big[\frac{r^2 \rho^n}{n} - \frac{\rho^{n+2}}{n+2}\Big]_0^r.$$

The bracket at $\rho = r$ equals $r^{n+2}\big(\frac{1}{n} - \frac{1}{n+2}\big) = r^{n+2} \cdot \frac{2}{n(n+2)}$, so with $\sigma_{n-1} = n\omega_n$,

$$T(B_r) = \frac{n\omega_n}{2n} \cdot \frac{2}{n(n+2)} \, r^{n+2} = \frac{\omega_n}{n(n+2)} \, r^{n+2} = T_n \, r^{n+2}. \tag{24}$$

The last equality is the definition $T_n = T(B_1) = \omega_n / \big(n(n+2)\big)$, recovered by setting $r = 1$. This is exactly the constant whose value will make the leading constant in Lemma 2.15 collapse to 1.

Scaling. If $t > 0$ and $w_t(x) := t^2 u_\Omega(x/t)$ on $t\Omega$, then $-\Delta w_t(x) = t^2 \cdot t^{-2} (-\Delta u_\Omega)(x/t) = 1$ and $w_t = 0$ on $\partial(t\Omega)$, so $u_{t\Omega} = w_t$ by uniqueness, and the change of variables $x = ty$ gives $T(t\Omega) = \int_{t\Omega} w_t \, dx = \int_\Omega t^2 u_\Omega(y) \, t^n \, dy = t^{n+2} T(\Omega)$. Likewise, if u realises the minimum in (23) on Ω, then $x \mapsto u(x/t)$ is admissible on $t\Omega$ and the Rayleigh quotient scales by t^{-2} (gradients carry an extra t^{-2} relative to values, and the Jacobian t^n cancels between numerator and denominator), whence $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$. Therefore the combination

$$\Psi(\Omega) := T(\Omega)\, \lambda_1(\Omega)^{(n+2)/2} \tag{25}$$

satisfies $\Psi(t\Omega) = t^{n+2}\, t^{-2\cdot(n+2)/2}\, \Psi(\Omega) = t^{n+2}\, t^{-(n+2)} \Psi(\Omega) = \Psi(\Omega)$: it is scale invariant. In particular every ball B_r is a dilate of B_1, so $\Psi(B_r) = \Psi(B_1) = T_n L_n^{(n+2)/2}$ for every ball. We abbreviate $\beta := (n + 2)/2$; with this notation $\Psi = T \lambda_1^\beta$.

2.4.2. *The Kohler–Jobin rearrangement.*

Theorem 2.10 (Kohler–Jobin). *For every open $\Omega \subset \mathbb{R}^n$ of finite measure with $\lambda_1(\Omega) < \infty$ and every ball $B \subset \mathbb{R}^n$,*

$$T(\Omega)\, \lambda_1(\Omega)^\beta \;\geq\; T(B)\, \lambda_1(B)^\beta, \qquad \beta = \tfrac{n+2}{2}. \tag{26}$$

Equivalently $\Psi(\Omega) \geq \Psi(B_1) = T_n L_n^\beta$, with equality if and only if Ω is a ball (up to translation and a Lebesgue-null modification).

Strategy. Both sides of (26) have the same scaling and $\Psi(B) = \Psi(B_1)$ for every ball; so it suffices to compare Ω with a single, well-chosen ball. The idea is to rearrange the first eigenfunction u of Ω into a radial decreasing function u^* on a ball B in such a way that

(A) the Dirichlet energy is *exactly preserved*, $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$, and
(B) the L^2 norm *does not decrease*, $\int_B (u^*)^2 \geq \int_\Omega u^2$.

Granting (A)–(B), the Rayleigh principle (23) on B gives

$$\lambda_1(B) \;\leq\; \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \;\leq\; \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} = \lambda_1(\Omega), \tag{27}$$

the last equality because u is the first eigenfunction of Ω.

Reading the chain (27). The first inequality is the variational characterisation (23) applied on B to the admissible competitor $u^* \in H_0^1(B)$: the minimum is no larger than the quotient at any one competitor. The second inequality replaces the numerator by an equal quantity (property (A), equality) and the denominator by a smaller-or-equal one (property (B), $\int_B (u^*)^2 \geq \int_\Omega u^2$): increasing the denominator of a positive fraction decreases it, so the quotient over B is $\leq$ the quotient over Ω. The final equality is the definition of u as a minimiser of (23) on Ω, so its Rayleigh quotient equals $\lambda_1(\Omega)$. Hence $\lambda_1(B) \leq \lambda_1(\Omega)$.

The decisive design choice is which geometric quantity of the superlevel sets the ball B is built to match. Schwarz symmetrisation matches *volume* and yields the Faber–Krahn inequality directly; the Kohler–Jobin rearrangement instead matches a *torsional functional*, the *modified torsional rigidity* introduced next, and this is exactly what allows the Dirichlet energy to be preserved (not merely decreased) while the L^2 norm grows. The relation $\lambda_1(B) \leq \lambda_1(\Omega)$ *at equal torsion* of B and Ω is then the inequality (26), as the assembly at the end of Part 1 makes precise.

Standing regularity facts (cited, not proved).

(R1) *(Regularity of the eigenfunction.)* The first eigenfunction u of Ω may be taken positive, solves $-\Delta u = \lambda u$ in Ω with $\lambda = \lambda_1(\Omega)$, and satisfies $u \in L^\infty(\Omega)$ (Moser iteration) and $u \in C^\infty(\Omega)$ in the interior: indeed $-\Delta u = \lambda u$ with $u \in L^\infty$ gives $u \in C^{1,\alpha}_{\mathrm{loc}}$ by Schauder, and bootstrapping the equation gives $u \in C^\infty(\Omega)$ ([4, Prop. 2.4], [14]). In particular Sard's theorem applies to u in the interior, so a.e. $t \in (0, M)$ is a regular value. Put $M := \|u\|_{L^\infty(\Omega)} < \infty$.

Why this is enough. Positivity is the Krein–Rutman/maximum-principle fact that the first eigenfunction has a sign; we fix the sign $u > 0$. The interior smoothness lets us apply, for a.e. level t, the divergence theorem on $\{u > t\}$ (whose boundary $\{u = t\}$ is then a smooth hypersurface, by the implicit function theorem at points where $\nabla u \neq 0$, which is a.e. on a regular level set). Sard guarantees a.e. level is regular, so all level-set statements below (”for a.e. t”) are legitimate.

(R2) *(Coarea formula.)* For $u \in W^{1,1}_{\mathrm{loc}}$ and Borel $g \geq 0$, $\int_\Omega g\, |\nabla u|\, dx = \int_0^M \big(\int_{\{u=t\}} g\, d\mathcal{H}^{n-1}\big)\, dt$ (Federer; see Maggi, *Sets of Finite Perimeter*, Thm. 13.1). Taking $g = \mathbf{1}_\Omega\, h(u)/\,|\nabla u|$ on $\{\nabla u \neq 0\}$ converts a bulk integral of $h(u)$ weighted by $|\nabla u|^{-1}$ into a level integral, which is the mechanism behind all the ”Cavalieri+coarea” computations to follow.

(R3) *(Saint–Venant inequality.)* For any open bounded $A \subset \mathbb{R}^n$, $T(A) \leq T(A^\#)$ where $A^\#$ is the ball with $|A^\#| = |A|$; in the scale-free form we use it,

$$T(A) \;\leq\; T_n\Big(\frac{|A|}{\omega_n}\Big)^{(n+2)/n}, \tag{28}$$

with equality iff A is a ball. ([4, (1.7) with $p = 2$].) The exponent is dictated by scaling: from $T(A^\#) = T_n(|A^\#|\,/\omega_n \cdot \omega_n/\omega_n) \cdots$, more directly $T(A^\#) = T_n R^{n+2}$ with $|A^\#| = \omega_n R^n$, i.e. $R = (|A|\,/\omega_n)^{1/n}$ and $T(A^\#) = T_n(|A|\,/\omega_n)^{(n+2)/n}$, which is (28).

(R4) *(Distribution function.)* The distribution function $\mu_u(t) := |\{u > t\}|$ is non-increasing; together with the coarea formula this gives, for a.e. regular value t, that $\int_{\{u=t\}} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ is finite and $\mu_u'(t) = -\int_{\{u=t\}} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ (Brothers–Ziemer). Thus μ_u is locally absolutely continuous on the set of regular values, the fact we need to differentiate under the integral and to integrate by the change of variables $t = \varphi(\tau)$.

Two preliminary identities. We record the variational form of T and the level-set flux identity for the eigenfunction; both are elementary. The first says torsion is the maximum of a concave energy and will let us define a *constrained* torsion (the modified torsion) by maximising the same energy over a smaller class. The second computes, for the eigenfunction, the boundary flux through each level set in closed form; it is what verifies the integrability condition that makes the modified torsion finite.

Lemma 2.11 (Variational torsion and the eigenfunction flux)**.**

(i) *For open A of finite measure, with $F(w) := \int_A w\, dx - \frac{1}{2}\int_A |\nabla w|^2\, dx$,*

$$T(A) = 2 \max_{w \in H^1_0(A)} F(w), \tag{29}$$

the maximiser being the torsion function u_A; equivalently $T(A) = \max\{(\int_A w)^2/\int_A |\nabla w|^2 :$ $0 \neq w \in H^1_0(A)\}$.

(ii) *The first eigenfunction u satisfies, for a.e. $t \in (0, M)$,*

$$\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{n-1} = \lambda \int_{\{u>t\}} u\, dx. \tag{30}$$

Proof. (i): The functional F is strictly concave on $H_0^1(A)$ because $w \mapsto \frac{1}{2}\int |\nabla w|^2$ is strictly convex (its Hessian is the positive-definite Dirichlet form $\int \nabla \cdot \nabla$) while $w \mapsto \int w$ is linear; and F is coercive because, by the Poincaré inequality on the finite-measure set A, $\int_A w \le |A|^{1/2}\,(\int_A w^2)^{1/2} \le C(A)\,(\int_A |\nabla w|^2)^{1/2}$, so the quadratic term dominates the linear one as $\|\nabla w\|_2 \to \infty$. A strictly concave coercive functional on a Hilbert space has a unique maximiser, characterised by its Euler–Lagrange equation. Computing the first variation, for $\phi \in H_0^1(A)$,

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} F(w+\varepsilon\phi) = \int_A \phi\,dx - \int_A \nabla w \cdot \nabla\phi\,dx = 0 \quad \Longleftrightarrow \quad -\Delta w = 1 \text{ weakly},$$

so the maximiser is the torsion function u_A. Testing $-\Delta u_A = 1$ against $u_A \in H_0^1(A)$ itself gives the key normalisation

$$\int_A |\nabla u_A|^2\,dx = \int_A u_A\,dx = T(A). \tag{31}$$

Substituting (31) into F,

$$F(u_A) = \int_A u_A\,dx - \tfrac{1}{2}\int_A |\nabla u_A|^2\,dx = T(A) - \tfrac{1}{2}T(A) = \tfrac{1}{2}T(A),$$

which is (29) after multiplying by 2. For the second (quotient) form: for $0 \ne w \in H_0^1(A)$ and $c > 0$, $F(cw) = c\int_A w - \frac{c^2}{2}\int_A |\nabla w|^2$ is a downward parabola in c, maximised at $c_\star = \int_A w / \int_A |\nabla w|^2$ with value $F(c_\star w) = \frac{1}{2}(\int_A w)^2/\int_A |\nabla w|^2$; taking the supremum over directions w and using (29) gives $T(A) = \max\{(\int_A w)^2/\int_A |\nabla w|^2\}$. This is the standard Rayleigh-type characterisation of T ([4, Prop. 2.2] with $p=2$).

(ii): Fix a regular value t, so that $\{u > t\}$ is open with smooth boundary $\{u = t\}$ on which $\nabla u \ne 0$. On $\{u=t\}$ the function u decreases outward (we cross from $\{u>t\}$ to $\{u<t\}$), so the outward unit normal to $\{u>t\}$ is $\nu = -\nabla u/|\nabla u|$, whence the normal derivative is

$$\partial_\nu u = \nabla u \cdot \nu = \nabla u \cdot \Big(-\frac{\nabla u}{|\nabla u|}\Big) = -\frac{|\nabla u|^2}{|\nabla u|} = -|\nabla u|\,.$$

Integrate the equation $-\Delta u = \lambda u$ over $\{u>t\}$ and apply the divergence theorem to the vector field ∇u:

$$\lambda \int_{\{u>t\}} u\,dx = \int_{\{u>t\}} (-\Delta u)\,dx = -\int_{\{u=t\}} \partial_\nu u\,d\mathcal{H}^{n-1} = \int_{\{u=t\}} |\nabla u|\,d\mathcal{H}^{n-1},$$

the middle equality being $\int_{\{u>t\}} \Delta u = \int_{\partial\{u>t\}} \partial_\nu u$ (Gauss–Green), and the last equality using $\partial_\nu u = -|\nabla u|$. Rearranged, this is (30). $\square$

The modified torsional rigidity. The engine of the technique is the following functional, which assigns to a nonnegative function a torsion-type quantity that is sensitive to the *level structure* of the function, not just to its domain. Concretely, ordinary torsion (29) maximises F over *all* of $H_0^1(\Omega)$; the modified torsion maximises F only over functions of the form $g \circ u$, i.e. functions constant on the level sets of u. This constraint is exactly what carries the level structure of u into the functional, and it is what the rearrangement will match between Ω and the ball.

Definition 2.12 (Modified torsional rigidity)**.** Let $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$, $u \ge 0$, $M = \|u\|_\infty$, with distribution function $\mu(t) = |\{u>t\}|$. Assume the integrability condition

$$t \;\longmapsto\; \frac{\mu(t)}{\int_{\{u=t\}} |\nabla u|\,d\mathcal{H}^{n-1}} \;\in\; L^\infty(0,M) \tag{32}$$

For the first eigenfunction this holds: by (30) the ratio in (32) equals $\mu(t)/(\lambda \int_{\{u>t\}} u)$. On $[M/2, M)$ the bound $\int_{\{u>t\}} u \geq t\,\mu(t)$ gives ratio $\leq 1/(\lambda t) \leq 2/(\lambda M)$; on $(0, M/2]$ the monotonicity $\int_{\{u>t\}} u \geq \int_{\{u>M/2\}} u > 0$ together with $\mu(t) \leq |\Omega|$ gives ratio $\leq |\Omega|/(\lambda \int_{\{u>M/2\}} u)$. Hence the ratio is bounded on $(0, M)$ (cf. [4, Lem. 3.2]). With $\mathcal{L} := \{g \in \mathrm{Lip}([0, M]) : g(0) = 0\}$, define the *modified torsional rigidity of* Ω *along* u

$$\mathrm{T}_{\mathrm{mod}}(\Omega; u) := 2 \sup_{g\in\mathcal{L}} F\big(g \circ u\big), \qquad F(w) = \int_\Omega w\,dx - \tfrac{1}{2}\int_\Omega |\nabla w|^2\,dx. \tag{33}$$

Unpacking the verification of (32)*.* Substituting the flux identity (30), $\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{n-1} = \lambda \int_{\{u>t\}} u$, into the denominator turns the ratio in (32) into $\mu(t)/\big(\lambda \int_{\{u>t\}} u\,dx\big)$. We bound this on two ranges of t:

- For $t \in [M/2, M)$: since $u > t$ on $\{u > t\}$, we have $\int_{\{u>t\}} u\,dx \geq t\,|\{u > t\}| = t\,\mu(t)$, so
$$\frac{\mu(t)}{\lambda \int_{\{u>t\}} u} \leq \frac{\mu(t)}{\lambda\, t\, \mu(t)} = \frac{1}{\lambda t} \leq \frac{1}{\lambda\,(M/2)} = \frac{2}{\lambda M},$$
using $t \geq M/2$ in the last step (and $\mu(t) > 0$ for $t < M$, so the cancellation is legitimate).
- For $t \in (0, M/2]$: the numerator is bounded by $\mu(t) \leq |\Omega|$ (monotonicity of μ and $\mu \leq |\Omega|$), while the denominator integral is non-increasing in t (its domain $\{u > t\}$ shrinks as t grows), so for $t \leq M/2$, $\int_{\{u>t\}} u \geq \int_{\{u>M/2\}} u > 0$ (positive because u is positive on the nonempty open set $\{u > M/2\}$). Hence the ratio is $\leq |\Omega|/\big(\lambda \int_{\{u>M/2\}} u\big)$, a finite constant.

Taking the larger of the two bounds shows the ratio lies in $L^\infty(0, M)$.

The pointwise bound $\mathrm{T}_{\mathrm{mod}} \leq T$*.* Since each competitor $g \circ u$ with $g \in \mathcal{L}$ lies in $H^1_0(\Omega)$ (a Lipschitz function of an H^1_0 function vanishing at 0 is again H^1_0, by the chain rule for Lipschitz compositions), the class $\{g \circ u : g \in \mathcal{L}\}$ is a subset of $H^1_0(\Omega)$. Maximising F over a smaller set cannot exceed the maximum over the whole space, so by (29),

$$\mathrm{T}_{\mathrm{mod}}(\Omega; u) = 2 \sup_{g\in\mathcal{L}} F(g \circ u) \;\leq\; 2 \max_{w\in H^1_0(\Omega)} F(w) = T(\Omega). \tag{34}$$

This inequality, used repeatedly below, is the single channel through which the Saint–Venant inequality (28) enters the argument.

Lemma 2.13 (Closed form of $\mathrm{T}_{\mathrm{mod}}$)**.** *Under the hypotheses of Definition 2.12, writing* $P(t) := \int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{n-1}$*, the supremum in* (33) *is uniquely attained by the nondecreasing* g_0 *with* $g_0'(t) = \mu(t)/P(t)$*, and*

$$\mathrm{T}_{\mathrm{mod}}(\Omega; u) = \int_0^M \frac{\mu(t)^2}{P(t)}\,dt. \tag{35}$$

Proof. Reduction to nondecreasing g*.* Given $g \in \mathcal{L}$, set $\tilde{g}(t) := \int_0^t |g'(s)|\,ds$. Then $\tilde{g} \in \mathcal{L}$ (it is Lipschitz with $\tilde{g}(0) = 0$ and $\tilde{g}' = |g'|$ a.e.) and $\tilde{g}$ is nondecreasing. We compare $F(\tilde{g} \circ u)$ with $F(g \circ u)$:

- *Gradient term unchanged.* By the chain rule $\nabla(g \circ u) = g'(u)\nabla u$, so $|\nabla(g \circ u)|^2 = g'(u)^2\,|\nabla u|^2$; similarly $|\nabla(\tilde{g} \circ u)|^2 = \tilde{g}'(u)^2\,|\nabla u|^2 = |g'(u)|^2\,|\nabla u|^2 = g'(u)^2\,|\nabla u|^2$. The two gradient terms coincide pointwise, hence $\int_\Omega |\nabla(\tilde{g} \circ u)|^2 = \int_\Omega |\nabla(g \circ u)|^2$.
- *Linear term does not decrease.* For every t, $g(t) = \int_0^t g' \leq \int_0^t |g'| = \tilde{g}(t)$, so $\tilde{g} \geq g$ pointwise, and since $u \geq 0$, $\tilde{g}(u) \geq g(u)$, whence $\int_\Omega \tilde{g}(u) \geq \int_\Omega g(u)$.

Therefore $F(\tilde g\circ u)\ge F(g\circ u)$, and the supremum in (33) is unchanged if we restrict to nondecreasing $g\in\mathcal{L}$.

Reduction to a one-dimensional integral. Let $g\in\mathcal{L}$ be nondecreasing. Writing $g(t)=\int_0^t g'(s)\,ds$ and applying Cavalieri's principle (the layer-cake/Fubini identity $\int_\Omega h(u)\,dx=\int_0^M h'(t)\mu(t)\,dt$ valid for h with $h(0)=0$, which is the integral against $-d\mu$ written via the distribution function),

$$\int_\Omega g(u)\,dx=\int_0^M g'(t)\,\mu(t)\,dt.$$

For the gradient term, the chain rule gives $|\nabla(g\circ u)|^2=g'(u)^2\,|\nabla u|^2=g'(u)\cdot\big(g'(u)\,|\nabla u|\big)\,|\nabla u|$; applying the coarea formula (R2) with $h:=g'(u)^2\,|\nabla u|$ (so that $h\,|\nabla u|^{-1}=g'(u)^2\,|\nabla u|$, matching the bulk integrand) – more directly, coarea with the Borel function $g'(u)^2\,|\nabla u|$ slices the integral by level sets, on each of which $g'(u)^2=g'(t)^2$ is constant:

$$\int_\Omega |\nabla(g\circ u)|^2\;dx=\int_\Omega g'(u)^2\,|\nabla u|^2\;dx=\int_0^M g'(t)^2\Big(\int_{\{u=t\}}|\nabla u|\;d\mathcal{H}^{n-1}\Big)\,dt=\int_0^M g'(t)^2\,P(t)\,dt.$$

(Here coarea is used in the form $\int_\Omega \Phi(u)\,|\nabla u|\;dx=\int_0^M\Phi(t)P(t)\,dt$ with $\Phi=g'(\cdot)^2$, taking $\Phi(u)\,|\nabla u|=g'(u)^2\,|\nabla u|=|\nabla(g\circ u)|^2/|\nabla u|\cdot|\nabla u|$, i.e. slicing $g'(u)^2$ against the factor $|\nabla u|$ that coarea supplies.) Hence

$$F(g\circ u)=\int_\Omega g(u)\,dx-\tfrac12\int_\Omega|\nabla(g\circ u)|^2\;dx=\int_0^M\Big[g'(t)\,\mu(t)-\tfrac12\,g'(t)^2\,P(t)\Big]\,dt.$$

Pointwise Young maximisation. The integrand depends on g only through the value $s:=g'(t)\ge0$ at each fixed t, through the quadratic $q_t(s):=s\,\mu(t)-\frac12 s^2P(t)$. Since $P(t)>0$ for a.e. regular t (the boundary flux of a nonconstant function), q_t is a downward parabola in s; setting $q_t'(s)=\mu(t)-sP(t)=0$ gives the unconstrained maximiser $s_\star=\mu(t)/P(t)\ge0$ (which is admissible since it is nonnegative), with maximum value

$$\max_{s\ge0}q_t(s)=q_t(s_\star)=\frac{\mu(t)}{P(t)}\cdot\mu(t)-\tfrac12\frac{\mu(t)^2}{P(t)^2}\,P(t)=\frac{\mu(t)^2}{P(t)}-\frac{\mu(t)^2}{2P(t)}=\frac{\mu(t)^2}{2P(t)}.$$

Integrating the pointwise bound $q_t(g'(t))\le\max_{s\ge0}q_t(s)=\mu(t)^2/(2P(t))$ over t,

$$F(g\circ u)=\int_0^M q_t\big(g'(t)\big)\,dt\;\le\;\int_0^M\frac{\mu(t)^2}{2P(t)}\,dt,$$

with equality if and only if $g'(t)=s_\star=\mu(t)/P(t)$ for a.e. t. By the integrability condition (32), the function $t\mapsto\mu(t)/P(t)$ is in $L^\infty(0,M)$, so $g_0(t):=\int_0^t\mu(s)/P(s)\,ds$ is Lipschitz with $g_0(0)=0$, i.e. $g_0\in\mathcal{L}$, and it attains the bound. Multiplying through by 2 (recall the factor 2 in (33)),

$$\mathrm{T}_{\mathrm{mod}}(\Omega;u)=2\sup_{g\in\mathcal{L}}F(g\circ u)=2\int_0^M\frac{\mu(t)^2}{2P(t)}\,dt=\int_0^M\frac{\mu(t)^2}{P(t)}\,dt,$$

which is (35), attained uniquely (the parabola has a unique maximiser) by g_0. □

The next proposition is the isoperimetric statement that makes the rearrangement decrease the Rayleigh quotient. It is the only place where the Saint–Venant inequality enters: it converts the algebraic bound $\mathrm{T}_{\mathrm{mod}}\le T$ into the *geometric* statement that the torsion-matched ball is smaller than Ω.

Proposition 2.14 (Isoperimetric inequality for $\mathrm{T}_{\mathrm{mod}}$). *Let $u\in H_0^1(\Omega)\cap L^\infty$, $u\ge0$, satisfy (32). If $B\subset\mathbb{R}^n$ is a ball with $T(B)=\mathrm{T}_{\mathrm{mod}}(\Omega;u)$, then $|B|\le|\Omega|$, with equality iff Ω is a ball and u is (a translate of) a radial function.*

Proof. By hypothesis $T(B) = \mathrm{T}_{\mathrm{mod}}(\Omega; u)$, and by (34) (the remark after Definition 2.12) $\mathrm{T}_{\mathrm{mod}}(\Omega; u) \le T(\Omega)$; chaining,

$$T(B) = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \;\le\; T(\Omega).$$

The Saint–Venant inequality (28) bounds the right-hand side from above by the ball value at equal volume: $T(\Omega) \le T_n(|\Omega|/\omega_n)^{(n+2)/n}$. The left-hand side is computed exactly by (24): writing $|B| = \omega_n R^n$ with R the radius of B, $T(B) = T_n R^{n+2} = T_n(|B|/\omega_n)^{(n+2)/n}$. Substituting,

$$T_n\Big(\frac{|B|}{\omega_n}\Big)^{(n+2)/n} = T(B) \le T(\Omega) \le T_n\Big(\frac{|\Omega|}{\omega_n}\Big)^{(n+2)/n}.$$

Cancelling the positive constant T_n and the positive normalisation $\omega_n^{-(n+2)/n}$, and raising to the positive power $n/(n+2)$ (a strictly increasing operation on $[0,\infty)$),

$$|B| \;\le\; |\Omega|.$$

Equality case. If $|B| = |\Omega|$, then both inequalities in the chain are equalities. Equality in the right inequality is equality in Saint–Venant (28), which by (R3) forces Ω to be a ball. Equality $T(B) = T(\Omega)$ together with $T(B) = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \le T(\Omega)$ forces $\mathrm{T}_{\mathrm{mod}}(\Omega; u) = T(\Omega)$; comparing (33) with (29), the constrained supremum equals the unconstrained maximum, so the maximiser $g_0 \circ u$ of the modified problem must coincide with the unconstrained maximiser, the torsion function u_Ω. On a ball u_Ω is radial (by (24), $u_{B_r}(x) = (r^2 - |x|^2)/(2n)$), so $g_0 \circ u = u_\Omega$ is radial; as g_0 is a strictly increasing bijection on the range of u (its derivative μ/P is a.e. positive), u itself is radial up to translation. This is [4, Prop. 3.8]. □

The Kohler–Jobin rearrangement. We now build u^*. The plan: instead of rearranging u so that its superlevel sets become balls of *equal volume* (Schwarz), we make them balls of *equal modified torsion.* We first record, for each level t, the modified torsion of the truncated function on the corresponding superlevel set, then invert this monotone quantity to parametrise the levels of u^*.

For $t \in [0, M]$ set $\Omega_t = \{u > t\}$ and let

$$\mathcal{T}(t) := \mathrm{T}_{\mathrm{mod}}\big(\Omega_t; (u-t)_+\big). \tag{36}$$

Since $|\{(u-t)_+ > s\}| = \mu(t+s)$ and the coarea factor of $(u-t)_+$ at level s is $P(t+s)$, formula (35) gives the explicit expression

$$\mathcal{T}(t) = \int_t^M \frac{\mu(\tau)^2}{P(\tau)}\, d\tau, \tag{37}$$

so that $\mathcal{T} \in \mathrm{Lip}_{\mathrm{loc}}([0, M))$, $\mathcal{T}(M) = 0$, $\mathcal{T}(0) = \mathrm{T}_{\mathrm{mod}}(\Omega; u) =: T_\star$, and for a.e. t

$$\mathcal{T}'(t) = -\frac{\mu(t)^2}{P(t)} < 0. \tag{38}$$

Thus $\mathcal{T}$ is a strictly decreasing bijection $[0, M] \to [0, T_\star]$.

Derivation of (37) *and its consequences.* Apply Lemma 2.13 to the function $v := (u-t)_+$ on the set $\Omega_t = \{u > t\}$. The superlevel sets of v are $\{v > s\} = \{(u-t)_+ > s\} = \{u > t+s\} = \Omega_{t+s}$, so the distribution function of v is $\mu_v(s) = |\Omega_{t+s}| = \mu(t+s)$; and on a regular level $\{v = s\} = \{u = t+s\}$ we have $|\nabla v| = |\nabla u|$ (translation by the constant t does not change the gradient on $\{u > t\}$), so the coarea factor of v is $P_v(s) = \int_{\{u=t+s\}} |\nabla u| \, d\mathcal{H}^{n-1} = P(t+s)$. The top level of v is $M - t$. Hence by (35),

$$\mathcal{T}(t) = \mathrm{T}_{\mathrm{mod}}(\Omega_t; v) = \int_0^{M-t} \frac{\mu_v(s)^2}{P_v(s)}\, ds = \int_0^{M-t} \frac{\mu(t+s)^2}{P(t+s)}\, ds = \int_t^M \frac{\mu(\tau)^2}{P(\tau)}\, d\tau,$$

the last step being the substitution $\tau = t + s$. The integrand $\mu(\tau)^2/P(\tau)$ is, by (32) and $\mu \le |\Omega|$, locally integrable, so $\mathcal{T}$ is locally absolutely continuous (indeed locally Lipschitz away from $t = M$), and the fundamental theorem of calculus gives, for a.e. t, $\mathcal{T}'(t) = -\mu(t)^2/P(t)$, which is (38); it is < 0 because $\mu(t) > 0$ for $t < M$ and $P(t) > 0$. The boundary values $\mathcal{T}(M) = \int_M^M = 0$ and $\mathcal{T}(0) = \int_0^M \mu^2/P = \mathrm{T}_{\mathrm{mod}}(\Omega; u) =: T_\star$ follow by setting $t = M$ and $t = 0$ in (37) and comparing with (35). A continuous, strictly decreasing function from $[0, M]$ onto $[\mathcal{T}(M), \mathcal{T}(0)] = [0, T_\star]$ is a bijection, with a strictly decreasing continuous inverse.
Definition of u^*. Let $B = B_{R_\star}$ be the ball with $T(B) = T_\star$, i.e. $R_\star = (T_\star/T_n)^{1/(n+2)}$. For $\tau \in [0, T_\star]$ let $R(\tau)$ be the radius with $T(B_{R(\tau)}) = \tau$, i.e.

$$R(\tau) = (\tau/T_n)^{1/(n+2)}, \qquad \left|B_{R(\tau)}\right| = \omega_n R(\tau)^n = \omega_n\,(\tau/T_n)^{n/(n+2)}. \tag{39}$$

The radius formula in (39) is the inversion of $T(B_R) = T_n R^{n+2}$ from (24): setting $T_n R^{n+2} = \tau$ and solving gives $R = (\tau/T_n)^{1/(n+2)}$; the volume then follows from $|B_R| = \omega_n R^n$, substituting this R.

Let $\varphi := \mathcal{T}^{-1} : [0, T_\star] \to [0, M]$ (so $\varphi(0) = M$, $\varphi(T_\star) = 0$, $\varphi' < 0$). We define u^* on B to be the radial decreasing function whose value on the sphere $\{|x| = R(\tau)\}$ is $\psi(\tau)$, where $\psi : [0, T_\star] \to [0, \infty)$ is the locally Lipschitz (monotone, absolutely continuous) function determined by

$$\big(-\psi'(\tau)\big)\,\mu^*\big(\psi(\tau)\big) = \big(-\varphi'(\tau)\big)\,\mu\big(\varphi(\tau)\big), \qquad \psi(T_\star) = 0, \tag{40}$$

μ^* denoting the distribution function of u^*. Concretely the superlevel set $\{u^* > \psi(\tau)\} = B_{R(\tau)}$ has, by construction, torsional rigidity exactly τ, the value of the *modified* torsion that the corresponding superlevel set $\Omega_{\varphi(\tau)}$ of u carries. In particular $\{u^* > 0\} = B = B_{R_\star}$, so

$$T(B) = T_\star = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \quad \le \quad T(\Omega). \tag{41}$$

Since $\mu^*(\psi(\tau)) = \left|B_{R(\tau)}\right| = \omega_n R(\tau)^n$, equation (40) integrates to an explicit ψ; we shall only need the two qualitative consequences (A), (B), proved next.

Why this construction is well posed and what (40) *encodes.* The map $\tau \mapsto R(\tau)$ assigns to each torsion level τ the radius of the ball of that torsion; the map $\tau \mapsto \varphi(\tau) = \mathcal{T}^{-1}(\tau)$ assigns to each torsion level τ the height t of u whose tail $(u - t)_+$ carries modified torsion τ. We then declare u^* to take the value $\psi(\tau)$ on the sphere of radius $R(\tau)$; this defines u^* as a radial decreasing function once ψ is fixed, since $R(\tau)$ increases with τ while we will see $\psi(\tau)$ decreases. The relation (40) fixes ψ by matching, at each τ, the *differential of the distribution function* of u^* to that of u: both sides equal $\frac{d}{d\tau}\big(\text{volume of the } \tau\text{-th superlevel set}\big) \cdot (-1)$ rewritten in the level variable, and this is precisely the identity that makes the Dirichlet integrands coincide in Lemma 2.15. The chain $T(B) = T_\star = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \le T(\Omega)$ in (41) is the special case $\tau = T_\star$ (so $\varphi(T_\star) = 0$, $\Omega_0 = \Omega$) of the construction, combined with (34).

Lemma 2.15 (Dirichlet energy is preserved). $\displaystyle\int_B |\nabla u^*|^2\,dx = \int_\Omega |\nabla u|^2\,dx.$

Proof. The original side. By the coarea formula (R2) applied with $g \equiv |\nabla u|$ on each level (so $\int_\Omega |\nabla u|^2 = \int_\Omega |\nabla u| \cdot |\nabla u| = \int_0^M \int_{\{u=t\}} |\nabla u|\,d\mathcal{H}^{n-1}\,dt = \int_0^M P(t)\,dt$), and then the change of variable $t = \varphi(\tau)$ (the one-dimensional area formula for the locally Lipschitz strictly decreasing $\varphi = \mathcal{T}^{-1}$, with $dt = \varphi'(\tau)\,d\tau = -(-\varphi'(\tau))\,d\tau$ and reversed limits $t : 0 \to M$ corresponding to

$\tau : T_\star \to 0$; Lemma 2.5 of [4]),

$$\int_\Omega |\nabla u|^2 \, dx = \int_0^M P(t)\, dt = \int_0^{T_\star} P(\varphi(\tau))\,(-\varphi'(\tau))\, d\tau. \tag{42}$$

(The two sign flips – reversing the limits of integration and $dt = -(-\varphi')\, d\tau$ – cancel, leaving the positive integrand $P(\varphi)(-\varphi')$ over $[0, T_\star]$.)

Differentiating the identity $\mathcal{T}(\varphi(\tau)) = \tau$ (valid because $\varphi = \mathcal{T}^{-1}$) by the chain rule gives $\mathcal{T}'(\varphi(\tau))\,\varphi'(\tau) = 1$, hence $\varphi'(\tau) = 1/\mathcal{T}'(\varphi(\tau))$; substituting $\mathcal{T}'(\varphi) = -\mu(\varphi)^2/P(\varphi)$ from (38),

$$-\varphi'(\tau) = \frac{1}{-\mathcal{T}'(\varphi(\tau))} = \frac{1}{\mu(\varphi(\tau))^2/P(\varphi(\tau))} = \frac{P(\varphi(\tau))}{\mu(\varphi(\tau))^2}. \tag{43}$$

This is the key relation. We use it to eliminate the factor $P(\varphi)$ from (42): from (43), $P(\varphi) = \mu(\varphi)^2\,(-\varphi')$, so

$$P(\varphi)\,(-\varphi') = \mu(\varphi)^2\,(-\varphi')\cdot(-\varphi') = \mu(\varphi)^2\,(-\varphi')^2.$$

Substituting into (42),

$$\int_\Omega |\nabla u|^2 \, dx = \int_0^{T_\star} \mu(\varphi(\tau))^2\,(-\varphi'(\tau))^2\, d\tau. \tag{44}$$

The rearranged side, computed in full. We now establish the radial analogue (51), spelling out the radial-gradient expansion that the source compresses. The function u^* is radial decreasing, say $u^*(x) = h(|x|)$ for a decreasing $h : [0, R_\star] \to [0, \infty)$ with $h(R(\tau)) = \psi(\tau)$ and $h(R_\star) = \psi(T_\star) = 0$. On the sphere $\{|x| = R(\tau)\}$ the gradient of a radial function is purely radial, $\nabla u^*(x) = h'(|x|)\,\frac{x}{|x|}$, so

$$|\nabla u^*| = \left|h'(|x|)\right| = -h'(R(\tau)) \qquad \text{(constant on the sphere; } h \text{ decreasing, so } h' \le 0). \tag{45}$$

Differentiating $h(R(\tau)) = \psi(\tau)$ in τ by the chain rule, $h'(R(\tau))\,R'(\tau) = \psi'(\tau)$, whence

$$-h'(R(\tau)) = \frac{-\psi'(\tau)}{R'(\tau)} = \frac{-\psi'(\tau)}{R'(\tau)}, \qquad \text{i.e.} \qquad |\nabla u^*|\,\big|_{|x|=R(\tau)} = \frac{-\psi'(\tau)}{R'(\tau)}, \tag{46}$$

using $R'(\tau) > 0$ (the radius increases with the torsion level τ) and $-\psi'(\tau) \ge 0$ (we verify ψ is decreasing in Lemma 2.16; here we only use $|\nabla u^*| = |\psi'|/R'$). This is the announced identity $|\nabla u^*| = (-\psi')/R'$ at the sphere of radius $R(\tau)$.

Now compute the two ingredients of the coarea slicing on B. The sphere $\{u^* = \psi(\tau)\} = \{|x| = R(\tau)\}$ has area $\mathcal{H}^{n-1}(\{|x| = R(\tau)\}) = n\omega_n R(\tau)^{n-1}$ (the surface area of a sphere of radius $R(\tau)$, with $n\omega_n = \sigma_{n-1}$ the unit sphere area). The distribution function of u^* at the level $\psi(\tau)$ is the volume of the corresponding superlevel ball,

$$\mu^*(\psi(\tau)) = \left|B_{R(\tau)}\right| = \omega_n R(\tau)^n. \tag{47}$$

The coarea factor of u^* at level $\psi(\tau)$ is

$$P^*(\psi(\tau)) = \int_{\{u^*=\psi(\tau)\}} |\nabla u^*|\, d\mathcal{H}^{n-1} = \underbrace{\frac{-\psi'(\tau)}{R'(\tau)}}_{|\nabla u^*|}\cdot\underbrace{n\omega_n R(\tau)^{n-1}}_{\text{sphere area}} = \frac{n\omega_n R(\tau)^{n-1}}{R'(\tau)}\,(-\psi'(\tau)), \tag{48}$$

the gradient being constant on the sphere, so it pulls out of the surface integral, multiplying the area. We also need $R'(\tau)$ explicitly. From $R(\tau) = (\tau/T_n)^{1/(n+2)}$,

$$R'(\tau) = \frac{1}{n+2}\,(\tau/T_n)^{1/(n+2)-1}\,\frac{1}{T_n} = \frac{1}{(n+2)T_n}\,(\tau/T_n)^{-(n+1)/(n+2)} = \frac{R(\tau)}{(n+2)\,\tau}, \tag{49}$$

where the last step uses $(\tau/T_n)^{-(n+1)/(n+2)} = (\tau/T_n)^{1/(n+2)} \cdot (\tau/T_n)^{-1} = R(\tau)\, T_n/\tau$, so $R'(\tau) = \frac{1}{(n+2)T_n} \cdot R(\tau)\, T_n/\tau = R(\tau)/((n+2)\tau)$.

We now write the Dirichlet energy of u^* via coarea (R2) (slicing $\int_B |\nabla u^*|^2 = \int_B |\nabla u^*| \cdot |\nabla u^*|$ by levels of u^*, then changing variable to τ via $u^* = \psi(\tau)$, $du^* = \psi'(\tau)\, d\tau$):

$$\int_B |\nabla u^*|^2 \, dx = \int_0^{M^*} \Big(\int_{\{u^*=\ell\}} |\nabla u^*| \, d\mathcal{H}^{n-1} \Big) |\nabla u^*| \big|_{\{u^*=\ell\}} \, d\ell = \int_0^{T_\star} P^*(\psi(\tau)) \cdot \frac{-\psi'(\tau)}{R'(\tau)} \, (-\psi'(\tau))\, d\tau,$$

where $M^* = \psi(0)$ is the maximum of u^*, the change of variable $\ell = \psi(\tau)$ carries $d\ell = \psi'(\tau)\, d\tau$ with the sign and limit flips cancelling (exactly as in (42)), and the integrand factor $|\nabla u^*|\,|_{\{u^*=\psi(\tau)\}} = (-\psi')/R'$ is (46). Substituting $P^*(\psi(\tau))$ from (48),

$$\int_B |\nabla u^*|^2 \, dx = \int_0^{T_\star} \frac{n\omega_n R(\tau)^{n-1}}{R'(\tau)} (-\psi') \cdot \frac{-\psi'}{R'(\tau)} (-\psi')\, d\tau = \int_0^{T_\star} \frac{n\omega_n R(\tau)^{n-1}}{R'(\tau)^2} \, (-\psi'(\tau))^2 \, d\tau.$$

The constant collapses to 1 *exactly because* $T_n = \omega_n/(n(n+2))$. Substitute $R'(\tau) = R(\tau)/((n+2)\tau)$ from (49):

$$\frac{n\omega_n R^{n-1}}{R'^2} = \frac{n\omega_n R^{n-1}}{\big(R/((n+2)\tau)\big)^2} = n\omega_n R^{n-1} \cdot \frac{(n+2)^2\tau^2}{R^2} = n\omega_n (n+2)^2\, \tau^2\, R^{n-3}.$$

Now express R^{n-3} through $R^n = |B_R|/\omega_n = \mu^*(\psi)/\omega_n$ and through τ. From $\tau = T_n R^{n+2}$ we get $R^{n+2} = \tau/T_n$, so

$$\tau^2 R^{n-3} = \tau^2\, R^{n+2}\, R^{-5} \quad \text{is awkward; instead use} \quad \tau = T_n R^{n+2} \;\Rightarrow\; \tau^2 = T_n^2 R^{2(n+2)}.$$

Hence

$$n\omega_n (n+2)^2\, \tau^2 R^{n-3} = n\omega_n (n+2)^2\, T_n^2 R^{2n+4} R^{n-3} = n\omega_n (n+2)^2 T_n^2\, R^{3n+1}.$$

This still contains R; the cancellation is cleaner if we keep one factor of τ and one of μ^*. Write instead $\tau = T_n R^{n+2}$ and $\mu^*(\psi(\tau)) = \omega_n R^n$, so that

$$\tau^2 R^{n-3} = \tau \cdot \tau\, R^{n-3} = \tau \cdot (T_n R^{n+2})\, R^{n-3} = \tau\, T_n\, R^{2n-1},$$

and

$$n\omega_n (n+2)^2\, \tau^2 R^{n-3} = (n+2)^2\, \tau\, \big(n\omega_n T_n\big)\, R^{2n-1}.$$

Using $T_n = \omega_n/(n(n+2))$, the bracket is $n\omega_n T_n = n\omega_n \cdot \omega_n/(n(n+2)) = \omega_n^2/(n+2)$, so

$$n\omega_n (n+2)^2\, \tau^2 R^{n-3} = (n+2)^2\, \tau\, \frac{\omega_n^2}{n+2}\, R^{2n-1} = (n+2)\, \tau\, \omega_n^2\, R^{2n-1}.$$

Finally substitute $\tau = T_n R^{n+2} = \frac{\omega_n}{n(n+2)} R^{n+2}$:

$$(n+2)\, \tau\, \omega_n^2 R^{2n-1} = (n+2) \cdot \frac{\omega_n}{n(n+2)} R^{n+2} \cdot \omega_n^2 R^{2n-1} = \frac{\omega_n^3}{n}\, R^{3n+1}.$$

On the other hand $\mu^*(\psi)^2 = (\omega_n R^n)^2 = \omega_n^2 R^{2n}$, and we must check this equals the coefficient just computed; that is, the claim of the source is that the rearranged Dirichlet integrand equals $\mu^*(\psi)^2(-\psi')^2$. We verify:

$$\frac{n\omega_n R^{n-1}}{R'^2} \stackrel{?}{=} \mu^*(\psi)^2 = \omega_n^2 R^{2n}.$$

Indeed, from (49), $R' = R/((n+2)\tau)$ and $\tau = T_n R^{n+2} = \frac{\omega_n}{n(n+2)} R^{n+2}$, so

$$R' = \frac{R}{(n+2)\, \frac{\omega_n}{n(n+2)} R^{n+2}} = \frac{R \cdot n(n+2)}{(n+2)\, \omega_n R^{n+2}} = \frac{n}{\omega_n}\, R^{-(n+1)},$$

whence $R'^2 = \frac{n^2}{\omega_n^2} R^{-2(n+1)}$ and

$$\frac{n\omega_n R^{n-1}}{R'^2} = n\omega_n R^{n-1} \cdot \frac{\omega_n^2}{n^2} R^{2(n+1)} = \frac{\omega_n^3}{n} R^{n-1+2n+2} = \frac{\omega_n^3}{n} R^{3n+1}.$$

This matches the coefficient computed above, confirming the algebra; it is *not* literally $\omega_n^2 R^{2n} = \mu^*(\psi)^2$ unless we restore the factor that the change of variable carries. The clean statement is obtained by reinstating the coarea/level bookkeeping: writing the rearranged Dirichlet energy in the same variables as (44) – i.e. expressing it through $\mu^*(\psi)$ and $(-\psi')$ rather than through R – one finds, exactly as for the original side (where $P(\varphi)(-\varphi') = \mu(\varphi)^2(-\varphi')^2$ followed from $-\varphi' = P(\varphi)/\mu(\varphi)^2$), the radial relation $P^*(\psi)(-\psi') = \mu^*(\psi)^2(-\psi')^2$. We derive this last identity directly. From (48) and the computation $\frac{n\omega_n R^{n-1}}{R'} = \mu^*(\psi)^2/(-\psi') \cdot (-\psi') \cdots$, more transparently: the radial analogue of (43) is the statement that the ball $B_{R(\tau)}$ has torsion τ, i.e. $T(B_{R(\tau)}) = \tau$, which on differentiation in τ gives $1 = \frac{d}{d\tau} T(B_{R(\tau)})$. By the same flux/coarea computation that produced (38) for $\mathcal{T}$ – now for the *ordinary* torsion of a ball, whose level sets are the spheres – the "modified torsion equals torsion" relation on the ball yields the radial counterpart of (43),

$$-\psi'(\tau) = \frac{P^*(\psi(\tau))}{\mu^*(\psi(\tau))^2}, \qquad \text{equivalently} \qquad P^*(\psi)(-\psi') = \mu^*(\psi)^2(-\psi')^2. \tag{50}$$

Granting (50), the coarea expression for the rearranged Dirichlet energy becomes

$$\int_B |\nabla u^*|^2 \, dx = \int_0^{T_\star} P^*(\psi(\tau))\, (-\psi'(\tau))\, d\tau = \int_0^{T_\star} \mu^*(\psi(\tau))^2\, (-\psi'(\tau))^2\, d\tau, \tag{51}$$

the exact radial analogue of (44); this is the specialisation to $p = 2$ of the identity (4.4) of [4], obtained there by the same coarea computation. (The first equality in (51) is coarea on B exactly as in the first equality of (42), written in the τ variable; the second is (50).)

Verifying (50) *concretely.* Substitute the explicit $P^*(\psi)$ and $\mu^*(\psi)$ into the claimed identity. From (48) with $|\nabla u^*| = (-\psi')/R'$ and from (47),

$$\frac{P^*(\psi)}{\mu^*(\psi)^2} = \frac{(-\psi')\, n\omega_n R^{n-1}/R'}{(\omega_n R^n)^2} = (-\psi')\, \frac{n\omega_n R^{n-1}}{R'\, \omega_n^2 R^{2n}} = (-\psi')\, \frac{n}{\omega_n R^{n+1} R'}.$$

With $R' = \frac{n}{\omega_n} R^{-(n+1)}$ (computed just above), $\omega_n R^{n+1} R' = \omega_n R^{n+1} \cdot \frac{n}{\omega_n} R^{-(n+1)} = n$, so

$$\frac{P^*(\psi)}{\mu^*(\psi)^2} = (-\psi')\, \frac{n}{n} = (-\psi'),$$

which is exactly (50). (Equivalently this re-derives $-\psi' = P^*(\psi)/\mu^*(\psi)^2$ from scratch, using only the explicit ball geometry; the appeal to the "torsion of the ball" identity above is just the conceptual reason it holds.) Thus (51) is justified.

Matching the two sides. Now the defining relation (40), $\mu^*(\psi)(-\psi') = \mu(\varphi)(-\varphi')$, makes the integrands of (51) and (44) *equal*, by squaring:

$$\mu^*(\psi(\tau))^2(-\psi'(\tau))^2 = \big[\mu^*(\psi(\tau))(-\psi'(\tau))\big]^2 = \big[\mu(\varphi(\tau))(-\varphi'(\tau))\big]^2 = \mu(\varphi(\tau))^2(-\varphi'(\tau))^2,$$

the middle equality being (40) squared. Integrating over $[0, T_\star]$, the right-hand side of (51) equals that of (44), i.e. $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$. This is the asserted equality, and we emphasise it is an *exact* equality, not merely an inequality: the Dirichlet energy is preserved. □

Lemma 2.16 (L^2 norm does not decrease)**.** *$\displaystyle\int_B (u^*)^2\,dx \;\geq\; \int_\Omega u^2\,dx$, and more generally $\int_B f(u^*)\,dx \geq \int_\Omega f(u)\,dx$ for every convex $f : [0,\infty) \to [0,\infty)$ with $f(0) = 0$, strictly if f is strictly convex unless Ω is a ball and u radial.*

Proof. The point is the comparison of *distribution functions* along the rearrangement. We proceed in three steps: a level-by-level volume comparison, the consequent monotonicity $\psi \geq \varphi$, and the convexity chain.

Step 1: volume comparison (52). Apply Proposition 2.14 not to u but to the truncations. For each $\tau \in [0, T_\star]$, the construction declares that the ball $B_{R(\tau)}$ has $T(B_{R(\tau)}) = \tau$; and by (36)–(37) and $\varphi = \mathcal{T}^{-1}$, $\tau = \mathcal{T}(\varphi(\tau)) = \mathrm{T}_{\mathrm{mod}}(\Omega_{\varphi(\tau)}; (u - \varphi(\tau))_+)$. Thus $B_{R(\tau)}$ is a ball whose ordinary torsion equals the modified torsion of the pair $(\Omega_{\varphi(\tau)}, (u - \varphi(\tau))_+)$. Proposition 2.14, applied with domain $\Omega_{\varphi(\tau)}$ and reference function $(u - \varphi(\tau))_+$ (which satisfies (32), being a truncation of the eigenfunction – the flux identity (30) for the tail follows from that for u), yields $\left|B_{R(\tau)}\right| \leq \left|\Omega_{\varphi(\tau)}\right|$. Writing both volumes through the distribution functions,

$$\mu^*(\psi(\tau)) = \left|B_{R(\tau)}\right| \;\leq\; \left|\Omega_{\varphi(\tau)}\right| = \mu(\varphi(\tau)), \qquad \tau \in [0, T_\star], \tag{52}$$

where the first equality is (47) and the last is the definition $\mu(t) = |\Omega_t|$ with $t = \varphi(\tau)$.

Step 2: $-\psi' \geq -\varphi'$ and $\psi \geq \varphi$. Solve the defining relation (40) for $-\psi'$:

$$-\psi'(\tau) = \frac{\mu(\varphi(\tau))}{\mu^*(\psi(\tau))}\,(-\varphi'(\tau)) \;\geq\; -\varphi'(\tau) \qquad \text{for a.e. } \tau, \tag{53}$$

the inequality because the prefactor $\mu(\varphi)/\mu^*(\psi) \geq 1$ by (52) (numerator $\geq$ denominator), and $-\varphi'(\tau) > 0$. Integrating (53) from τ up to $T_\star$ and using the common terminal value $\psi(T_\star) = \varphi(T_\star) = 0$ (so that $\psi(\tau) = \psi(\tau) - \psi(T_\star) = \int_\tau^{T_\star} (-\psi'(s))\,ds$, and likewise for φ),

$$\psi(\tau) = \int_\tau^{T_\star} (-\psi'(s))\,ds \;\geq\; \int_\tau^{T_\star} (-\varphi'(s))\,ds = \varphi(\tau), \qquad \tau \in [0, T_\star], \tag{54}$$

the inequality being the integral of the pointwise inequality (53). Thus $\psi \geq \varphi$ on $[0, T_\star]$: at each torsion level, the rearranged function u^* takes a *higher* value than u does at the matched level. (This is the analytic content of the sign remark: smaller superlevel balls force higher level values.)

Step 3: the convexity chain. Let $f : [0,\infty) \to [0,\infty)$ be convex with $f(0) = 0$. Then f is locally Lipschitz, f' exists a.e. and is nondecreasing (convexity), and $f(\xi) = \int_0^\xi f'(t)\,dt$ for $\xi \geq 0$. We compute $\int_\Omega f(u)$ by Cavalieri and convert to the τ variable, then bound by monotonicity of

f' and re-convert:

$$\begin{aligned}
\int_\Omega f(u)\,dx &= \int_0^M f'(t)\,\mu(t)\,dt && \text{(Cavalieri: } f(u)=\textstyle\int_0^u f',\ \text{slice by } \mu(t)=|\{u>t\}|) \\
&= \int_0^{T_\star} f'(\varphi(\tau))\,(-\varphi'(\tau))\,\mu(\varphi(\tau))\,d\tau && \text{(change of variable } t=\varphi(\tau)\text{, as in (42))} \\
&= \int_0^{T_\star} f'(\varphi(\tau))\,(-\psi'(\tau))\,\mu^*(\psi(\tau))\,d\tau && \text{(by (40): } \mu(\varphi)(-\varphi')=\mu^*(\psi)(-\psi')) \\
&\le \int_0^{T_\star} f'(\psi(\tau))\,(-\psi'(\tau))\,\mu^*(\psi(\tau))\,d\tau && \text{(by (54), } \psi\ge\varphi\text{, and } f' \text{ nondecreasing, so } f'(\varphi)\le f'(\psi)) \\
&= \int_0^M f'(\ell)\,\mu^*(\ell)\,d\ell && \text{(change of variable } \ell=\psi(\tau)\text{, reverse of step 2)} \\
&= \int_B f(u^*)\,dx && \text{(Cavalieri on } B \text{ for } u^*\text{, reverse of step 1).}
\end{aligned}$$

The single inequality uses only that f' is nondecreasing and $\psi \ge \varphi$, while $(-\psi') \ge 0$ and $\mu^*(\psi) \ge 0$ keep the multiplied factors nonnegative so the inequality is preserved after multiplication and integration. The two changes of variable are the area formula for the monotone Lipschitz functions φ, ψ, with the sign and limit flips cancelling exactly as before. Taking $f(s) = s^2$ (convex, $f(0) = 0$, $f'(s) = 2s$ nondecreasing) gives the principal claim $\int_B (u^*)^2 \ge \int_\Omega u^2$.

Equality case. Suppose f is strictly convex and equality holds. Then equality holds in the single inequality step, i.e. $f'(\varphi(\tau)) = f'(\psi(\tau))$ for a.e. τ in the support of the integrand. Strict convexity makes f' strictly increasing, so $f'(\varphi) = f'(\psi)$ forces $\varphi(\tau) = \psi(\tau)$ a.e.; then (40) with $\psi = \varphi$ gives $\mu^*(\psi) = \mu(\varphi)$ a.e., i.e. equality in (52), $|B_{R(\tau)}| = |\Omega_{\varphi(\tau)}|$ for a.e. τ. By the equality case of Proposition 2.14 (equivalently the equality case of Saint–Venant), this makes a.e. superlevel set $\Omega_t = \{u > t\}$ a ball. The sets Ω_t are nested ($\Omega_s \subset \Omega_t$ for $s > t$); writing $\Omega_t = B(c_t, r_t)$, the inclusion of nested balls forces $|c_s - c_t| \le r_t - r_s$ on the centres and radii. As $t \downarrow 0$ the radii r_t increase to a limit $r_\star$ (monotone bounded by $|\Omega|$ through the volume), and the centre estimate $|c_s - c_t| \le r_t - r_s \to 0$ makes $\{c_t\}$ Cauchy, converging to some $c_\star$. Since the Ω_t increase to $\{u > 0\} = \Omega$ up to a null set, $\Omega = B(c_\star, r_\star)$: a ball. Then u, having all superlevel sets concentric balls, is radial about $c_\star$, consistently with the equality case of Proposition 2.14. □

Proof of Theorem 2.10. Apply the construction to the first eigenfunction u of Ω. It is a valid reference function: by (R1) it is positive, bounded, smooth in the interior, and by Lemma 2.11(ii) (the flux identity (30)) it satisfies the integrability condition (32), as verified in detail after Definition 2.12. By Lemmas 2.15 and 2.16 the rearranged $u^* \in H_0^1(B)$ satisfies $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$ (equality, property (A)) and $\int_B (u^*)^2 \ge \int_\Omega u^2$ (property (B)). Feeding these into the Rayleigh principle on B (23) – as spelt out at (27) – gives the chain

$$\lambda_1(B) \underset{\text{(i)}}{\le} \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \underset{\text{(ii)}}{\le} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} \underset{\text{(iii)}}{=} \lambda_1(\Omega),$$

where (i) is (23) on B at the competitor u^*, (ii) replaces the numerator by an equal value (Lemma 2.15) and increases the denominator (Lemma 2.16), thereby decreasing the positive quotient, and (iii) is the eigenfunction property of u on Ω.

It remains to assemble $\Psi(B) \le \Psi(\Omega)$. Multiply $\lambda_1(B) \le \lambda_1(\Omega)$ – both nonnegative – by the factor $T(B)^\beta > 0$ after raising to the power $\beta > 0$ (a strictly increasing operation on $[0, \infty)$),

$\lambda_1(B)^\beta \le \lambda_1(\Omega)^\beta$, and note $T(B) \le T(\Omega)$ from (41). Then

$$\Psi(B) = T(B)\,\lambda_1(B)^\beta \;\le\; T(\Omega)\,\lambda_1(B)^\beta \;\le\; T(\Omega)\,\lambda_1(\Omega)^\beta = \Psi(\Omega),$$

the first inequality using $T(B) \le T(\Omega)$ (and $\lambda_1(B)^\beta \ge 0$), the second using $\lambda_1(B)^\beta \le \lambda_1(\Omega)^\beta$ (and $T(\Omega) \ge 0$). Finally, by scale invariance (25), $\Psi(B) = \Psi(B_1) = T_n L_n^\beta$ for the ball B, and likewise the right side of (26) for an arbitrary ball is $\Psi(B_1)$; so

$$T(B_1)L_n^\beta = \Psi(B_1) = \Psi(B) \le \Psi(\Omega) = T(\Omega)\lambda_1(\Omega)^\beta,$$

which is exactly (26).

Equality. If (26) holds with equality, then every inequality in the chain above is an equality; in particular the inequality $\int_B (u^*)^2 \ge \int_\Omega u^2$ from Lemma 2.16 with the strictly convex $f(s) = s^2$ is an equality. By the equality case of Lemma 2.16 this forces Ω to be a ball (and u radial). Conversely, if Ω is a ball then $\Psi(\Omega) = \Psi(B_1)$ by scale invariance, giving equality. This is the equality characterisation claimed in Theorem 2.10. □

3. An internal, computable proof of the Fusco–Julin inequality

This section gives a self-contained, computable proof of the Fusco–Julin sharp quantitative isoperimetric inequality used in §2.2 (Theorem 2.2 and Corollary 2.3); the dimensional constant $C_{\mathrm{FJ}}(n)$ will be exhibited as a finite composition of explicit dimensional constants, with no compactness selection and no contradiction step. The argument proceeds in three blocks. In §3.1 we record the finite-perimeter radial Gauss–Green identity that singles out the normal-oscillation index $\beta(E)$, and we reduce the problem to the small-deficit regime. In §3.2 we prove a Fuglede-type estimate with displayed constants: a balanced $W^{1,\infty}$-small radial graph is controlled, about one fixed centre, in both volume asymmetry and radial normal oscillation by the perimeter deficit. In §3.3 we run the Fusco–Julin regularising selection, prove the resulting set is an additive perimeter almost-minimiser via quantitative annular trapping and localisation of the potential centres, certify computability of the graph-entry constant by a finite spectral truncation argument, and close the global theorem on arbitrary finite-perimeter sets through the Fusco–Julin confinement step. The combined-asymmetry form of [6], used as Theorem 2.2, then follows.

Notation warning. Throughout this section, the symbol β denotes the *radial-normal oscillation index* of Definition 3.1, that is, $\beta(E)^2 := \inf_z \mathcal{O}_z(E)$. This is *not* the boundary-perimeter density function $\beta(t)$ used elsewhere in the manuscript. The two objects share a letter for historical reasons but live in different sections and never appear together in the same display.

Definition 3.1 (Radial normal oscillation)**.** Let $E \subset \mathbb{R}^n$ be a set of finite perimeter with $|E| = |B_\rho|$. For $z \in \mathbb{R}^n$ set $e_z(x) := (x - z)/|x - z|$ for $x \ne z$ and

$$\mathcal{O}_z(E) := \frac{1}{2}\int_{\partial^* E} |\nu_E - e_z|^2 \, d\mathcal{H}^{n-1}.$$

The Fusco–Julin oscillation index is $\beta(E)^2 := \inf_{z\in\mathbb{R}^n} \mathcal{O}_z(E)$; the unnormalised perimeter deficit is $\mathcal{D}(E) := P(E) - P(B_\rho)$.

3.1. **Radial Gauss–Green identity and reduction to small deficit.**

This section spells out the identities which explain the choice of $\beta(E)$. The calculation is elementary, but it is important that it be written with the correct factor 1/2, the correct scaling, and no hidden smoothness assumption.

3.1.1. *A finite-perimeter radial Gauss–Green identity.*

Lemma 3.2 (Radial divergence field). *Fix $z \in \mathbb{R}^n$, and define $e_z(x) = (x-z)/|x-z|$ for $x \neq z$. Then*

$$\operatorname{div} e_z(x) = \frac{n-1}{|x-z|} \qquad (x \neq z). \tag{2.1}$$

Moreover $x \mapsto |x-z|^{-1}$ is locally integrable on $\mathbb{R}^n$.

Proof. By translation it suffices to take $z = 0$. For $x \neq 0$, write

$$e_0(x) = \frac{x}{|x|}.$$

For each $i \in \{1, \dots, n\}$,

$$\partial_i\left(\frac{x_i}{|x|}\right) = \frac{1}{|x|} - \frac{x_i^2}{|x|^3}.$$

Summing in i gives

$$\operatorname{div} e_0 = \frac{n}{|x|} - \frac{|x|^2}{|x|^3} = \frac{n-1}{|x|}.$$

Finally,

$$\int_{B_r(z)} \frac{dx}{|x-z|} = n\omega_n \int_0^r s^{n-2}\, ds = \frac{n\omega_n}{n-1} r^{n-1} < \infty,$$

because $n \geq 2$. $\square$

Proposition 3.3 (Radial Gauss–Green formula). *Let $E \subset \mathbb{R}^n$ be a bounded set of finite perimeter. Then for every $z \in \mathbb{R}^n$,*

$$\int_{\partial^* E} \nu_E \cdot e_z \, d\mathcal{H}^{n-1} = (n-1) \int_E \frac{dx}{|x-z|}. \tag{2.2}$$

Consequently,

$$\mathcal{O}_z(E) = P(E) - (n-1) \int_E \frac{dx}{|x-z|}. \tag{2.3}$$

Proof. The field e_z is singular at z, so we first remove that singularity. Choose a smooth function $\chi_\varepsilon : [0,\infty) \to [0,1]$ satisfying

$$\chi_\varepsilon(r) = 0 \quad (0 \leq r \leq \varepsilon), \qquad \chi_\varepsilon(r) = 1 \quad (r \geq 2\varepsilon), \qquad |\chi_\varepsilon'(r)| \leq \frac{2}{\varepsilon}.$$

Set

$$X_\varepsilon(x) := \chi_\varepsilon(|x-z|) e_z(x).$$

Since E is bounded, the finite-perimeter Gauss–Green formula, after an irrelevant compactly supported cut-off outside a ball containing E, gives

$$\int_{\partial^* E} X_\varepsilon \cdot \nu_E \, d\mathcal{H}^{n-1} = \int_E \operatorname{div} X_\varepsilon \, dx. \tag{2.4}$$

On $\mathbb{R}^n \setminus \{z\}$,

$$\operatorname{div} X_\varepsilon = \chi_\varepsilon(|x-z|) \frac{n-1}{|x-z|} + \chi_\varepsilon'(|x-z|).$$

The second term is supported in $B_{2\varepsilon}(z) \setminus B_\varepsilon(z)$, and hence

$$\left| \int_E \chi_\varepsilon'(|x-z|)\, dx \right| \leq \frac{2}{\varepsilon} |B_{2\varepsilon}| = 2^{n+1} \omega_n \varepsilon^{n-1} \longrightarrow 0.$$

The first term converges in $L^1(E)$ to $(n-1)|x-z|^{-1}$ by Lemma 3.2 and dominated convergence. On the boundary side, $X_\varepsilon \to e_z$ at every point other than z, and $\partial^* E \cap \{z\}$ has $\mathcal{H}^{n-1}$-measure zero. Since $|X_\varepsilon| \le 1$, dominated convergence applies there as well. Letting $\varepsilon \downarrow 0$ in (2.4) proves (2.2).

Finally,

$$\frac{1}{2}|\nu_E - e_z|^2 = \frac{1}{2}\big(|\nu_E|^2 + |e_z|^2 - 2\nu_E \cdot e_z\big) = 1 - \nu_E \cdot e_z$$

for $\mathcal{H}^{n-1}$-almost every $x \in \partial^* E$. Integration and (2.2) give (2.3). □

3.1.2. *Scaling and the large-deficit branch.*

Lemma 3.4 (Scaling)**.** *Let $\lambda > 0$, $a \in \mathbb{R}^n$, and $F = a + \lambda E$. If $|E| = |B_\rho|$, then $|F| = |B_{\lambda\rho}|$ and*

$$P(F) - P(B_{\lambda\rho}) = \lambda^{n-1}\big(P(E) - P(B_\rho)\big), \tag{2.5}$$

$$\beta(F)^2 = \lambda^{n-1}\beta(E)^2. \tag{2.6}$$

Proof. Perimeter scales as λ^{n-1}. Also the map $x = a + \lambda y$ sends $\partial^* E$ to $\partial^* F$, preserves the unit normal, and sends $e_z(y)$ to $e_{a+\lambda z}(x)$. Therefore

$$\mathcal{O}_{a+\lambda z}(F) = \lambda^{n-1}\mathcal{O}_z(E).$$

Taking the infimum over z proves (2.6). □

Lemma 3.5 (Only small deficits require a subtle proof)**.** *If $|E| = |B_\rho|$ and $\mathcal{D}(E) \ge \eta P(B_\rho)$ for some $\eta > 0$, then*

$$\beta(E)^2 \le 2(1 + \eta^{-1})\mathcal{D}(E). \tag{2.7}$$

Proof. For any z, $|\nu_E - e_z|^2 \le 4$, and thus

$$\beta(E)^2 \le \mathcal{O}_z(E) \le 2P(E).$$

Since $P(E) = P(B_\rho) + \mathcal{D}(E)$ and $P(B_\rho) \le \eta^{-1}\mathcal{D}(E)$, we obtain

$$\beta(E)^2 \le 2P(E) \le 2(1 + \eta^{-1})\mathcal{D}(E).$$

□

Lemma 4.33 explains the architecture of every effective proof: one only has to convert a sufficiently small perimeter deficit into a geometric regime in which the normal oscillation can be estimated explicitly.

3.2. **Nearly-spherical sets: an explicit Fuglede-type estimate.** This attempt is complete. We assume from the start that the set is a small radial graph over a ball and prove a same-centre strong estimate by direct calculation. The balance assumptions remove the constant and translation modes. Their presence is not an inconvenience of the proof: a translation of a ball has zero deficit and cannot be controlled as a graph about an incorrect centre.

3.2.1. *Radial graphs, volume, and barycentre.* Let $S = S^{n-1}$, with surface measure $d\sigma$. Recall

$$|S| = n\omega_n. \tag{3.1}$$

Let $u \in W^{1,\infty}(S)$ satisfy $1 + u > 0$, and define

$$E_u := \{r\theta : \theta \in S,\ 0 \le r < 1 + u(\theta)\}. \tag{3.2}$$

Throughout this section the norm is understood in the concrete form

$$\|u\|_{W^{1,\infty}(S)} := \max\{\|u\|_{L^\infty(S)}, \|\nabla_\tau u\|_{L^\infty(S)}\}.$$

Thus $\|u\|_{W^{1,\infty}} \le \delta$ separately implies $|u| \le \delta$ and $|\nabla_\tau u| \le \delta$ almost everywhere. For brevity write

$$a(\theta) := 1 + u(\theta), \qquad s(\theta) := |\nabla_\tau u(\theta)|,$$

$$A := \int_S s^2 \, d\sigma, \qquad B := \int_S u^2 \, d\sigma. \tag{3.3}$$

Lemma 3.6 (Volume and barycentre formulae)**.** *For E_u defined in* (3.2)*,*

$$|E_u| = \frac{1}{n}\int_S a^n \, d\sigma, \tag{3.4}$$

and

$$\int_{E_u} x \, dx = \frac{1}{n+1}\int_S \theta a^{n+1} \, d\sigma. \tag{3.5}$$

In particular, E_u has the volume and barycentre of B_1 precisely when

$$\int_S (a^n - 1) \, d\sigma = 0, \qquad \int_S \theta a^{n+1} \, d\sigma = 0. \tag{3.6}$$

Proof. Use polar coordinates $x = r\theta$, $dx = r^{n-1} \, dr \, d\sigma(\theta)$:

$$\begin{aligned} |E_u| &= \int_S \int_0^{a(\theta)} r^{n-1} \, dr \, d\sigma(\theta) \\ &= \frac{1}{n}\int_S a(\theta)^n \, d\sigma(\theta). \end{aligned}$$

Similarly,

$$\begin{aligned} \int_{E_u} x \, dx &= \int_S \int_0^{a(\theta)} r\theta \, r^{n-1} \, dr \, d\sigma(\theta) \\ &= \frac{1}{n+1}\int_S \theta a(\theta)^{n+1} \, d\sigma(\theta). \end{aligned}$$

For $u = 0$, the two right sides equal $|B_1|$ and 0, respectively, because $\int_S \theta \, d\sigma = 0$. This proves (3.6). ∎

3.2.2. *Perimeter and fixed-centre normal oscillation.*

Lemma 3.7 (Geometry of the graph)**.** *For $\mathcal{H}^{n-1}$-almost every $\theta \in S$, the point $x = a(\theta)\theta$ of ∂E_u has outer normal*

$$\nu_{E_u}(a\theta) = \frac{a\theta - \nabla_\tau u}{\sqrt{a^2 + s^2}}, \tag{3.7}$$

and surface Jacobian

$$J_u(\theta) = a^{n-2}\sqrt{a^2 + s^2}. \tag{3.8}$$

Consequently,

$$P(E_u) = \int_S a^{n-2}\sqrt{a^2 + s^2} \, d\sigma, \tag{3.9}$$

and the normal oscillation about the graph centre 0 *is*

$$\mathcal{O}_0(E_u) = \int_S a^{n-2}\big(\sqrt{a^2+s^2}-a\big)\,d\sigma \tag{3.10}$$
$$= \int_S \frac{a^{n-3}s^2}{\sqrt{1+s^2/a^2}+1}\,d\sigma.$$

Proof. Parametrise the boundary by

$$\Phi : S \to \mathbb{R}^n, \qquad \Phi(\theta) = a(\theta)\theta.$$

If $\tau \in T_\theta S$, then

$$D_\tau \Phi = a\tau + (D_\tau u)\theta.$$

The vector $a\theta - \nabla_\tau u$ is orthogonal to every such tangent vector, since

$$(a\theta - \nabla_\tau u)\cdot\big(a\tau + (D_\tau u)\theta\big) = aD_\tau u - aD_\tau u = 0.$$

It points outward and has length $\sqrt{a^2+s^2}$, proving (3.7).

In an orthonormal tangent basis $\tau_1,\dots,\tau_{n-1}$, the Gram matrix of $D\Phi$ has entries

$$D_{\tau_i}\Phi\cdot D_{\tau_j}\Phi = a^2\delta_{ij} + D_{\tau_i}uD_{\tau_j}u.$$

Writing $v \in \mathbb{R}^{n-1}$ for the tangent vector with components $v_i = D_{\tau_i}u$ (so $|v|^2 = s^2$), the $(n-1)\times(n-1)$ Gram matrix equals $a^2I + v\otimes v$. Since $v\otimes v$ is a rank-one symmetric operator with nonzero eigenvalue $|v|^2$ (in the direction of v) and vanishes on the $(n-2)$-dimensional orthogonal complement of v, the eigenvalues of $a^2I+v\otimes v$ are $a^2+|v|^2$ (with multiplicity one) and a^2 (with multiplicity $n-2$). Multiplying the eigenvalues yields the determinant identity

$$\det(a^2I + v\otimes v) = (a^2)^{n-2}(a^2+|v|^2),$$

which gives (3.8) and hence (3.9).

At $x = a\theta$, the vector $e_0(x)$ equals θ. By (3.7),

$$1 - \nu_{E_u}\cdot\theta = 1 - \frac{a}{\sqrt{a^2+s^2}}.$$

Multiplying by $J_u = a^{n-2}\sqrt{a^2+s^2}$, and recalling that $\mathcal{O}_0 = \int_{\partial E_u}(1-\nu\cdot e_0)\,d\mathcal{H}^{n-1}$, yields

$$\mathcal{O}_0(E_u) = \int_S a^{n-2}\big(\sqrt{a^2+s^2}-a\big)\,d\sigma.$$

Finally,

$$\sqrt{a^2+s^2} - a = a\left(\sqrt{1+\frac{s^2}{a^2}}-1\right) = \frac{s^2/a}{\sqrt{1+s^2/a^2}+1},$$

which proves the second line of (3.10). □

3.2.3. *Explicit constants.* For $0<\delta<1$ and any real exponent p, set

$$m_p(\delta) := \min_{t\in[1-\delta,1+\delta]} t^p, \qquad M_p(\delta) := \max_{t\in[1-\delta,1+\delta]} t^p. \tag{3.11}$$

Define

$$L_n(\delta) := \frac{m_{n-3}(\delta)}{1+\sqrt{1+\big(\delta/(1-\delta)\big)^2}}, \qquad U_n(\delta) := \frac{1}{2}M_{n-3}(\delta), \tag{3.12}$$

$$R_n(\delta) := \frac{n-1}{2}M_{n-3}(\delta)\big(1+(n-1)\delta\big), \tag{3.13}$$

$$C_0(\delta) := \frac{n-1}{2} M_{n-2}(\delta), \qquad C_1(\delta) := \frac{n}{2} M_{n-1}(\delta), \tag{3.14}$$

$$\eta_n(\delta) := \delta^2 \big(C_0(\delta)^2 + n C_1(\delta)^2 \big), \tag{3.15}$$

and

$$d_n(\delta) := L_n(\delta) - \frac{R_n(\delta)}{2n\big(1 - \eta_n(\delta)\big)}. \tag{3.16}$$

Theorem 3.8 (Explicit balanced nearly-spherical strong form)**.** *Let $n \geq 2$. Let E_u be defined by (3.2), assume*

$$|E_u| = |B_1|, \qquad \int_{E_u} x \, dx = 0, \tag{3.17}$$

and suppose

$$\|u\|_{W^{1,\infty}(S)} \leq \delta < 1, \qquad \eta_n(\delta) < 1, \qquad d_n(\delta) > 0. \tag{3.18}$$

Then

$$\beta(E_u)^2 \leq \frac{U_n(\delta)}{d_n(\delta)} \big(P(E_u) - P(B_1) \big). \tag{3.19}$$

More strongly, the same centre 0 satisfies

$$\int_{\partial^* E_u} \left| \nu_{E_u} - \frac{x}{|x|} \right|^2 d\mathcal{H}^{n-1} \leq \frac{2U_n(\delta)}{d_n(\delta)} \big(P(E_u) - P(B_1) \big), \tag{3.20}$$

and

$$|E_u \triangle B_1|^2 \leq \frac{M_{n-1}(\delta)^2 \omega_n}{2(1 - \eta_n(\delta)) d_n(\delta)} \big(P(E_u) - P(B_1) \big). \tag{3.21}$$

Proof. We break the proof into five explicit steps.

Step 1: an upper bound for the fixed-centre oscillation. By (3.10) and the elementary inequality

$$\sqrt{1+t} + 1 \geq 2 \qquad (t \geq 0),$$

we have

$$\begin{aligned} \mathcal{O}_0(E_u) &= \int_S \frac{a^{n-3} s^2}{\sqrt{1 + s^2/a^2} + 1} \, d\sigma \\ &\leq \frac{1}{2} M_{n-3}(\delta) \int_S s^2 \, d\sigma = U_n(\delta) A. \end{aligned} \tag{3.22}$$

Since $\beta(E_u)^2 = \inf_z \mathcal{O}_z(E_u) \leq \mathcal{O}_0(E_u)$, it will suffice to bound A by the perimeter deficit.

Step 2: separate the perimeter deficit. Using (3.9),

$$\begin{aligned} P(E_u) - P(B_1) &= \int_S \big(a^{n-1} - 1 \big) \, d\sigma \\ &\qquad + \int_S a^{n-2} \big(\sqrt{a^2 + s^2} - a \big) \, d\sigma \\ &=: I + G. \end{aligned} \tag{3.23}$$

Since $|u| \leq \delta$ and $s \leq \delta$,

$$1 - \delta \leq a \leq 1 + \delta, \qquad 0 \leq \frac{s}{a} \leq \frac{\delta}{1 - \delta}.$$

The exact identity

$$\sqrt{1+t}-1=\frac{t}{\sqrt{1+t}+1}$$

therefore gives

$$\begin{aligned} G &= \int_S \frac{a^{n-3}s^2}{\sqrt{1+s^2/a^2}+1}\,d\sigma \\ &\geq \frac{m_{n-3}(\delta)}{1+\sqrt{1+\big(\delta/(1-\delta)\big)^2}}\int_S s^2\,d\sigma \\ &= L_n(\delta)A. \end{aligned} \tag{3.24}$$

Step 3: estimate the part I by B. The volume constraint in (3.6) yields

$$\int_S (a^n-1)\,d\sigma=0.$$

Thus

$$I=\int_S f(u)\,d\sigma, \qquad f(q):=(1+q)^{n-1}-1-\frac{n-1}{n}\big((1+q)^n-1\big). \tag{3.25}$$

Direct differentiation gives

$$f(0)=0, \qquad f'(0)=0,$$

and

$$\begin{aligned} f''(q) &= (n-1)(n-2)(1+q)^{n-3}-(n-1)^2(1+q)^{n-2} \\ &= -(n-1)(1+q)^{n-3}\big(1+(n-1)q\big). \end{aligned} \tag{3.26}$$

Taylor's formula with integral remainder reads

$$f(q)=q^2\int_0^1 (1-t)f''(tq)\,dt.$$

If $|q|\leq\delta$, then

$$|f''(tq)|\leq (n-1)M_{n-3}(\delta)\big(1+(n-1)\delta\big).$$

Since $\int_0^1(1-t)\,dt=1/2$, we get

$$f(q)\geq -R_n(\delta)q^2.$$

Integrating this inequality with $q=u(\theta)$ proves

$$I\geq -R_n(\delta)B. \tag{3.27}$$

Step 4: remove the zero and first spherical harmonic modes. The first balance identity in (3.6) can be written as

$$0=n\int_S u\,d\sigma+\int_S\big(a^n-1-nu\big)\,d\sigma.$$

For $|q|\leq\delta$, Taylor's formula gives

$$|(1+q)^n-1-nq|\leq\frac{n(n-1)}{2}M_{n-2}(\delta)q^2.$$

After division by n, it follows that

$$\left|\int_S u\,d\sigma\right|\leq C_0(\delta)B. \tag{3.28}$$

Likewise the barycentre identity gives

$$0 = (n+1)\int_S \theta u \, d\sigma + \int_S \theta\big(a^{n+1} - 1 - (n+1)u\big)\, d\sigma.$$

The estimate

$$|(1+q)^{n+1} - 1 - (n+1)q| \le \frac{n(n+1)}{2} M_{n-1}(\delta) q^2$$

and $|\theta| = 1$ imply

$$\left| \int_S \theta u \, d\sigma \right| \le C_1(\delta) B. \tag{3.29}$$

Let u_0 be the orthogonal projection of u onto the constants and let u_1 be its projection onto the first spherical harmonics. Since $|S| = n\omega_n$,

$$\|u_0\|^2_{L^2(S)} = \frac{1}{n\omega_n}\left(\int_S u \, d\sigma\right)^2. \tag{3.30}$$

The functions $\theta_i/\sqrt{\omega_n}$, $i = 1, \dots, n$, form an orthonormal basis of the first harmonic space, because

$$\int_S \theta_i \theta_j \, d\sigma = \omega_n \delta_{ij}.$$

Hence

$$\|u_1\|^2_{L^2(S)} = \frac{1}{\omega_n}\left| \int_S \theta u \, d\sigma \right|^2. \tag{3.31}$$

From (3.28), (3.29), (3.30), and (3.31),

$$\|u_0\|_2^2 + \|u_1\|_2^2 \le \left(\frac{C_0(\delta)^2}{n\omega_n} + \frac{C_1(\delta)^2}{\omega_n} \right) B^2. \tag{3.32}$$

Since $|u| \le \delta$, we have

$$B = \int_S u^2 \, d\sigma \le \delta^2 |S| = \delta^2 n \omega_n.$$

Substitution in (3.32) gives

$$\|u_0\|_2^2 + \|u_1\|_2^2 \le \eta_n(\delta) B. \tag{3.33}$$

The eigenvalues of $-\Delta_S$ on spherical harmonics of degree k are

$$\lambda_k = k(k+n-2).$$

Thus every harmonic of degree $k \ge 2$ has eigenvalue at least $\lambda_2 = 2n$. The gradient energy has no contribution from the constant mode, has a nonnegative contribution from the degree-one mode, and therefore

$$\begin{aligned} A = \int_S |\nabla_\tau u|^2 \, d\sigma &\ge 2n\big(B - \|u_0\|_2^2 - \|u_1\|_2^2\big) \\ &\ge 2n(1 - \eta_n(\delta))B. \end{aligned} \tag{3.34}$$

Equivalently,

$$B \le \frac{A}{2n(1 - \eta_n(\delta))}. \tag{3.35}$$

Step 5: conclude. Putting (3.23), (3.24), (3.27), and (3.35) together gives

$$\begin{aligned} P(E_u) - P(B_1) &\geq L_n(\delta)A - R_n(\delta)B \\ &\geq \left[L_n(\delta) - \frac{R_n(\delta)}{2n(1-\eta_n(\delta))} \right] A \\ &= d_n(\delta)A. \end{aligned} \tag{3.36}$$

Therefore (3.22) implies both

$$\beta(E_u)^2 \leq \mathcal{O}_0(E_u) \leq \frac{U_n(\delta)}{d_n(\delta)} \big(P(E_u) - P(B_1) \big)$$

and, since the normal integral is $2\mathcal{O}_0(E_u)$, (3.20).

It remains only to prove (3.21). Along each ray from the origin, $E_u \triangle B_1$ has one-dimensional radial measure corresponding to the difference between a^n and 1. Consequently,

$$|E_u \triangle B_1| = \frac{1}{n} \int_S |a^n - 1| \, d\sigma. \tag{3.37}$$

By the mean-value theorem,

$$|a^n - 1| = |(1+u)^n - 1| \leq nM_{n-1}(\delta)|u|.$$

Using Cauchy–Schwarz and $|S| = n\omega_n$, we obtain

$$\begin{aligned} |E_u \triangle B_1|^2 &\leq M_{n-1}(\delta)^2 \left(\int_S |u| \, d\sigma \right)^2 \\ &\leq M_{n-1}(\delta)^2 (n\omega_n) B. \end{aligned} \tag{3.38}$$

Applying (3.35) and (3.36) to the final factor gives

$$|E_u \triangle B_1|^2 \leq \frac{M_{n-1}(\delta)^2 \omega_n}{2(1-\eta_n(\delta))d_n(\delta)} \big(P(E_u) - P(B_1) \big).$$

This is (3.21). □

Corollary 3.9 (A simple displayed constant)**.** *For every $n \geq 2$, set*

$$\delta_*(n) := \frac{1}{100n^3}. \tag{3.39}$$

If the hypotheses of Theorem 3.8 hold and $\|u\|_{W^{1,\infty}(S)} \leq \delta_*(n)$, then

$$\beta(E_u)^2 \leq 3\big(P(E_u) - P(B_1)\big), \tag{3.40}$$

$$\int_{\partial^* E_u} \left| \nu_{E_u} - \frac{x}{|x|} \right|^2 d\mathcal{H}^{n-1} \leq 6\big(P(E_u) - P(B_1)\big), \tag{3.41}$$

and

$$|E_u \triangle B_1|^2 \leq 3\omega_n \big(P(E_u) - P(B_1)\big). \tag{3.42}$$

Proof. We verify the numerical estimates without suppressing the elementary calculation. Let $\delta = \delta_*(n)$. Then

$$0 < \delta \leq \frac{1}{800}. \tag{3.43}$$

For the exponents $p = n-3, n-2, n-1$, if $p \geq 0$, then

$$(1+\delta)^p \leq e^{p\delta} \leq e^{1/(100n^2)} \leq e^{1/400} < 1.004.$$

The only negative exponent occurring is $p = n-3 = -1$ when $n = 2$; in that case

$$M_{-1}(\delta) = \frac{1}{1-\delta} \le \frac{800}{799} < 1.004.$$

Thus

$$M_{n-3}(\delta),\ M_{n-2}(\delta),\ M_{n-1}(\delta) < 1.004. \tag{3.44}$$

Similarly, if $n \ge 4$, Bernoulli's inequality gives

$$(1-\delta)^{n-3} \ge 1-(n-3)\delta \ge 1 - \frac{1}{100n^2} > 0.997.$$

For $n = 3$, $m_{n-3} = m_0 = 1$; for $n = 2$,

$$m_{-1}(\delta) = \frac{1}{1+\delta} > \frac{800}{801} > 0.997.$$

Hence

$$m_{n-3}(\delta) > 0.997. \tag{3.45}$$

Furthermore,

$$\frac{\delta}{1-\delta} \le \frac{1}{799},$$

and so

$$1 + \sqrt{1 + \left(\frac{\delta}{1-\delta}\right)^2} < 2.000002.$$

Using (3.45), we conclude that

$$L_n(\delta) > \frac{0.997}{2.000002} > 0.498, \qquad U_n(\delta) < \frac{1.004}{2} = 0.502. \tag{3.46}$$

From (3.14) and (3.44),

$$C_0(\delta) < 0.502n, \qquad C_1(\delta) < 0.502n.$$

Therefore

$$\begin{aligned} \eta_n(\delta) &< \delta^2(0.502^2 n^2 + 0.502^2 n^3) \\ &\le 0.504008\,\delta^2 n^3 = \frac{0.504008}{10000n^3} < 10^{-4}. \end{aligned} \tag{3.47}$$

Also

$$1 + (n-1)\delta \le 1 + \frac{1}{100n^2} \le 1.0025,$$

and hence, using (3.44) and (3.47),

$$\begin{aligned} \frac{R_n(\delta)}{2n(1-\eta_n(\delta))} &< \frac{n-1}{4n}\frac{1.004 \cdot 1.0025}{1-10^{-4}} \\ &< 0.252. \end{aligned} \tag{3.48}$$

Equations (3.46) and (3.48) yield

$$d_n(\delta) > 0.498 - 0.252 = 0.246 > 0.2. \tag{3.49}$$

It follows that

$$\frac{U_n(\delta)}{d_n(\delta)} < \frac{0.502}{0.246} < 3,$$

proving (3.40) and (3.41). Finally,

$$\frac{M_{n-1}(\delta)^2}{2(1-\eta_n(\delta))d_n(\delta)} < \frac{1.004^2}{2(1-10^{-4})0.246} < 3,$$

which proves (3.42). □

Corollary 3.10 (Translation and scaling)**.** *Let $\rho > 0$, $z \in \mathbb{R}^n$, and suppose*

$$E = z + \rho E_u,$$

where E_u satisfies the hypotheses of Corollary 3.9. Then the same centre z satisfies

$$\rho^{-n-1}|E \triangle B_\rho(z)|^2 + \int_{\partial^* E} \left|\nu_E - \frac{x-z}{|x-z|}\right|^2 d\mathcal{H}^{n-1} \le (3\omega_n + 6)\big(P(E) - P(B_\rho)\big). \tag{3.50}$$

Proof. Under $x = z + \rho y$, symmetric-difference measure scales by ρ^n, while perimeter and the boundary normal integral scale by ρ^{n-1}. Therefore (3.42) becomes

$$|E \triangle B_\rho(z)|^2 \le 3\omega_n \rho^{n+1}\big(P(E) - P(B_\rho)\big),$$

and (3.41) becomes

$$\int_{\partial^* E} \left|\nu_E - \frac{x-z}{|x-z|}\right|^2 d\mathcal{H}^{n-1} \le 6\big(P(E) - P(B_\rho)\big).$$

Adding the two inequalities proves (3.50). □

3.3. **Regularising selection and the general case.** This subsection sets up the selection apparatus used in the regularising construction. We first record the natural positive decomposition $\beta^2 = \mathcal{D} + \mathcal{C}$ and the basic estimates for the potential deficit $\mathcal{C}$; these definitions and the continuity lemma for the maximal radial potential M are reused throughout the regularising selection of Subsection 3.3.4.

A first, decomposition-based selection map (penalising $\mathcal{D}$ while preserving $\mathcal{C}$) is conceptually appealing, since it allows quotient preservation by an elementary minimisation. We do not develop it here, however, because the resulting selected set carries a perimeter-order error in its near-minimality estimate, originating in the sharp $(n-1)/n$-Hölder modulus of M in symmetric difference (Lemma 3.12); this obstructs the perimeter regularity step. We therefore proceed in Subsection 3.3.4 with the Fusco–Julin choice of penalising β^2 itself, which we quantify by an intermediate annular-separation step that produces a genuine additive perimeter almost-minimality.

3.3.1. *The potential deficit.* For every measurable $U \subset \mathbb{R}^n$ with finite measure, define

$$r_U := \left(\frac{|U|}{\omega_n}\right)^{1/n}, \qquad p(U) := P(B_{r_U}) = n\omega_n^{1/n}|U|^{(n-1)/n}, \tag{4.2}$$

$$M(U) := \sup_{z\in\mathbb{R}^n} \int_U \frac{dx}{|x-z|}, \tag{4.3}$$

and

$$\mathcal{D}(U) := P(U) - p(U), \qquad \mathcal{C}(U) := p(U) - (n-1)M(U). \tag{4.4}$$

For the empty set, all four quantities are set equal to zero.

Lemma 3.11 (Positivity and decomposition)**.** *For every finite-perimeter set U of finite measure,*

$$\mathcal{D}(U) \geq 0, \qquad \mathcal{C}(U) \geq 0, \tag{4.5}$$

and

$$\beta(U)^2 = \mathcal{D}(U) + \mathcal{C}(U), \tag{4.6}$$

where the radius in the definition of $\beta(U)$ is r_U. Under a dilation $U \mapsto a + \lambda U$,

$$\mathcal{D}(a + \lambda U) = \lambda^{n-1}\mathcal{D}(U), \qquad \mathcal{C}(a + \lambda U) = \lambda^{n-1}\mathcal{C}(U). \tag{4.7}$$

Proof. The isoperimetric inequality gives $P(U) \geq P(B_{r_U}) = p(U)$, proving $\mathcal{D}(U) \geq 0$. For each z, the radially decreasing rearrangement inequality applied to $x \mapsto |x - z|^{-1}$ gives

$$\int_U \frac{dx}{|x - z|} \leq \int_{B_{r_U}(z)} \frac{dx}{|x - z|} = n\omega_n \int_0^{r_U} s^{n-2}\, ds = \frac{p(U)}{n - 1}.$$

Taking the supremum in z proves $\mathcal{C}(U) \geq 0$.

By Proposition 3.3,

$$\begin{aligned}
\beta(U)^2 &= P(U) - (n - 1)M(U) \\
&= \big(P(U) - p(U)\big) + \big(p(U) - (n - 1)M(U)\big) \\
&= \mathcal{D}(U) + \mathcal{C}(U).
\end{aligned}$$

Finally, perimeter and p scale by λ^{n-1}; the change of variables $x = a + \lambda y$ in (4.3) gives $M(a + \lambda U) = \lambda^{n-1}M(U)$. This proves (4.7). ☐

3.3.2. *A continuity estimate for the maximal radial potential.* The following continuity estimate for M in symmetric difference is the load-bearing input from this subsection into the regularising construction of Subsection 3.3.4.

Lemma 3.12 (Continuity of the maximal radial potential)**.** *Let $U, V \subset \mathbb{R}^n$ have finite measure. Then*

$$|M(U) - M(V)| \leq K_n\, |U \triangle V|^{(n-1)/n}, \qquad K_n := \frac{n}{n - 1}\omega_n^{1/n}. \tag{4.9}$$

Proof. Fix $z \in \mathbb{R}^n$ and set $A = U \triangle V$. Since the integrand is nonnegative,

$$\left| \int_U \frac{dx}{|x - z|} - \int_V \frac{dx}{|x - z|} \right| \leq \int_A \frac{dx}{|x - z|}.$$

Among all sets of measure $|A|$, the final integral is maximised by the ball centred at z. If $|A| = \omega_n r^n$, then

$$\begin{aligned}
\int_A \frac{dx}{|x - z|} &\leq \int_{B_r(z)} \frac{dx}{|x - z|} \\
&= n\omega_n \int_0^r s^{n-2}\, ds = \frac{n}{n - 1}\omega_n^{1/n}|A|^{(n-1)/n}.
\end{aligned}$$

This bound is independent of z. Taking the supremum over z once with U and once with V proves (4.9). ☐

3.3.3. *Density and measure closeness give Hausdorff closeness.* The last general-purpose ingredient of the selection apparatus is a deterministic upgrade from a two-sided density estimate plus symmetric-difference smallness to Hausdorff closeness. It is used in Subsection 3.3.4 to convert the density output of the regularising selection into the annular trapping needed for the additive perimeter almost-minimality theorem.

Lemma 3.13 (Density and measure closeness give Hausdorff closeness)**.** *Let $G \subset \mathbb{R}^n$ be a bounded finite-perimeter set with $|G| = |B_1|$, represented so that its topological boundary agrees with the support of its perimeter measure. Assume there exist $c_d > 0$ and $0 < r_d \leq 1/4$ such that*

$$|G \cap B_r(x)| \geq c_d r^n, \qquad |B_r(x) \setminus G| \geq c_d r^n \tag{4.31}$$

for every $x \in \partial G$ and every $0 < r \leq r_d$. Let $z \in \mathbb{R}^n$, and put

$$m := |G \triangle B_1(z)|.$$

If

$$m < \min\{c_d r_d^n, \omega_n (r_d/2)^n\}, \tag{4.32}$$

then

$$d_H\big(\partial G, \partial B_1(z)\big) \leq C_H(n, c_d)\, m^{1/n}, \tag{4.33}$$

where one may take

$$C_H(n, c_d) := \max\{2c_d^{-1/n},\, 4\omega_n^{-1/n}\}. \tag{4.34}$$

Proof. Translations do not affect the assertion, so take $z = 0$.

Boundary points of G are close to the sphere. Let $x \in \partial G$, and set $d = \operatorname{dist}(x, \partial B_1)$. If $d = 0$, the desired estimate for x is immediate, so suppose $d > 0$. If $d > 2r_d$, then $B_{r_d}(x)$ is entirely contained either in B_1 or in $\mathbb{R}^n \setminus B_1$. In the first case the second density estimate in (4.31) gives

$$m \geq |B_{r_d}(x) \setminus G| \geq c_d r_d^n;$$

in the second case the first density estimate gives the same conclusion. Both contradict (4.32). Thus $d \leq 2r_d$. The ball $B_{d/2}(x)$ is again entirely on one side of ∂B_1. Apply the appropriate density inequality at radius $d/2 \leq r_d$:

$$m \geq c_d (d/2)^n.$$

It follows that

$$\operatorname{dist}(x, \partial B_1) \leq 2c_d^{-1/n} m^{1/n}. \tag{4.35}$$

Boundary points of the sphere are close to G. Let $y \in \partial B_1$, and set $d = \operatorname{dist}(y, \partial G)$. If $d = 0$, the desired estimate for y is immediate, so suppose $d > 0$. If $d > 2r_d$, the connected ball $B_{2r_d}(y)$, which is disjoint from ∂G, lies, up to a null set, entirely in G or entirely in its complement. The two balls

$$B_{r_d/2}\big((1 - r_d/2)y\big) \subset B_1 \cap B_{2r_d}(y),$$
$$B_{r_d/2}\big((1 + r_d/2)y\big) \subset (\mathbb{R}^n \setminus B_1) \cap B_{2r_d}(y)$$

show, in either alternative, that

$$m \geq \omega_n (r_d/2)^n,$$

contrary to (4.32). Hence $d \leq 2r_d$. Now $B_{d/2}(y)$ is disjoint from ∂G, so it lies, up to a null set, wholly in G or wholly in its complement. The same inner/outer ball construction, with radius $d/4$, gives

$$m \geq \omega_n (d/4)^n.$$

Therefore

$$\operatorname{dist}(y, \partial G) \le 4\omega_n^{-1/n} m^{1/n}. \tag{4.36}$$

Combining (4.35) and (4.36) proves (4.33). □

3.3.4. *Regularising selection by penalising the true excess.* A first attempt at selection – penalising $\mathcal{D}$ while preserving $\mathcal{C}$ – is conceptually attractive but, as we noted at the start of this subsection, fails to produce additive perimeter almost-minimality. The original Fusco–Julin construction [6] instead preserves β^2 itself. At first sight this appears worse, because $\beta^2 = P - (n-1)M$ contains the perimeter and the singular potential M. The point is that the argument is performed in two stages:

$$\begin{aligned} &\text{penalisation of } \beta^2 \Longrightarrow \text{area quasi-minimality and density,} \\ &\text{density plus small deficit} \Longrightarrow \text{annular trapping and centre separation,} \\ &\text{centre separation} \Longrightarrow \text{additive perimeter almost-minimality.} \end{aligned} \tag{4.49}$$

The last implication is where the maximiser in M becomes regular enough: on all local variations touching the boundary, the kernel $|x-y|^{-1}$ is uniformly bounded because every maximising centre y lies strictly inside the trapped annulus.

For this subsection set

$$b(U) := \beta(U)^2 = P(U) - (n-1)M(U). \tag{4.50}$$

Fix $R > 3$, a container B_R, a number $\Lambda > n$, and set

$$a := \frac{1}{4}. \tag{4.51}$$

For a bounded input one may translate the configuration without changing b or $\mathcal{D}$. Lemma 3.28 below supplies bounded inputs from arbitrary finite-perimeter sets.

Lemma 3.14 (The volume penalty baseline)**.** *For every finite-perimeter $U \subset B_R$, put*

$$H_\Lambda(U) := P(U) + \Lambda\big||U| - \omega_n\big| - P(B_1). \tag{4.52}$$

Then

$$H_\Lambda(U) \ge \mathcal{D}(U) + (\Lambda - n)\big||U| - \omega_n\big| \ge 0. \tag{4.53}$$

Proof. Write $m = |U|$ and $\alpha = (n-1)/n$. By the isoperimetric inequality,

$$P(U) \ge p(m) = n\omega_n^{1/n} m^\alpha.$$

If $m \ge \omega_n$, then $p(m) \ge p(\omega_n) = P(B_1)$, and

$$p(m) + \Lambda(m - \omega_n) - P(B_1) \ge \Lambda(m - \omega_n) \ge (\Lambda - n)(m - \omega_n).$$

If $0 \le m \le \omega_n$, set $t = m/\omega_n$. Since $0 \le t \le 1$ and $\alpha < 1$, $t^\alpha \ge t$. Therefore

$$\begin{aligned} p(m) + \Lambda(\omega_n - m) - P(B_1) &= \omega_n\{n(t^\alpha - 1) + \Lambda(1-t)\} \\ &\ge (\Lambda - n)(\omega_n - m). \end{aligned}$$

Subtracting $p(m)$ from $P(U)$ gives (4.53). □

Theorem 3.15 (Regularising single-set FJ selection)**.** *Let $E_0 \subset B_R$ satisfy*

$$|E_0| = \omega_n, \qquad b_0 := b(E_0) > 0, \qquad d_0 := \mathcal{D}(E_0), \qquad q_0 := \frac{d_0}{b_0}. \tag{4.54}$$

Assume

$$q_0 \le \frac{1}{16}, \qquad d_0 \le \frac{\Lambda - n}{2}\,\omega_n. \tag{4.55}$$

Then the functional

$$\mathcal{I}_{E_0}(U) := P(U) + \Lambda\big||U| - \omega_n\big| + a\,|b(U) - b_0|, \qquad U \subset B_R, \tag{4.56}$$

has a minimiser F. Every minimiser satisfies

$$\mathcal{D}(F) \le d_0, \qquad \big||F| - \omega_n\big| \le \frac{d_0}{\Lambda - n}, \tag{4.57}$$

$$|b(F) - b_0| \le 4d_0, \qquad b(F) \ge (1 - 4q_0)b_0 \ge \frac{3}{4}b_0, \tag{4.58}$$

and

$$\frac{\mathcal{D}(F)}{b(F)} \le \frac{q_0}{1 - 4q_0} \le \frac{4}{3}q_0. \tag{4.59}$$

If $s = (|F|/\omega_n)^{1/n}$ and $\widetilde{F} = s^{-1}F$, then

$$|\widetilde{F}| = \omega_n, \qquad \frac{\mathcal{D}(\widetilde{F})}{b(\widetilde{F})} = \frac{\mathcal{D}(F)}{b(F)}, \qquad \mathcal{D}(\widetilde{F}) \le 2d_0. \tag{4.60}$$

Proof. For existence, let U_j be a minimising sequence. Comparison with E_0 bounds $P(U_j)$, since all terms in (4.56) except the perimeter are nonnegative. After passing to a subsequence, $U_j \to F$ in $L^1(B_R)$ and $P(F) \le \liminf_j P(U_j)$. Lemma 3.12 implies $M(U_j) \to M(F)$. For fixed c, the function

$$t \longmapsto t + a|t - c|$$

is nondecreasing when $0 < a < 1$. Applying this with $c = (n-1)M(U_j) + b_0$, which converges to $(n-1)M(F) + b_0$, proves lower semicontinuity of $P(U) + a|b(U) - b_0|$. The volume term is continuous; hence F is a minimiser.

Minimality against E_0, followed by subtraction of $P(B_1)$, gives

$$H_\Lambda(F) + a|b(F) - b_0| \le d_0. \tag{4.61}$$

Lemma 3.14 yields (4.57); since $a = 1/4$, it also yields $|b(F) - b_0| \le 4d_0$. Dividing the latter estimate by b_0 proves (4.58), and then

$$\frac{\mathcal{D}(F)}{b(F)} \le \frac{d_0}{(1 - 4q_0)b_0} = \frac{q_0}{1 - 4q_0},$$

which proves (4.59).

From (4.57) and (4.55), $|F| \ge \omega_n/2$, hence $s \ge 2^{-1/n}$. Both $\mathcal{D}$ and b scale by s^{n-1}; therefore the quotient in (4.60) is scale invariant. Finally,

$$\mathcal{D}(\widetilde{F}) = s^{1-n}\mathcal{D}(F) \le 2^{(n-1)/n}d_0 \le 2d_0.$$

□

Proposition 3.16 (First regularisation stage: area quasi-minimality)**.** *Let F be a minimiser in Theorem 3.15. If*

$$F \triangle V \Subset B_r(x), \tag{4.62}$$

then

$$P(F;B_r(x)) \le 3P(V;B_r(x)) + 2\Lambda|F \triangle V|. \tag{4.63}$$

Consequently, for

$$r_a := \min\Big\{1, \frac{n}{4\Lambda}\Big\}, \tag{4.64}$$

and $0<r\le r_a$,

$$P(F;B_r(x)) \le 7P(V;B_r(x)). \tag{4.65}$$

Proof. First suppose $B_r(x) \Subset B_R$, so that V itself is an admissible competitor in (4.56). Minimality and the triangle inequality for the last two terms in (4.56) give

$$P(F) \le P(V) + \Lambda|F \triangle V| + a|b(F) - b(V)|. \tag{4.66}$$

If $b(F) \ge b(V)$, select a centre y_V realising $b(V)$; if $b(V) > b(F)$, use a centre y_F realising $b(F)$. In either case the oscillation integrand

$$1 - \nu_U(z) \cdot \frac{z-y}{|z-y|}$$

lies between 0 and 2. Since $F = V$ outside $B_r(x)$,

$$|b(F) - b(V)| \le 2\{P(F;B_r(x)) + P(V;B_r(x))\}. \tag{4.67}$$

Insert $a = 1/4$ into (4.66), cancel the perimeters outside $B_r(x)$, and move $\frac{1}{2}P(F;B_r(x))$ to the left. This proves (4.63).

Let $A = F \triangle V$. Since $A \subset B_r(x)$, the isoperimetric inequality and $P(A) \le P(F;B_r(x)) + P(V;B_r(x))$ give

$$|A| \le \frac{r}{n}\{P(F;B_r(x)) + P(V;B_r(x))\}. \tag{4.68}$$

For $r \le n/(4\Lambda)$, inserting this estimate in (4.63) gives

$$P(F;B_r(x)) \le 3P(V;B_r(x)) + \frac{1}{2}\{P(F;B_r(x)) + P(V;B_r(x))\}.$$

Rearrangement proves (4.65).

If $B_r(x)$ is not contained in the container, apply the preceding argument to $W = V \cap B_R$, which is an admissible competitor because $F \subset B_R$. Since $B_R \subset V \cup B_R$, the isoperimetric inequality gives

$$P(B_R) \le P(V \cup B_R).$$

The perimeter lattice inequality

$$P(V \cap B_R) + P(V \cup B_R) \le P(V) + P(B_R)$$

then implies $P(W) \le P(V)$. The same cancellation outside $B_r(x)$ shows $P(W;B_r(x)) \le P(V;B_r(x))$, while $|F \triangle W| \le |F \triangle V|$. Thus (4.63) and (4.65) also hold for balls meeting ∂B_R. □

Lemma 3.17 (Explicit density from the coarse stage)**.** *Let F satisfy* (4.65) *and take the representative whose boundary is* $\operatorname{spt}|D\mathbf{1}_F|$*. Then, for every $x \in \partial F$ and $0<r\le r_a$,*

$$|F \cap B_r(x)| \ge c_a r^n, \qquad |B_r(x) \setminus F| \ge c_a r^n, \qquad c_a := \frac{\omega_n}{8^n}. \tag{4.69}$$

Proof. Set $m(r) = |F \cap B_r(x)|$. For almost every r, comparison with the set obtained by removing F from $B_r(x)$, using an arbitrarily thin collar to meet the compact-containment condition, gives

$$P(F; B_r(x)) \le 7m'(r).$$

The perimeter of $F \cap B_r(x)$ is at most $P(F; B_r(x)) + m'(r) \le 8m'(r)$. Thus the isoperimetric inequality gives

$$n\omega_n^{1/n} m(r)^{(n-1)/n} \le 8m'(r)$$

for almost every r. Since x belongs to the support of the perimeter, $m(r) > 0$ for every $r > 0$, and integration of

$$\frac{d}{dr} m(r)^{1/n} \ge \frac{\omega_n^{1/n}}{8}$$

from 0 to r yields the first inequality in (4.69). Apply the same argument to the complement of F, whose local perimeter satisfies the identical comparison inequality, to obtain the second one. □

Lemma 3.18 (Quantitative localisation of potential centres)**.** *Let*

$$K_n := \frac{n}{n-1}\omega_n^{1/n}$$

be as in Lemma 3.12, and define the computable positive number

$$\begin{aligned} \Gamma_n &:= \int_{B_1} \frac{dx}{|x|} - \int_{B_1} \frac{dx}{|x - \frac{1}{4}e_1|} > 0, \\ m_c(n) &:= \left(\frac{\Gamma_n}{4K_n}\right)^{n/(n-1)}. \end{aligned} \tag{4.70}$$

Let G have finite measure and let $B_s(z)$, $s > 0$, satisfy

$$\frac{|G \triangle B_s(z)|}{s^n} \le m_c(n). \tag{4.71}$$

Then every maximiser y_G in the definition of $M(G)$ satisfies

$$|y_G - z| < \frac{s}{4}. \tag{4.72}$$

Proof. The function $y \mapsto \int_G |x-y|^{-1}\, dx$ is continuous, tends to zero as $|y| \to \infty$, and therefore admits a maximiser. By translation and scaling it is enough to take $s = 1$, $z = 0$. The potential of B_1 is a strictly radially decreasing function of the centre distance; consequently, if $|y| \ge 1/4$,

$$\int_{B_1} \frac{dx}{|x|} - \int_{B_1} \frac{dx}{|x-y|} \ge \Gamma_n.$$

Maximality of y_G, and the rearrangement estimate used in Lemma 3.12, give

$$\begin{aligned} \Gamma_n &\le \left| \int_{B_1} \frac{dx}{|x|} - \int_G \frac{dx}{|x|} \right| + \left| \int_G \frac{dx}{|x-y_G|} - \int_{B_1} \frac{dx}{|x-y_G|} \right| \\ &\le 2K_n |G \triangle B_1|^{(n-1)/n} \le \frac{\Gamma_n}{2}, \end{aligned}$$

a contradiction. Scaling gives (4.72). □

Theorem 3.19 (Second regularisation stage: additive almost-minimality)**.** *Let F be selected by Theorem 3.15, and let $\widetilde{F} = s^{-1}F$ be its volume normalisation. Assume that a constructive quantitative isoperimetric inequality supplies*

$$|\widetilde{F} \triangle B_1(z)|^2 \le C_{\mathrm{iso}}(n)\mathcal{D}(\widetilde{F}). \tag{4.73}$$

Let $C_H = C_H(n, c_a)$ be the constant in Lemma 3.13, with density radius

$$\widetilde{r}_a := \frac{2}{3} r_a,$$

and set

$$\begin{aligned} \overline{m} &:= \frac{1}{2} \min\left\{c_a \widetilde{r}_a^{\,n},\ \omega_n (\widetilde{r}_a/2)^n,\ (8C_H)^{-n},\ m_c(n)\right\}, \\ d_{\mathrm{ann}} &:= \min\left\{\frac{\overline{m}^{\,2}}{2C_{\mathrm{iso}}(n)},\ \frac{\Lambda - n}{2}\omega_n\right\}, \\ r_{\mathrm{reg}} &:= \min\left\{r_a,\ \frac{1}{32},\ \frac{1}{2}\left(\frac{m_c(n)}{2\omega_n}\right)^{1/n}\right\}. \end{aligned} \tag{4.74}$$

If $d_0 \le d_{\mathrm{ann}}$, then F is an $(\Lambda_{\mathrm{reg}}, r_{\mathrm{reg}})$-almost minimiser of perimeter in the standard volume-error sense: for every competitor as in (4.62) with $0 < r \le r_{\mathrm{reg}}$,

$$\begin{aligned} P(F; B_r(x)) &\le P(V; B_r(x)) + \Lambda_{\mathrm{reg}} |F \triangle V| \\ &\le P(V; B_r(x)) + \Omega_{\mathrm{reg}} r^n, \\ \Lambda_{\mathrm{reg}} &:= \frac{4}{3}\big(\Lambda + n - 1\big), \qquad \Omega_{\mathrm{reg}} := \omega_n \Lambda_{\mathrm{reg}}. \end{aligned} \tag{4.75}$$

The normalised set $\widetilde{F}$ satisfies the same estimate with radius $(2/3) r_{\mathrm{reg}}$ and constant at most $(3/2)\Lambda_{\mathrm{reg}}$ in the volume-error form.

Proof. By (4.60) and (4.73),

$$m := |\widetilde{F} \triangle B_1(z)| \le (2C_{\mathrm{iso}}(n) d_0)^{1/2} \le \overline{m}. \tag{4.76}$$

Proposition 3.16 and Lemma 3.17 are invariant under scaling; since (4.57) gives

$$2^{-1/n} \le s \le (3/2)^{1/n} \le \frac{3}{2},$$

the normalised set has density coefficient c_a at least up to radius $\widetilde{r}_a$. Lemma 3.13 and (4.74) yield

$$d_H(\partial \widetilde{F}, \partial B_1(z)) \le \frac{1}{8}. \tag{4.77}$$

After scaling back, every point of ∂F has distance from $z_F := sz$ between $7s/8$ and $9s/8$. Moreover (4.76) and Lemma 3.18 give

$$|y_F - z_F| < \frac{s}{4} \tag{4.78}$$

for every potential centre of F.

First take $V \subset B_R$ as in the statement. If $P(F; B_r(x)) = 0$, (4.75) is immediate. Otherwise $B_r(x)$ meets ∂F. If $A = F \triangle V$, then

$$\frac{|V \triangle B_s(z_F)|}{s^n} \le m + \frac{|A|}{s^n} \le \frac{1}{2} m_c(n) + \omega_n (2r)^n \le m_c(n).$$

Lemma 3.18 therefore gives $|y_V - z_F| < s/4$ for every potential centre of V. For $w \in A$, choose a boundary point of F in $B_r(x)$. Using $s \ge 1/2$, $r \le 1/32$, (4.77), and either $y = y_F$ or $y = y_V$, we obtain

$$|w - y| \ge s\left(1 - \frac{1}{8} - \frac{1}{4}\right) - 2r \ge \frac{1}{4}. \tag{4.79}$$

Evaluating $M(F)$ at a maximising centre of F and $M(V)$ at a maximising centre of V, and using (4.79), gives

$$|M(F) - M(V)| \le 4|F \triangle V|. \tag{4.80}$$

Let $X = P(F; B_r(x)) - P(V; B_r(x))$. If $X \le 0$, there is nothing to prove. If $X > 0$, then

$$|b(F) - b(V)| \le X + 4(n-1)|F \triangle V|.$$

Insert this in (4.66), cancel outside perimeters, and use $a = 1/4$:

$$(1-a)X \le \{\Lambda + 4a(n-1)\}|F \triangle V| \le \{\Lambda + n - 1\}\omega_n r^n.$$

Since $1 - a = 3/4$, this proves both inequalities in (4.75). Finally, scaling a competitor for $\widetilde{F}$ by s multiplies perimeter by s^{n-1} and symmetric-difference volume by s^n; thus the volume-error almost-minimality constant is multiplied by $s \le 3/2$, while its admissible radius is at least $(2/3)r_{\mathrm{reg}}$.

For a competitor V not contained in B_R, replace it first by $V \cap B_R$. The perimeter lattice argument at the end of Proposition 3.16 does not increase either the local perimeter or the symmetric-difference error; hence the same estimate holds without requiring $B_r(x) \Subset B_R$. □

3.3.5. *Reducing graph entry to one local flatness number.* Theorems 3.15 and 3.19 have removed the Fusco–Julin-specific singularity. We now reduce the remaining regularity work to a single local epsilon-regularity number for perimeter almost-minimisers. Existence of such a number follows from the regularity theorem of Tamanini, in the form presented for example in [15, Chapters 24–26]. For the present purpose, existence alone is not enough: the harmonic-approximation lemma in that presentation is proved by compactness. The lemma below replaces that one use of compactness by a finite spectral truncation and thereby certifies computability of the required number.

Definition 3.20 (Meaning of computable in this section)**.** A dimensional constant is called *computable* if it can be selected by a finite list of explicit inequalities from rational numbers, ω_k, elementary operations, and finitely many Dirichlet eigenvalues and eigenfunction bounds on Euclidean balls. These spectral data are computable: in one dimension they are trigonometric, and in higher dimensions they are given by spherical harmonics and zeros of Bessel functions, which admit certified rational bracketing. Numerical evaluation of the resulting expression is not required.

Definition 3.21 (Admissible flatness number)**.** Fix $n \ge 2$ and $0 < \ell < 1/8$. A number $\varepsilon_{\mathrm{fl}}(n, \ell) \in (0, 1/16)$ is called *admissible* if the following statement holds.

Let G be represented so that $\partial G = \operatorname{spt}|D\mathbf{1}_G|$, and suppose that, in every ball of radius at most r_*,

$$P(G; B_t(y)) \le P(V; B_t(y)) + \Lambda_*|G \triangle V| \quad \text{whenever } G \triangle V \Subset B_t(y). \tag{4.81}$$

Let $x \in \partial G$, $e \in S^{n-1}$, and $0 < 16\rho \le r_*$. If

$$\begin{gathered} \Lambda_*\rho \le \varepsilon_{\mathrm{fl}}(n, \ell), \\ \partial G \cap B_{8\rho}(x) \subset \{y : |(y - x) \cdot e| \le \varepsilon_{\mathrm{fl}}(n, \ell)\rho\}, \\ x - 4\rho e \in G^{(1)}, \qquad x + 4\rho e \in G^{(0)}, \end{gathered} \tag{4.82}$$

then $\partial G \cap B_\rho(x)$ is a single Lipschitz graph in direction e, and its Lipschitz constant is at most ℓ. Here $G^{(1)}$ and $G^{(0)}$ denote the points of density one and zero of G, respectively.

Lemma 3.22 (Constructive harmonic approximation)**.** *Let $m \geq 1$, $B = B_1^m$, and $D = B_{1/2}^m$. For every $\tau \in (0,1)$ there is a computable number $\sigma_{\mathrm{har}}(m,\tau) > 0$ with the following property. If $u \in W^{1,2}(B)$ satisfies*

$$\int_B |\nabla u|^2 \leq 1, \qquad \left| \int_B \nabla u \cdot \nabla \varphi \right| \leq \sigma_{\mathrm{har}}(m,\tau)\, \|\nabla \varphi\|_{L^\infty(B)} \quad \textit{for every } \varphi \in C_c^\infty(B),$$

then there is a harmonic function v on D such that

$$\int_D |\nabla v|^2 \leq 1, \qquad \int_D |u - v|^2 \leq \tau.$$

Proof. Let v be the harmonic replacement of u in D, and put $w = u - v \in W_0^{1,2}(D)$. Then

$$\int_D |\nabla v|^2 \leq \int_D |\nabla u|^2 \leq 1, \qquad \int_D |\nabla w|^2 \leq 1.$$

Let $(\lambda_j, \phi_j)_{j \geq 1}$ be an $L^2(D)$-orthonormal Dirichlet eigenbasis for $-\Delta$ on D, ordered so that λ_j is nondecreasing, and set

$$L_j := \|\nabla \phi_j\|_{L^\infty(D)}.$$

Each ϕ_j, extended by zero outside D, is Lipschitz and can be mollified inside B without increasing its gradient bound in the limit. Thus the almost-harmonicity assumption gives

$$\left| \int_D \nabla w \cdot \nabla \phi_j \right| = \left| \int_D \nabla u \cdot \nabla \phi_j \right| \leq \sigma_{\mathrm{har}}(m,\tau) L_j.$$

Writing $w = \sum_{j \geq 1} a_j \phi_j$, this says

$$|a_j| \leq \sigma_{\mathrm{har}}(m,\tau) \frac{L_j}{\lambda_j}.$$

Choose a finite $N = N(m,\tau)$ such that $\lambda_{N+1}^{-1} \leq \tau/2$, put

$$A_N := \sum_{j=1}^N \left(\frac{L_j}{\lambda_j} \right)^2, \qquad \sigma_{\mathrm{har}}(m,\tau) := \min\left\{ 1, \sqrt{\frac{\tau}{2A_N}} \right\},$$

with the evident convention if $A_N = 0$. The low modes and high modes then satisfy

$$\sum_{j=1}^N a_j^2 \leq \frac{\tau}{2},$$

$$\sum_{j>N} a_j^2 \leq \lambda_{N+1}^{-1} \sum_{j>N} \lambda_j a_j^2 \leq \frac{\tau}{2} \int_D |\nabla w|^2 \leq \frac{\tau}{2}.$$

Therefore $\int_D |u - v|^2 = \int_D |w|^2 \leq \tau$. Definition 3.20 makes every choice in this construction computable. ☐

Proposition 3.23 (Computability certificate for graph entry)**.** *For every $n \geq 2$ and $0 < \ell < 1/8$, there exists a computable admissible flatness number $\varepsilon_{\mathrm{fl}}(n,\ell)$ in the sense of Definition 3.21.*

Proof. The local regularity proof for (Λ_*, r_*)-almost minimisers is the Tamanini iteration presented in [15, Theorems 24.1, 24.2, 25.1, 25.3, and 26.3]. After scaling a ball $B_\rho(x)$ to a unit cylinder, the error enters only through the dimensionless number $\Lambda_* \rho$. The proof has the following four inputs:

(i) the slab and phase hypotheses, which orient a cylinder;
(ii) the Caccioppoli, or reverse Poincare, inequality converting height flatness to excess;
(iii) Lipschitz approximation and harmonic approximation;
(iv) tilt improvement followed by geometric iteration.

For each input, the dimensional constant that enters is explicit and standard:

- In (i), only volume comparison: the slab and phase hypotheses determine upper/lower-density regions in a cylinder of normalised height, with constants depending only on the dimension and on the chosen contraction factor of the cylinder.
- In (ii), the relevant constant is the one in the relative isoperimetric inequality on a half-ball (Maggi [15, Thm. 12.17]), applied to slices of the boundary above and below the cylinder axis.
- In (iii), the Lipschitz-approximation constant is obtained from Maggi [15, Thm. 23.7]; the harmonic-approximation step (which is the only originally-compactness step in the standard Tamanini argument) is replaced here by Lemma 3.22, whose constant is the explicit one from finite spectral truncation.
- In (iv), tilt improvement combines the harmonic-decay rate $\theta^{2\alpha}$ (with $\alpha \in (0, 1/2)$ and $\theta \in (0, 1/16)$ chosen below) with the Caccioppoli constant of (ii) and the harmonic-approximation constant of (iii); the geometric iteration sums to a fixed multiple of ℓ.

The two phase points in (4.82), together with the absence of boundary outside the slab, identify the upper and lower phases in the working cylinder. The comparison proof of (ii) then gives, after a fixed contraction of that cylinder, an inequality of the form

$$\mathrm{exc} \le C(n) \left(\varepsilon_{\mathrm{fl}}(n, \ell)^2 + \Lambda_* \rho\right),$$

where the numerical value of $C(n)$ is obtained by carrying out that comparison estimate. The constants in (i), (ii), and the Lipschitz-approximation part of (iii) arise from relative isoperimetry, Fubini/Chebyshev estimates, and explicit comparison competitors. The tilt estimate in (iv) consists of inserting the harmonic decay estimate into (ii), choosing a fixed contraction factor, and taking minima of finitely many preceding smallness conditions. These operations meet Definition 3.20.

The compactness step in the printed chain is the harmonic-approximation lemma used in the tilt argument: approximate harmonicity is converted into closeness to a harmonic function by contradiction and the compact embedding $W^{1,2} \Subset L^2$. Replace that invocation, after one fixed interior contraction of the cylinder, by Lemma 3.22 with $m = n - 1$. Losing the fixed factor $1/2$ only changes the contraction factor in (iv): harmonic decay is then applied on the smaller disk and the next cylinder is chosen inside it.

For completeness, the parameter-selection algorithm is as follows. Fix a contraction factor $\theta \in (0, 1/16)$. Starting with the desired slope ℓ, choose the target harmonic error τ so that harmonic decay plus the Caccioppoli error is at most $\theta^{2\alpha}$ times the preceding excess, for some fixed $\alpha \in (0, 1/2)$. Choose the corresponding $\sigma_{\mathrm{har}}(n-1, \tau)$ by Lemma 3.22. Then take $\varepsilon_{\mathrm{fl}}(n, \ell)$ to be the minimum of the finitely many height, Lipschitz-approximation, Caccioppoli, and tilt smallness requirements with that σ_{har}. Iteration gives a geometric series for the changes of tangent plane, whose sum is made at most ℓ by the initial choice of the minimum. This proves the single-graph conclusion in Definition 3.21.

Every operation in this backward choice is finite, except for summing a geometric series with specified ratio. Hence $\varepsilon_{\mathrm{fl}}(n,\ell)$ is computable, although we do not evaluate it. $\square$

Lemma 3.24 (Orientation of a trapped unit-volume set)**.** *Set*

$$h_{\mathrm{or}}(n) := \min\left\{\frac{1}{32}, \frac{1}{4n}\right\}.$$

Let G be a finite-perimeter set with $|G| = \omega_n$, $\partial G = \operatorname{spt}|D\mathbf{1}_G|$, and

$$d_H(\partial G, \partial B_1(z)) \leq h \leq h_{\mathrm{or}}(n). \tag{4.83}$$

Then, up to null sets,

$$B_{1-h}(z) \subset G \subset B_{1+h}(z). \tag{4.84}$$

Proof. The open connected regions $B_{1-h}(z)$ and $\mathbb{R}^n \setminus \overline{B_{1+h}(z)}$ do not meet ∂G. Since the boundary is the support of the perimeter measure, the characteristic function of G is almost everywhere constant on each of these regions. The exterior region cannot belong to G, because $|G| < \infty$. If the interior region did not belong to G, then

$$|G| \leq |B_{1+h} \setminus B_{1-h}| = \omega_n\big((1+h)^n - (1-h)^n\big).$$

Since $nh \leq 1/4$, Bernoulli's inequality and $(1+h)^n \leq e^{nh} \leq e^{1/4} < 3/2$ give

$$(1+h)^n - (1-h)^n < \frac{3}{2} - \frac{3}{4} < 1. \tag{4.85}$$

This contradicts $|G| = \omega_n$, proving (4.84). $\square$

Lemma 3.25 (Recentering a small radial graph)**.** *Set*

$$C_{\mathrm{rec}}(n) := 32\big(1 + n2^n\big). \tag{4.86}$$

Let $|E| = \omega_n$ and suppose that, about a centre z,

$$\partial E = \{z + (1 + v(\theta))\theta : \theta \in S^{n-1}\}, \qquad \|v\|_{W^{1,\infty}} \leq \kappa \leq \frac{1}{256(1 + n2^n)}. \tag{4.87}$$

If c_E is the barycentre of E, then E is a radial graph about c_E, with graph function u satisfying

$$\|u\|_{W^{1,\infty}} \leq C_{\mathrm{rec}}(n)\kappa. \tag{4.88}$$

Proof. Translating, take $z = 0$, and put $a = c_E$. Polar integration and $|E| = \omega_n$ give

$$a = \frac{1}{(n+1)\omega_n} \int_{S^{n-1}} (1+v)^{n+1}\theta \, d\sigma.$$

Since $\int_{S^{n-1}} \theta \, d\sigma = 0$, the mean value theorem and $|S^{n-1}| = n\omega_n$ yield

$$|a| \leq n(1+\kappa)^n \kappa \leq n2^n\kappa. \tag{4.89}$$

Put

$$\eta := \kappa + |a| \leq (1 + n2^n)\kappa \leq \frac{1}{256}.$$

The map

$$\Theta(\theta) := \frac{(1 + v(\theta))\theta - a}{|(1 + v(\theta))\theta - a|}$$

is quantitatively close to the identity. Indeed, writing $p = (1+v)\theta - a$, $R = |p|$, and letting τ be a unit tangent vector, one has

$$|R-1| \le \eta, \qquad |\Theta - \theta| \le \frac{2\eta}{1-\eta} \le 3\eta, \qquad |Dp[\tau] - \tau| \le 2\kappa \le 2\eta.$$

Since

$$D\Theta[\tau] = R^{-1}(I - \Theta \otimes \Theta) Dp[\tau],$$

and $|\Theta \otimes \Theta - \theta \otimes \theta| \le 2|\Theta - \theta|$, it follows that

$$|D\Theta[\tau] - \tau| \le \frac{2\eta}{1-\eta} + \frac{\eta}{1-\eta} + 6\eta \le 16\eta \le \frac{1}{16}.$$

Thus Θ is a local bi-Lipschitz map. The homotopy

$$\Theta_t(\theta) := \frac{(1+tv(\theta))\theta - ta}{|(1+tv(\theta))\theta - ta|}, \qquad 0 \le t \le 1,$$

is well-defined because its denominator is at least $1-\eta > 0$; therefore Θ has degree one. A degree-one local homeomorphism of S^{n-1} is a homeomorphism. Thus the translated boundary is a radial graph about a, whose radius is

$$1 + u(\Theta(\theta)) = |(1+v(\theta))\theta - a|.$$

Moreover, the displayed derivative bound gives $\|D\Theta^{-1}\| \le 2$, and differentiation of the radius gives

$$|DR[\tau]| \le (1+\kappa)|\Theta - \theta| + \kappa \le 7\eta.$$

Consequently

$$\|u\|_{L^\infty} \le \eta, \qquad \|\nabla_\tau u\|_{L^\infty} \le 14\eta.$$

Together with (4.89), these inequalities imply the larger stated bound (4.88). ☐

Theorem 3.26 (Effective graph entry from an admissible flatness number)**.** *Let G have unit volume, take the representative for which $\partial G = \operatorname{spt}|D\mathbf{1}_G|$, and suppose that it satisfies* (4.81) *with parameters Λ_*, r_*. Put*

$$\kappa_* := \frac{\delta_*(n)}{C_{\mathrm{rec}}(n)}, \qquad \ell_* := \frac{\kappa_*}{64}, \qquad \varepsilon_* := \varepsilon_{\mathrm{fl}}(n, \ell_*), \tag{4.90}$$

where $\varepsilon_{\mathrm{fl}}(n,\ell_*)$ is admissible, and define

$$\begin{aligned} \rho_* &:= \min\left\{\frac{r_*}{16}, \frac{\varepsilon_*}{512}, \frac{\varepsilon_*}{16(1+\Lambda_*)}, \frac{\kappa_*}{2048}, \frac{1}{64}\right\}, \\ h_* &:= \min\left\{\frac{\varepsilon_*\rho_*}{16}, \frac{\rho_*}{16}, h_{\mathrm{or}}(n)\right\}. \end{aligned} \tag{4.91}$$

If

$$d_H(\partial G, \partial B_1(z)) \le h_*, \tag{4.92}$$

then, after translation by the barycentre of G,

$$\partial G = \{(1+u(\theta))\theta : \theta \in S^{n-1}\}, \qquad \|u\|_{W^{1,\infty}} \le \delta_*(n). \tag{4.93}$$

Proof. After translating, take $z=0$. The last restriction in (4.91) permits direct application of Lemma 3.24, which fixes the inner and outer phases.

Fix $x \in \partial G$, choose $y \in \partial B_1$ with $|x-y| \le h_*$, and put $e = y$. If $p \in \partial G \cap B_{8\rho_*}(x)$, choose $\bar p \in \partial B_1$ with $|p - \bar p| \le h_*$. Then $|\bar p - y| \le 8\rho_* + 2h_* \le 9\rho_*$, and the elementary sphere identity

$$|(\bar p - y)\cdot y| = \frac{1}{2}|\bar p - y|^2$$

gives

$$|(p-x)\cdot e| \le 2h_* + \frac{81}{2}\rho_*^2 \le \varepsilon_*\rho_*. \tag{4.94}$$

Moreover, $x - 4\rho_* e \in B_{1-h_*}$ and $x + 4\rho_* e \notin B_{1+h_*}$, because $2h_* < 4\rho_*$. Thus the phase assumptions in (4.82) hold. Finally, $\Lambda_*\rho_* \le \varepsilon_*$ by (4.91). Definition 3.21 implies that $\partial G \cap B_{\rho_*}(x)$ is a single graph of slope at most ℓ_*, for every $x \in \partial G$.

It remains to pass from local graphs to one radial graph. The annular trapping makes the graph direction uniformly transverse to radial rays. Indeed, if $q \in \partial G \cap B_{\rho_*}(x)$, then

$$\left|\frac{q}{|q|} - e\right| \le 4(\rho_* + h_*) \le \frac{17}{256} < \frac{1}{4}.$$

At almost every point of the Lipschitz graph through x, choose its unit normal ν with $\nu\cdot e > 0$. The slope bound gives

$$|\nu - e| \le 2\ell_*, \qquad \left|\nu - \frac{q}{|q|}\right| \le 4(\ell_* + \rho_* + h_*) < \frac{1}{4}.$$

Consequently radial projection from each graph patch to S^{n-1} is locally one-to-one. To see this without an implicit smoothness argument, suppose that two points of one patch have the form $q_i = t_i\theta$, $t_1 \ne t_2$. Since $|\theta - e| < 1/4$, one has $\theta\cdot e > 3/4$. On the other hand, the graph slope estimate, applied to $q_2 - q_1 = (t_2 - t_1)\theta$, would give

$$|\theta\cdot e| \le \ell_*|\theta - (\theta\cdot e)e| \le \ell_* < \frac{1}{8},$$

a contradiction. In graph coordinates radial projection is continuous and locally injective; invariance of domain makes it open. Its image on ∂G is nonempty and compact, hence also closed; connectedness of S^{n-1} shows that it is onto. Finally, two global points on the same ray would both lie within h_* of the same spherical point and hence at distance less than $2h_* < \rho_*$, putting them in one graph patch and yielding the same contradiction. Thus

$$\partial G = \{(1+v(\theta))\theta : \theta \in S^{n-1}\}.$$

Here $\|v\|_{L^\infty} \le h_*$. For a radial graph its normal satisfies

$$\frac{|\nabla_\tau v|}{1+v} = \frac{|\nu - (\nu\cdot\theta)\theta|}{\nu\cdot\theta}, \qquad \theta = \frac{q}{|q|}.$$

The preceding estimate, $1 + v \le 1 + h_* < 2$, and $\nu\cdot\theta > 1/2$ imply

$$\|\nabla_\tau v\|_{L^\infty} \le 16(\ell_* + \rho_* + h_*) < \frac{\kappa_*}{3}.$$

Furthermore,

$$\|v\|_{L^\infty} \le h_* \le \frac{\rho_*}{16} \le \frac{\kappa_*}{32768}.$$

In particular, whether the $W^{1,\infty}$-norm is taken as a maximum or as a sum of these two quantities,

$$\|v\|_{W^{1,\infty}} \le \kappa_*. \tag{4.95}$$

Since $\delta_*(n) \le 1/8$, the definition of κ_* yields $\kappa_* \le 1/[256(1 + n2^n)]$. Apply Lemma 3.25 and (4.90) to obtain (4.93). □

Proposition 3.27 (Bounded closure through regularised selection)**.** *Let $\varepsilon_{\mathrm{fl}}(n,\ell_*)$ be the computable admissible number supplied by Proposition 3.23. In Theorem 3.26 take*

$$\Lambda_* := \frac{3}{2}\Lambda_{\mathrm{reg}}, \qquad r_* := \frac{2}{3}r_{\mathrm{reg}},$$

and let h_* be given by (4.91)*. Define*

$$d_{\mathrm{ent}} := \min\left\{ d_{\mathrm{ann}},\ \frac{1}{2C_{\mathrm{iso}}(n)}\left(\frac{h_*}{C_H(n,c_a)}\right)^{2n}\right\}. \tag{4.96}$$

Then, in the bounded class used above,

$$\beta(E)^2 \le C_{\mathrm{reg}}(n,R,\Lambda)\mathcal{D}(E), \tag{4.97}$$

where one may take

$$C_{\mathrm{reg}} := \max\left\{16,\ 2\left(1+\frac{P(B_1)}{d_{\mathrm{ent}}}\right)\right\}. \tag{4.98}$$

Proof. Let $d = \mathcal{D}(E)$ and $b = \beta(E)^2$. If $d > d_{\mathrm{ann}}$, Lemma 4.33 gives

$$b \le 2\{P(B_1)+d\} \le 2\left(1+\frac{P(B_1)}{d_{\mathrm{ent}}}\right)d.$$

If $d_{\mathrm{ent}} < d \le d_{\mathrm{ann}}$, the same conclusion follows with d_{ent} in place of d_{ann}. If $d/b \ge 1/16$, then $b \le 16d$. It remains to exclude

$$d \le d_{\mathrm{ent}}, \qquad \frac{d}{b} < \frac{1}{16}.$$

Apply Theorem 3.15 and Theorem 3.19. Equations (4.33) and (4.96) place $\widetilde{F}$, represented by $\partial\widetilde{F} = \operatorname{spt}|D\mathbf{1}_{\widetilde{F}}|$, within the Hausdorff threshold h_*. The perimeter comparisons and the normal oscillation are unchanged by this representative choice, and Theorem 3.26 places it in the radial-graph regime of Corollary 3.9; therefore

$$b(\widetilde{F}) \le 3\mathcal{D}(\widetilde{F}), \qquad \frac{\mathcal{D}(\widetilde{F})}{b(\widetilde{F})} \ge \frac{1}{3}.$$

On the other hand, (4.59) gives

$$\frac{\mathcal{D}(\widetilde{F})}{b(\widetilde{F})} \le \frac{d/b}{1-4d/b} < \frac{1}{12},$$

a contradiction. □

3.3.6. *Computable confinement and the global conclusion.*

Lemma 3.28 (Computable Fusco–Julin confinement)**.** *There are computable constants*

$$R_{\mathrm{red}}(n) > 3, \qquad A_{\mathrm{red}}(n) > 0, \qquad B_{\mathrm{red}}(n) > 0 \tag{4.99}$$

such that, for every finite-perimeter set $E \subset \mathbb{R}^n$ with $|E| = \omega_n$, there is a finite-perimeter set $E_0 \subset B_{R_{\mathrm{red}}(n)}$, after a translation, satisfying

$$|E_0| = \omega_n, \qquad b(E) \le b(E_0) + A_{\mathrm{red}}(n)\mathcal{D}(E), \qquad \mathcal{D}(E_0) \le B_{\mathrm{red}}(n)\mathcal{D}(E). \tag{4.100}$$

Proof. The geometric construction is Lemma 3.2 of Fusco–Julin [6, Lemma 3.2], which in turn repeats the coordinate-slicing construction of [8, Lemma 5.1]. We explain why the constants in that construction are computable.

The slicing construction performs one cut in each coordinate direction. At one step, the coarea formula

$$\int_a^b \mathcal{H}^{n-1}(E \cap \{x_i = t\})\, dt \le P(E)$$

integrates the slice perimeters over a fixed-dimensional interval $[a, b]$; an averaging choice of two cutting hyperplanes $t_1, t_2 \in [a, b]$ satisfying $\mathcal{H}^{n-1}(E \cap \{x_i = t_1\}) + \mathcal{H}^{n-1}(E \cap \{x_i = t_2\}) \le 2P(E)/(b-a)$ retains the central slab, does not increase perimeter, and removes at most a dimensional multiple of d in volume. Repeating this operation exactly n times gives a set G, contained after translation in a rectangular box of dimensional diameter $L_{\mathrm{red}}(n)$, for which

$$G \subset E, \qquad P(G) \le P(E), \qquad \omega_n - |G| \le C_{\mathrm{cut}}(n)d. \tag{4.102}$$

The cutting width $L_{\mathrm{red}}(n)$ and the dimensional constant $C_{\mathrm{cut}}(n)$ are obtained by finitely many applications of the displayed slicing inequalities in [8, Lemma 5.1] and are therefore computable; in particular, $C_{\mathrm{cut}}(n)$ is bounded above by an explicit polynomial in n (we do not need its numerical value).

Now fix the computable threshold

$$d_{\mathrm{cut}} := \frac{\omega_n}{2C_{\mathrm{cut}}(n)}, \tag{55}$$

so that for every E with $d := \mathcal{D}(E) < d_{\mathrm{cut}}$ the slicing construction yields $|G| \ge \omega_n/2$. If instead $d \ge d_{\mathrm{cut}}$, take $E_0 = B_1$, so that $\mathcal{D}(E_0) = 0$ and, from $b(E) \le P(E) \le P(B_1) + d$,

$$b(E) \le \left(1 + \frac{P(B_1)}{d_{\mathrm{cut}}}\right) d = \left(1 + \frac{2C_{\mathrm{cut}}(n)P(B_1)}{\omega_n}\right) d, \qquad \mathcal{D}(E_0) = 0. \tag{4.101}$$

This dispatches the large-deficit branch, with computable constant.

Because $G \subset E$, evaluation of $M(E)$ at a maximising centre for G gives $M(E) \ge M(G)$. Therefore

$$\begin{aligned} b(E) - b(G) &\le P(E) - P(G) \\ &\le d + P(B_1) - P(G) \\ &\le d + P(B_1)\left[1 - \left(\frac{|G|}{\omega_n}\right)^{(n-1)/n}\right] \\ &\le C_1(n)d. \end{aligned} \tag{4.103}$$

The third line is the isoperimetric inequality applied to G; the last line follows from (4.102) and the mean value theorem on $[1/2, 1]$. Hence $C_1(n)$ is computable.

Set $\lambda = (\omega_n/|G|)^{1/n}$ and $E_0 = \lambda G$, followed by translation of its containing box into a ball. Then $1 \le \lambda \le 2^{1/n}$, so one may take

$$R_{\mathrm{red}}(n) := \max\{4, 2^{1/n} L_{\mathrm{red}}(n)\}.$$

Moreover, since b scales by λ^{n-1} and is nonnegative,

$$b(E) \le b(G) + C_1(n)d \le b(E_0) + C_1(n)d.$$

Finally,

$$\begin{aligned} \mathcal{D}(E_0) &= \lambda^{n-1}P(G) - P(B_1) \\ &\le \{\lambda^{n-1} - 1\}P(B_1) + \lambda^{n-1}d \le C_2(n)d, \end{aligned} \tag{4.104}$$

again by (4.102) and the mean value theorem on $[1/2, 1]$. Combining the small- and large-deficit branches gives (4.100), with computable A_{red} and B_{red}. $\square$

Theorem 3.29 (A truly quantitative Fusco–Julin inequality)**.** *For every $n \geq 2$, there is a computable dimensional constant $C_{\mathrm{FJ}}^{\mathrm{comp}}(n) > 0$ such that every finite-perimeter $E \subset \mathbb{R}^n$ with $|E| = |B_\rho|$ satisfies*

$$\beta(E)^2 \leq C_{\mathrm{FJ}}^{\mathrm{comp}}(n)\mathcal{D}(E). \tag{4.105}$$

Proof. By scaling it suffices to consider $|E| = \omega_n$. Apply Lemma 3.28 and take

$$R = R_{\mathrm{red}}(n), \qquad \Lambda = 2n$$

in Proposition 3.27. Since E_0 belongs to the bounded class of that proposition,

$$\begin{aligned} b(E) &\leq b(E_0) + A_{\mathrm{red}}(n)\mathcal{D}(E) \\ &\leq C_{\mathrm{reg}}\big(n, R_{\mathrm{red}}(n), 2n\big)\mathcal{D}(E_0) + A_{\mathrm{red}}(n)\mathcal{D}(E) \\ &\leq \big\{A_{\mathrm{red}}(n) + B_{\mathrm{red}}(n)C_{\mathrm{reg}}\big(n, R_{\mathrm{red}}(n), 2n\big)\big\}\, \mathcal{D}(E). \end{aligned}$$

Thus one may set

$$C_{\mathrm{FJ}}^{\mathrm{comp}}(n) := A_{\mathrm{red}}(n) + B_{\mathrm{red}}(n)C_{\mathrm{reg}}\big(n, R_{\mathrm{red}}(n), 2n\big). \tag{4.106}$$

Every constant on the right is computable by Lemma 3.28, Proposition 3.23, and the displayed formulae preceding Proposition 3.27. The scaling laws in Lemma 3.4 give the assertion for general $\rho > 0$. $\square$

Corollary 3.30 (Original Fusco–Julin formulation)**.** *For $|E| = |B_\rho|$, define the scale-invariant same-centre index*

$$\mathcal{A}_{\mathrm{FJ}}(E) := \inf_{z\in\mathbb{R}^n} \left\{ \frac{|E \triangle B_\rho(z)|}{\rho^n} + \left(\frac{1}{\rho^{n-1}} \int_{\partial^* E} \left| \nu_E - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1} \right)^{1/2} \right\}. \tag{4.107}$$

There is a computable dimensional constant $\widehat{C}_{\mathrm{FJ}}^{\mathrm{comp}}(n) > 0$ such that every finite-perimeter set $E \subset \mathbb{R}^n$ with $|E| = |B_\rho|$ satisfies

$$\mathcal{A}_{\mathrm{FJ}}(E)^2 \leq \widehat{C}_{\mathrm{FJ}}^{\mathrm{comp}}(n)\, \rho^{1-n}\mathcal{D}(E). \tag{4.108}$$

Proof. Write $\beta_{\mathrm{FJ}}(E)^2 := \rho^{1-n}\beta(E)^2$ for the scale-invariant normal index, with β as in Definition 3.1. For $z \in \mathbb{R}^n$, abbreviate the centred potential deficit

$$\mathcal{C}_z(E) := P(B_\rho) - (n-1) \int_E \frac{dx}{|x-z|},$$

which is nonnegative for every z by the bathtub principle, as in the proof of Lemma 3.11. By Proposition 3.3 (applied with the prescribed radius ρ),

$$\mathcal{O}_z(E) = P(E) - (n-1) \int_E \frac{dx}{|x-z|} = \mathcal{D}(E) + \mathcal{C}_z(E), \tag{4.110}$$

exactly as in the decomposition (4.6). In particular

$$\beta(E)^2 = \inf_{z\in\mathbb{R}^n} \mathcal{O}_z(E) = \mathcal{D}(E) + \inf_{z\in\mathbb{R}^n} \mathcal{C}_z(E),$$

so $\beta(E)^2 \geq \mathcal{D}(E)$ and a fortiori

$$\rho^{1-n}\mathcal{D}(E) \leq \rho^{1-n}\beta(E)^2 = \beta_{\mathrm{FJ}}(E)^2. \tag{4.111}$$

Step 1: a bathtub shell bound for the symmetric difference. We claim that, for every $z \in \mathbb{R}^n$ and every finite-perimeter $E \subset \mathbb{R}^n$ with $|E| = |B_\rho|$,

$$|E \triangle B_\rho(z)|^2 \le C_{\mathrm{bt}}(n)\, \rho^{n+1}\, \mathcal{C}_z(E), \qquad C_{\mathrm{bt}}(n) := \frac{8n\omega_n}{n-1}. \tag{4.112}$$

Set $A_+ := B_\rho(z) \setminus E$ and $A_- := E \setminus B_\rho(z)$. Since $|E| = |B_\rho(z)|$ we have $|A_+| = |A_-| =: V$, and therefore

$$|E \triangle B_\rho(z)| = 2V.$$

Applying (4.110) to the ball $B_\rho(z)$ (for which $\mathcal{O}_z(B_\rho(z)) = 0$ and $\mathcal{D}(B_\rho(z)) = 0$) gives $\mathcal{C}_z(B_\rho(z)) = 0$, i.e.

$$P(B_\rho) = (n-1) \int_{B_\rho(z)} \frac{dx}{|x-z|} = (n-1)\, n\omega_n \int_0^\rho s^{n-2}\, ds.$$

Subtracting from the defining expression for $\mathcal{C}_z(E)$,

$$\mathcal{C}_z(E) = (n-1)\Big[\int_{A_+} \frac{dx}{|x-z|} - \int_{A_-} \frac{dx}{|x-z|}\Big].$$

On $A_+ \subset B_\rho(z)$ the integrand $|x-z|^{-1}$ is at least ρ^{-1}; on $A_- \subset \mathbb{R}^n \setminus B_\rho(z)$ it is at most ρ^{-1}. This shows $\mathcal{C}_z(E) \ge 0$ directly. To extract a quantitative gap, we apply the bathtub principle in each shell. Among all measurable $F_+ \subset B_\rho(z)$ of measure V, the quantity $\int_{F_+} |x-z|^{-1}\, dx$ is minimised by the inner shell $\{r' \le |x-z| \le \rho\}$ with $\omega_n(\rho^n - (r')^n) = V$; among all measurable $F_- \subset \mathbb{R}^n \setminus B_\rho(z)$ of measure V, the same quantity is maximised by the outer shell $\{\rho \le |x-z| \le r\}$ with $\omega_n(r^n - \rho^n) = V$. Both statements are layer-cake rearrangements with the radially decreasing kernel $|x-z|^{-1}$. Consequently,

$$\int_{A_+} \frac{dx}{|x-z|} \ge n\omega_n \int_{r'}^{\rho} s^{n-2}\, ds = \frac{n\omega_n}{n-1}\big(\rho^{n-1} - (r')^{n-1}\big),$$

$$\int_{A_-} \frac{dx}{|x-z|} \le n\omega_n \int_{\rho}^{r} s^{n-2}\, ds = \frac{n\omega_n}{n-1}\big(r^{n-1} - \rho^{n-1}\big),$$

and therefore

$$\mathcal{C}_z(E) \ge n\omega_n\big[2\rho^{n-1} - (r')^{n-1} - r^{n-1}\big]. \tag{4.113}$$

Set $s := V/(\omega_n\rho^n)$, so that $r^{n-1} = \rho^{n-1}(1+s)^{(n-1)/n}$ and $(r')^{n-1} = \rho^{n-1}(1-s)^{(n-1)/n}$; since $|E| = |B_\rho|$, $V \le |E| = \omega_n\rho^n$, so $s \in [0,1]$. Define $h(s) := 2 - (1+s)^{(n-1)/n} - (1-s)^{(n-1)/n}$. A direct differentiation gives $h(0) = h'(0) = 0$ and

$$h''(s) = \frac{n-1}{n^2}\big[(1+s)^{-(n+1)/n} + (1-s)^{-(n+1)/n}\big] \ge \frac{n-1}{n^2} \qquad (s \in [0,1)),$$

since $(1-s)^{-(n+1)/n} \ge 1$ for $s \in [0,1)$. Two integrations yield

$$h(s) \ge \frac{n-1}{2n^2}\, s^2 \qquad (s \in [0,1]).$$

Inserting this lower bound into (4.113) gives

$$\mathcal{C}_z(E) \ge n\omega_n\rho^{n-1} \cdot \frac{n-1}{2n^2}\, s^2 = \frac{(n-1)\omega_n\rho^{n-1}}{2n} \cdot \frac{V^2}{\omega_n^2\rho^{2n}} = \frac{n-1}{2n\omega_n} \cdot \frac{V^2}{\rho^{n+1}}.$$

Using $|E \triangle B_\rho(z)| = 2V$,

$$|E \triangle B_\rho(z)|^2 = 4V^2 \le \frac{8n\omega_n}{n-1}\, \rho^{n+1}\, \mathcal{C}_z(E),$$

which is (4.112).

Step 2: comparison at any near-minimising centre. Fix $\varepsilon > 0$ and choose $z_\varepsilon \in \mathbb{R}^n$ with $\mathcal{O}_{z_\varepsilon}(E) \le \beta(E)^2 + \varepsilon\rho^{n-1}$. By (4.110), $\mathcal{C}_{z_\varepsilon}(E) \le \beta(E)^2 + \varepsilon\rho^{n-1}$, so (4.112) gives

$$|E\triangle B_\rho(z_\varepsilon)|^2 \le C_{\mathrm{bt}}(n)\,\rho^{n+1}\big(\beta(E)^2 + \varepsilon\rho^{n-1}\big) = C_{\mathrm{bt}}(n)\,\rho^{2n}\big(\beta_{\mathrm{FJ}}(E)^2 + \varepsilon\big),$$

hence

$$\frac{|E\triangle B_\rho(z_\varepsilon)|}{\rho^n} \le \sqrt{C_{\mathrm{bt}}(n)}\,\big(\beta_{\mathrm{FJ}}(E)^2 + \varepsilon\big)^{1/2}. \tag{4.114}$$

The normal-oscillation term at the same centre satisfies

$$\left(\frac{1}{\rho^{n-1}}\int_{\partial^* E}\big|\nu_E - e_{z_\varepsilon}\big|^2\,d\mathcal{H}^{n-1}\right)^{1/2} = \big(\rho^{1-n}\cdot 2\mathcal{O}_{z_\varepsilon}(E)\big)^{1/2} \le \sqrt{2}\big(\beta_{\mathrm{FJ}}(E)^2 + \varepsilon\big)^{1/2}. \tag{4.115}$$

(The factor $\sqrt{2}$ appears because the normalisation in $\mathcal{A}_{\mathrm{FJ}}$ integrates the full $|\nu_E - e_z|^2$, while $\mathcal{O}_z$ carries a factor $1/2$; compare Definition 3.1 and (4.107).) Adding (4.114) and (4.115) and infimising over admissible z,

$$\mathcal{A}_{\mathrm{FJ}}(E) \le \big(\sqrt{C_{\mathrm{bt}}(n)} + \sqrt{2}\big)\big(\beta_{\mathrm{FJ}}(E)^2 + \varepsilon\big)^{1/2}.$$

Letting $\varepsilon \downarrow 0$,

$$\mathcal{A}_{\mathrm{FJ}}(E) \le \big(\sqrt{C_{\mathrm{bt}}(n)} + \sqrt{2}\big)\,\beta_{\mathrm{FJ}}(E). \tag{4.116}$$

Step 3: assembling the comparison and the conclusion. Combining (4.111) and (4.116),

$$\mathcal{A}_{\mathrm{FJ}}(E) + \big(\rho^{1-n}\mathcal{D}(E)\big)^{1/2} \le C_{\mathrm{cmp}}(n)\,\beta_{\mathrm{FJ}}(E), \quad C_{\mathrm{cmp}}(n) := 1 + \sqrt{2} + \sqrt{C_{\mathrm{bt}}(n)}. \tag{4.117}$$

The constant $C_{\mathrm{cmp}}(n)$ is computable since $C_{\mathrm{bt}}(n) = 8n\omega_n/(n-1)$ is a finite combination of the volume of the unit ball and elementary dimensional factors. In particular, the proof has used only the radial Gauss–Green identity (Proposition 3.3), the bathtub-principle layer-cake comparison for the radially decreasing kernel $|x - z|^{-1}$, and the elementary concavity estimate $h(s) \ge \frac{(n-1)}{2n^2}s^2$ on $[0, 1]$; no compactness selection or contradiction argument is invoked.

Squaring (4.117) and applying Theorem 3.29,

$$\mathcal{A}_{\mathrm{FJ}}(E)^2 \le C_{\mathrm{cmp}}(n)^2\,\beta_{\mathrm{FJ}}(E)^2 = C_{\mathrm{cmp}}(n)^2\,\rho^{1-n}\beta(E)^2 \le C_{\mathrm{cmp}}(n)^2\,C_{\mathrm{FJ}}^{\mathrm{comp}}(n)\,\rho^{1-n}\mathcal{D}(E),$$

which is (4.108) with

$$\widehat{C}_{\mathrm{FJ}}^{\mathrm{comp}}(n) := \big(1 + \sqrt{2} + \sqrt{8n\omega_n/(n-1)}\big)^2\,C_{\mathrm{FJ}}^{\mathrm{comp}}(n). \tag{4.118}$$

Every constant on the right-hand side is computable: $C_{\mathrm{bt}}(n)$ is the explicit dimensional expression above, and $C_{\mathrm{FJ}}^{\mathrm{comp}}(n)$ is given by (4.106). □

4. Proof of the main theorem

With the technical inputs of §2 and the internal Fusco–Julin estimate of §3 now in place, we are ready to assemble Theorem 1.1. The argument runs in five movements. Section 4.1 establishes the exact level-set deficit identity, which writes $\delta_T(\Omega)$ as a reduced-boundary integral of the Cauchy–Schwarz defect D_H and the squared-perimeter isoperimetric defect D_I, and derives its profile-gap and bad-density consequences. Section 4.2 carries out the De Giorgi truncation that removes the boundedness hypothesis, taking the bounded Saint–Venant stability inequality (Theorem 4.11 below) as input. Sections 4.3–4.5 then prove that bounded stability by developing the affine moving-ball first variation, the endpoint trace, and the positive kinetic estimate.

Finally, §4.6 closes the bounded estimate and fits it together with the Kohler–Jobin transfer of §2.4 to produce Theorem 1.1.

4.1. **The exact level-set deficit identity.**

4.1.1. *Finite-perimeter level-set identities.* All integrals in this section are taken on De Giorgi reduced boundaries and all statements hold for almost every level; no smoothness of $\partial\Omega$, no graph parametrisation of the level sets, and no pointwise lower bound on $|\nabla u|$ are used. We keep the normalisation $|\Omega| = \omega_n$, so the ball of the same volume is B_1; the general case follows by scaling. Write

$$M := \|u\|_\infty, \qquad L := |\Omega| = \omega_n, \qquad m(t) := |\{u > t\}|,$$
$$\alpha(t) := \int_{\partial^* E_t} \frac{1}{|\nabla u|}\, d\mathcal{H}^{n-1}, \qquad \beta(t) := \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1}, \qquad c_n := n^2\omega_n^{2/n}.$$

By the Talenti pointwise comparison [18], $u^*(s) \le u^*_{B_1}(s) = (1-(s/\omega_n)^{2/n})/(2n)$, so $M \le 1/(2n) < \infty$; here u^* is the decreasing rearrangement of u.
Regularity and level conventions.

Lemma 4.1 (Critical set and level conventions)**.** *Let u be extended by* 0 *to* $\mathbb{R}^n$*. Then:*

(a) $|\{u = 0\} \cap \Omega| = 0$ *and* $|\{\nabla u = 0\} \cap \Omega| = 0$*;*
(b) *for every* $t \in (0, M)$*,* $|\{u = t\}| = 0$*; hence m is continuous and strictly decreasing on* $(0, M)$*;*
(c) *for a.e.* $t \in (0, M)$*:* E_t *has finite perimeter,* $\partial^* E_t \subset \Omega$ *up to* $\mathcal{H}^{n-1}$*-null sets,* $\nu_{E_t} = -\nabla u/|\nabla u|$ $\mathcal{H}^{n-1}$*-a.e. on* $\partial^* E_t$*, and* $\mathcal{H}^{n-1}(\partial^* E_t \cap \{\nabla u = 0\}) = 0$.

Proof. (a) Testing $-\Delta u = 1$ against $u^- \in H^1_0(\Omega)$ gives $\int |\nabla u^-|^2 = -\int u^- \le 0$, so $u \ge 0$ a.e.; the weak strong maximum principle on each connected component yields $u > 0$ a.e. in Ω [14], i.e. $|\{u = 0\} \cap \Omega| = 0$. Interior elliptic regularity gives $u \in W^{2,p}_{\mathrm{loc}}(\Omega)$ for all $p < \infty$; applying Stampacchia's theorem [17] to the weak derivatives $\partial_i u$ gives $D^2 u = 0$ a.e. on $\{\nabla u = 0\}$, while $\Delta u = -1$ a.e. Hence $|\{\nabla u = 0\} \cap \Omega| = 0$.

(b) For $t \in (0, M)$ the Sobolev chain rule applied to $(u - t)^+$ gives $\nabla u = 0$ a.e. on $\{u = t\}$; since $\{u = t\} \subset \{u > 0\} \subset \Omega$ up to a null set, part (a) gives $|\{u = t\}| = 0$. A jump of m at t would have size $|\{u = t\}| = 0$, so m is continuous on $(0, M)$. For $0 < a < b < M$, some component of Ω has essential supremum of u exceeding b; on it $\{a < u < b\}$ is a nonempty open set of positive measure, so $m(a) - m(b) = |\{a < u \le b\}| > 0$.

(c) By the BV coarea formula [15, Thm. 13.1], [1, Thm. 3.40], for every Borel $\varphi \ge 0$,

$$\int_\Omega |\nabla u|\, \varphi\, dx = \int_{-\infty}^{\infty} \Big(\int_{\partial^* E_t} \varphi\, d\mathcal{H}^{n-1} \Big)\, dt. \tag{56}$$

Taking $\varphi = \mathbf{1}_{(0,M)}(u)$ (which equals 1 on $\partial^* E_t$ precisely for $t \in (0, M)$) gives $\int_0^M P(E_t)\, dt = \int_{\{0<u<M\}} |\nabla u|\, dx \le \int_\Omega |\nabla u| < \infty$, the last bound because $|\nabla u| \in L^2(\Omega) \subset L^1(\Omega)$ on the bounded set Ω; hence $P(E_t) < \infty$ for a.e. t. For $t > 0$ the precise representative of u vanishes $\mathcal{H}^{n-1}$-a.e. on $\mathbb{R}^n \setminus \Omega$ (as $u \in H^1_0(\Omega)$), and for a.e. t one has $\partial^* E_t \subset \{u = t\}$ up to $\mathcal{H}^{n-1}$-null sets [1, 15]; therefore $\partial^* E_t \subset \Omega$ up to $\mathcal{H}^{n-1}$-null sets. The orientation $\nu_{E_t} = -\nabla u/|\nabla u|$ is the standard one for superlevel sets of Sobolev functions [1]. Finally, (56) with $\varphi = \mathbf{1}_{\{\nabla u = 0\}}$ gives $\int_0^M \mathcal{H}^{n-1}(\partial^* E_t \cap \{\nabla u = 0\})\, dt = \int_{\{\nabla u = 0\}} |\nabla u|\, dx = 0$, so the last assertion holds for a.e. t. □

Statements "for a.e. t" below refer to the full-measure set of levels furnished by Lemma 4.1(c).

Coarea form of $-m'$ and the flux identity.

Lemma 4.2 (Coarea form of $-m'$)**.** *For a.e.* $t \in (0, M)$*,* $-m'(t) = \alpha(t)$*. In particular* m *is locally absolutely continuous on* $(0, M)$ *and* $\alpha \in L^1_{\mathrm{loc}}((0, M))$*.*

Proof. Fix $0 < a < b < M$ and apply (56) to $\varphi = |\nabla u|^{-1} \mathbf{1}_{\{|\nabla u|>0\}} \mathbf{1}_{(a,b)}(u)$. By Lemma 4.1(a) the integrand on the left equals $\mathbf{1}_{\{a<u<b\}}$ up to a null set, so $m(a) - m(b) = \int_a^b \alpha(t)\, dt$. Local absolute continuity of m and $-m' = \alpha$ a.e. follow. □

Lemma 4.3 (Flux equals volume)**.** *For a.e.* $t \in (0, M)$*,* $\beta(t) = \int_{\partial^* E_t} |\nabla u|\, d\mathcal{H}^{n-1} = m(t)$*.*

Proof. For $\varepsilon > 0$ set $\phi_\varepsilon(s) := \min(1, \max(0, s/\varepsilon))$ and $\eta_\varepsilon := \phi_\varepsilon(u-t) \in H^1_0(\Omega)$. Testing $-\Delta u = 1$ against η_ε,

$$\int_\Omega \nabla u \cdot \nabla \eta_\varepsilon\, dx = \int_\Omega \eta_\varepsilon\, dx.$$

The Sobolev chain rule [1, Thm. 3.99] gives $\nabla \eta_\varepsilon = \varepsilon^{-1} \nabla u\, \mathbf{1}_{\{t<u<t+\varepsilon\}}$, so the left side equals $\varepsilon^{-1} \int_{\{t<u<t+\varepsilon\}} |\nabla u|^2$, which by (56) (with $\varphi = |\nabla u|\, \mathbf{1}_{(t,t+\varepsilon)}(u)$) equals $\varepsilon^{-1} \int_t^{t+\varepsilon} \beta(\tau)\, d\tau$. As $\varepsilon \downarrow 0$, $\eta_\varepsilon \to \mathbf{1}_{E_t}$ boundedly (Lemma 4.1(b)), so the right side tends to $m(t)$. Since $\beta \in L^1((0, M))$ by (56) and the Dirichlet identity $\int_0^M \beta = \int_\Omega |\nabla u|^2 = \int_\Omega u < \infty$, at every Lebesgue point of β the left side tends to $\beta(t)$. Hence $\beta(t) = m(t)$ for a.e. t. □

The rearranged primitive. Let $I(s) := \int_0^s u^*(\sigma)\, d\sigma$, $s \in [0, L]$.

Lemma 4.4 (Derivatives of the primitive)**.** I *is absolutely continuous on* $[0, L]$ *with* $I'(s) = u^*(s)$ *a.e.; on every compact subinterval of* $(0, L)$ *the rearrangement* u^* *is absolutely continuous and*

$$I''(s) = (u^*)'(s) = \frac{1}{m'(u^*(s))} = -\frac{1}{\alpha(u^*(s))} \qquad \text{for a.e. } s \in (0, L). \tag{57}$$

Proof. By the Talenti bound $u^* \in L^\infty(0, L) \subset L^1(0, L)$; the Lebesgue differentiation theorem gives $I' = u^*$ a.e. By Lemma 4.2, m is locally absolutely continuous with $m' = -\alpha$ a.e., and m is continuous and strictly decreasing on $(0, M)$ by Lemma 4.1(b), so $m(u^*(s)) = s$. The absolutely continuous inverse theorem for monotone functions [12, Thm. 0.4, §1] gives that u^* is absolutely continuous on compact subintervals of $(0, L)$ with $(u^*)'(s) = 1/m'(u^*(s))$ wherever $m'(u^*(s))$ exists and is nonzero. For a.e. $t \in (0, M)$ one has $0 < |E_t| < |\Omega|$, hence $P(E_t) > 0$ and $\alpha(t) > 0$; thus $\{m' = 0\} = \{\alpha = 0\}$ is null and carries no image measure ($|m(\{m' = 0\})| \le \int_{\{m'=0\}} |m'| = 0$). Since $m' = -\alpha$, (57) follows; absolute continuity of u^* on compact subintervals identifies the distributional and a.e. derivatives of $I' = u^*$. □

The exact deficit identity. For a.e. $t \in (0, M)$ the two level-set defects of the standing notation read, as functions of the level,

$$D_H(t) = \alpha(t)\beta(t) - P(E_t)^2, \qquad D_I(t) = P(E_t)^2 - c_n\, m(t)^{2-2/n},$$

consistent with §2.1 under $t = t(\rho)$ (so $P(B_\rho)^2 = c_n m(t)^{2-2/n}$). By Cauchy–Schwarz on $\partial^* E_t$ applied to $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$, $P(E_t)^2 \le \alpha(t)\beta(t)$, so $D_H(t) \ge 0$, with equality iff $|\nabla u|$ is $\mathcal{H}^{n-1}$-a.e. constant on $\partial^* E_t$. By the De Giorgi isoperimetric inequality [13, 15], $P(E_t) \ge n\omega_n^{1/n} m(t)^{1-1/n}$, so $D_I(t) \ge 0$, with equality iff E_t is equivalent to a ball.

Lemma 4.5 (Pointwise convexity formula)**.** *Set* $G(s) := I(s) + \dfrac{s^{1+2/n}}{(4 + 2n)\, \omega_n^{2/n}}$*. Then for a.e.* $s \in (0, L)$*, with* $t = u^*(s)$*,*

$$G''(s) = \frac{1}{c_n\, s^{1-2/n}} - \frac{1}{\alpha(t)}.$$

Proof. Direct differentiation gives $\frac{d^2}{ds^2}\frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{1}{c_n s^{1-2/n}}$; add (57). $\square$

Lemma 4.6 (Convexity identity). $\displaystyle\int_0^L s\,G''(s)\,ds = \int_0^M \frac{D_H(t)+D_I(t)}{c_n\,m(t)^{1-2/n}}\,dt.$

Proof. Substitute $s = m(t)$, $ds = -\alpha(t)\,dt$, in $\int_0^L s\,G''(s)\,ds$ (Lemma 4.1(b) removes jump terms, and $\{\alpha = 0\}$ carries no image measure):

$$\int_0^L s\,G''\,ds = \int_0^M m(t)\Big(\frac{1}{c_n m(t)^{1-2/n}} - \frac{1}{\alpha(t)}\Big)\alpha(t)\,dt = \int_0^M \Big(\frac{m(t)\alpha(t)}{c_n m(t)^{1-2/n}} - m(t)\Big)dt.$$

Use $m(t) = \beta(t)$ (Lemma 4.3) and add and subtract $P(E_t)^2/(c_n m(t)^{1-2/n})$:

$$\frac{m\alpha}{c_n m^{1-2/n}} - m = \frac{\alpha\beta - c_n m^{2-2/n}}{c_n m^{1-2/n}} = \frac{D_H + D_I}{c_n m^{1-2/n}}. \qquad \square$$

Lemma 4.7 (Endpoint convexity-gap formula). *With* $\Gamma(\Omega) := G'_-(L) - G(L)/L$*, one has* $\Gamma(\Omega) = \frac{1}{L}\int_0^L s\,G''(s)\,ds$.

Proof. $G \in W^{2,1}_{\mathrm{loc}}((0,L])$ and $G(0) = I(0) = 0$. Integration by parts on (ε, L) gives $\int_\varepsilon^L sG'' = L\,G'_-(L) - \varepsilon G'(\varepsilon) - G(L) + G(\varepsilon)$. By Lemma 4.4, $G'(s) = u^*(s) + s^{2/n}/(2n\omega_n^{2/n})$ is bounded, so $\varepsilon G'(\varepsilon) \to 0$ and $G(\varepsilon) \to 0$; divide by L. $\square$

Lemma 4.8 (Endpoint value). $\Gamma(\Omega) = \dfrac{2}{|\Omega|}\big(E(\Omega) - E(B_1)\big) = \dfrac{2}{\omega_n}\delta_T(\Omega)$.

Proof. At $s = L$, $u^*(s) \to 0$ as $s \uparrow L$ (Lemma 4.1 gives $m(t) \uparrow L$ as $t \downarrow 0$), so $G'_-(L) = L^{2/n}/(2n\omega_n^{2/n})$ and $I(L) = \int_\Omega u\,dx$. With $L = \omega_n$, using $\frac{1}{2n} - \frac{1}{4+2n} = \frac{1}{2n} - \frac{1}{2(n+2)} = \frac{1}{n(n+2)}$,

$$\Gamma(\Omega) = \frac{L^{2/n}}{2n\omega_n^{2/n}} - \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}} - \frac{1}{L}\int_\Omega u = \frac{1}{n(n+2)} - \frac{1}{\omega_n}\int_\Omega u.$$

For the unit-volume ball B_1, $u_{B_1}(x) = (1-|x|^2)/(2n)$ and

$$\int_{B_1} u_{B_1} = \frac{1}{2n}n\omega_n\int_0^1 (1-r^2)r^{n-1}\,dr = \frac{\omega_n}{2}\Big(\frac{1}{n} - \frac{1}{n+2}\Big) = \frac{\omega_n}{n(n+2)},$$

so $\frac{1}{\omega_n}\int_{B_1} u_{B_1} = \frac{1}{n(n+2)}$ and $\Gamma(B_1) = 0$ (the Pólya–Szegő identification [16, 2]). Subtracting, $\Gamma(\Omega) = \frac{1}{\omega_n}(\int_{B_1} u_{B_1} - \int_\Omega u) = \frac{2}{\omega_n}(E(\Omega) - E(B_1))$, since $E(\cdot) = -\frac12\int(\cdot)$. $\square$

Theorem 4.9 (Exact deficit identity on reduced-boundary level sets). *Let* $\Omega \subset \mathbb{R}^n$ *be open with* $|\Omega| = \omega_n$ *and* $u \in H^1_0(\Omega)$ *the torsion function. Then* (56) *and Lemmas 4.2, 4.3, 4.4 hold, and*

$$\boxed{2\,\delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t)+D_I(t)}{n^2\omega_n^{2/n}\,m(t)^{1-2/n}}\,dt,} \tag{58}$$

with $D_H(t), D_I(t) \geq 0$ *for a.e.* t*. By scaling, the corresponding identity* $\frac{2}{|\Omega|}(E(\Omega) - E(B)) = \frac{1}{|\Omega|}\int_0^{\|u\|_\infty}(D_H + D_I)/(c_n m^{1-2/n})\,dt$ *holds for every open* Ω *of finite measure.*

Proof. Combine Lemmas 4.6, 4.7, 4.8; the scaling statement follows since both sides are 0-homogeneous under $\Omega \mapsto \lambda\Omega$ after the displayed normalisation. $\square$

4.1.2. *Profile gap and bad-density estimate.*

Proposition 4.10 (Level-set budget and profile-gap bad set). *There are computable constants*

$$C_L = C_L(n,\rho_*), \qquad C_\tau = C_\tau(n,\rho_*), \qquad \kappa = \kappa(\rho_*) > 0,$$

such that, with

$$\rho_\delta := 1 - \kappa\sqrt{\delta_T(\Omega)}$$

and whenever $\rho_\delta \geq (1+\rho_*)/2$, one has

$$\int_{\rho_*}^{\rho_\delta} \big(D_H(t(\rho)) + D_I(t(\rho))\big)\, d\mu(\rho) \leq C_L\, \delta_T(\Omega), \tag{59}$$

$$0 \leq w(\rho) \leq \frac{1}{n} \quad \textit{for a.e. } \rho,$$

and, with

$$\tau_0 := \frac{\rho_*}{2n}, \qquad B_\tau := \{\rho \in [\rho_*, \rho_\delta] : w(\rho) < \tau_0\}, \qquad G_\tau := [\rho_*, \rho_\delta] \setminus B_\tau,$$

one has

$$|B_\tau| \leq C_\tau\, \delta_T(\Omega). \tag{60}$$

Proof. Recall $m(t) = |\{u > t\}|$ and $\alpha(t) = \int_{\partial^*\{u>t\}} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ from §4.1.1. Theorem 4.9 gives the exact deficit identity

$$2\,\delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2\omega_n^{2/n} m(t)^{1-2/n}}\, dt, \tag{61}$$

all integrals taken on the reduced boundaries $\partial^*\{u > t\}$ and holding for a.e. level.

Changing variables $t = t(\rho)$, $m(t(\rho)) = \omega_n\rho^n$, and $dt = d\mu(\rho)$ in decreasing-radius orientation gives

$$2\delta_T(\Omega) = \int_0^1 \frac{D_H(t(\rho)) + D_I(t(\rho))}{n^2\omega_n\, \rho^{n-2}}\, d\mu(\rho).$$

Since $n \geq 2$ and $0 < \rho \leq 1$,

$$\int_{\rho_*}^{\rho_\delta} \big(D_H(t(\rho)) + D_I(t(\rho))\big)\, d\mu(\rho) \leq 2n^2\omega_n\, \delta_T(\Omega),$$

which proves (59), for instance with $C_L = 2n^2\omega_n$.

It remains to prove the density bounds. Write

$$a(\rho) := w(\rho) = -t'(\rho), \qquad a_B(\rho) := \frac{\rho}{n}.$$

We next derive from (61) the profile derivative gap identity

$$0 \leq a(\rho) \leq a_B(\rho) \quad \text{for a.e. } \rho \in (0,1), \qquad \int_0^1 \rho^n\big(a_B(\rho) - a(\rho)\big)\, d\rho = \frac{2}{\omega_n}\delta_T(\Omega). \tag{62}$$

For completeness, the algebra is as follows. Coarea gives

$$a(\rho) = \frac{n\omega_n\rho^{n-1}}{\alpha(t(\rho))}.$$

For the ball profile the corresponding coarea coefficient is

$$\alpha_B(\rho) = n^2\omega_n\rho^{n-2}, \qquad \frac{n\omega_n\rho^{n-1}}{\alpha_B(\rho)} = \frac{\rho}{n} = a_B(\rho).$$

Since

$$D_H(t(\rho)) + D_I(t(\rho)) = \omega_n\rho^n\big(\alpha(t(\rho)) - \alpha_B(\rho)\big),$$

the nonnegativity of $D_H + D_I$ gives $0 \leq a \leq a_B$, and inserting the same formula in (61) gives the integral identity in (62).

Thus $0 \le w(\rho) \le \rho/n \le 1/n$. On

$$B_\tau = \Big\{\rho \in [\rho_*, \rho_\delta] : a(\rho) < \frac{\rho_*}{2n}\Big\}$$

one has

$$a_B(\rho) - a(\rho) \ge \frac{\rho}{n} - \frac{\rho_*}{2n} \ge \frac{\rho_*}{2n}.$$

Using $\rho^n \ge \rho_*^n$ on $[\rho_*, \rho_\delta]$ and (62),

$$\frac{\rho_*^{n+1}}{2n}|B_\tau| \le \int_{B_\tau} \rho^n (a_B - a)\, d\rho \le \frac{2}{\omega_n}\delta_T(\Omega).$$

Therefore (60) holds with

$$C_\tau = \frac{4n}{\omega_n \rho_*^{n+1}}.$$

Finally, choose for example $\kappa = (1-\rho_*)/4$. Then $\rho_\delta \ge (1+\rho_*)/2$ whenever $\delta_T(\Omega) \le 1$, and in the large-deficit part of the final theorem this smallness is absorbed into the constant. $\square$

4.2. **Removing the boundedness hypothesis.** The purpose of this section is to remove the boundedness hypothesis from the sharp Saint–Venant stability inequality. The idea is simple to state. We already possess a sharp quadratic stability estimate for sets *confined to a ball of a fixed radius*; we want the same estimate for arbitrary open sets of finite measure, with a constant depending only on the dimension. The obstruction is that a general competitor may have very long, thin tentacles reaching out to infinity. The remedy, following Brasco–De Philippis–Velichkov, is to amputate those tails: we will show that a set with small deficit carries almost all of its torsional energy and almost all of its mass inside a ball of dimensionally bounded radius, so that chopping it off there changes the deficit and the asymmetry by only a controlled amount. The bounded estimate, applied to the amputated set, then transfers back.

The input is the following *bounded* estimate, proved in the earlier part of the manuscript (Theorem 4.29), restated here in energy form.

Theorem 4.11 (Bounded sharp Saint–Venant)**.** *For every $n \ge 2$ and every $R \ge 2$ there is a constant $c_{\mathrm{SV}}(n,R) > 0$ such that*

$$E(\Omega) - E(B_1) \;\ge\; c_{\mathrm{SV}}(n,R)\,\mathcal{A}(\Omega)^2 \tag{63}$$

for every open set $\Omega \subset B_R$ with $|\Omega| = \omega_n$.

This is Theorem 4.29, restated in energy form: with the constant $C(n,R,\rho_*)$ there one may take $c_{\mathrm{SV}}(n,R) = 1/C(n,R,\rho_*)$ and $\delta_T(\Omega) = E(\Omega) - E(B_1)$, and the parameter ρ_* is fixed at, say, $\rho_* = \frac{3}{4}$.

Our goal is the global statement.

Theorem 4.12 (Global sharp Saint–Venant stability)**.** *There is a dimensional constant $c_{\mathrm{SV}}^{\mathrm{glob}}(n) > 0$ such that*

$$E(\Omega) - E(B_1) \;\ge\; c_{\mathrm{SV}}^{\mathrm{glob}}(n)\,\mathcal{A}(\Omega)^2 \tag{64}$$

for every open set $\Omega \subset \mathbb{R}^n$ with $|\Omega| = \omega_n$. Equivalently, after rescaling, for every open set Ω of finite measure, with B^* the ball of the same volume,

$$T(B^*) - T(\Omega) \;=\; 2\big(E(\Omega) - E(B^*)\big) \;\ge\; 2\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)\left(\tfrac{|\Omega|}{\omega_n}\right)^{(n+2)/n} \mathcal{A}(\Omega)^2. \tag{65}$$

The strategy is the constructive reduction of Brasco, De Philippis and Velichkov [3, §5]. From a set Ω with small Saint–Venant deficit we manufacture a *bounded* surrogate $\widetilde{\Omega}$, of the same volume, of dimensionally bounded diameter, with comparable deficit and with controlled loss of asymmetry; applying Theorem 4.11 to $\widetilde{\Omega}$ at a fixed dimensional radius and transferring the estimate back to Ω closes the small-deficit regime. The complementary large-deficit regime is trivial because $\mathcal{A} < 2$. The whole construction is explicit: no compactness, no minimising sequence and no contradiction argument enter.

The two technical ingredients are an L^∞ tail estimate for the torsion function (Lemma 4.15, proved by a De Giorgi iteration) and the truncation construction itself (Lemma 4.18). Both are proved here in full rather than cited. The reader who wishes to keep the logical skeleton in mind may note the order of dependence: a purely numerical convergence lemma (§4.2.2) feeds the De Giorgi L^∞ bound (§4.2.3), which feeds the energy-comparison inequality that drives the truncation (§4.2.5); the truncation, the trivial far branch (§4.2.6) and the transfer (§4.2.7) are then assembled in §4.2.8.

4.2.1. *Local notation and the tools used in this section.* Throughout, $n \geq 2$ and all sets are open subsets of $\mathbb{R}^n$ of finite Lebesgue measure. We write $u_\Omega \in H_0^1(\Omega)$ for the torsion function, i.e. the unique weak solution of $-\Delta u_\Omega = 1$ in Ω, $u_\Omega = 0$ on $\partial\Omega$, equivalently the minimiser of

$$J_\Omega(v) \;=\; \tfrac{1}{2}\int_\Omega |\nabla v|^2 \, dx - \int_\Omega v \, dx, \qquad v \in H_0^1(\Omega), \tag{66}$$

and $E(\Omega) = J_\Omega(u_\Omega) = \min_{H_0^1(\Omega)} J_\Omega$. Testing the equation with u_Ω gives the standard identities

$$\int_\Omega |\nabla u_\Omega|^2 \, dx = \int_\Omega u_\Omega \, dx = T(\Omega), \qquad E(\Omega) = -\tfrac{1}{2}\int_\Omega u_\Omega \, dx = -\tfrac{1}{2}\, T(\Omega). \tag{67}$$

We use $\omega_n = |B_1|$, $T_n = T(B_1) = \omega_n/(n(n+2))$, the Fraenkel asymmetry $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega \triangle (B_1 + x_0)| / \omega_n$ (for $|\Omega| = \omega_n$ this lies in $[0, 2)$), and the scale-invariant deficit

$$D(\Omega) \;=\; E(\Omega)\, |\Omega|^{-(n+2)/n} - E(B_1)\, |B_1|^{-(n+2)/n}\,, \tag{68}$$

so that $D(\Omega) = \omega_n^{-(n+2)/n}\big(E(\Omega) - E(B_1)\big) \geq 0$ when $|\Omega| = \omega_n$. The non-negativity of D is the *Saint–Venant inequality* (equivalently Talenti's comparison theorem): among sets of given volume the ball minimises E, i.e.

$$E(\Omega)\; |\Omega|^{-(n+2)/n} \;\geq\; E(B)\; |B|^{-(n+2)/n} \qquad \text{for every ball } B. \tag{69}$$

We treat (69) as a known classical result.

We will also need the elementary scaling law for E. If Ω' is open and $t > 0$ then $E(t\Omega') = t^{n+2}E(\Omega')$, with torsion function $u_{t\Omega'}(x) = t^2 u_{\Omega'}(x/t)$.

The external analytic facts we invoke without proof are:

- the *Sobolev inequality* $\|w\|_{L^{2^*}(\mathbb{R}^n)} \leq S_n \|\nabla w\|_{L^2(\mathbb{R}^n)}$ for $w \in H^1(\mathbb{R}^n)$ when $n \geq 3$ (with $2^* = 2n/(n-2)$), and its $n = 2$ surrogate explained in Remark 4.16;
- the *Saint–Venant/Talenti inequality* (69), used in the truncation argument and in the global L^∞ bound (70) below;
- Theorem 4.11, the bounded stability input;
- the *suboptimal* torsional stability bound $\mathcal{A}(\Omega)^4 \leq C(n)D(\Omega)$ (Lemma 4.17), used only to seed the truncation iteration; this seed is proved here from the level-set identity, the profile-gap identity, and the constructive Fraenkel inequality (Theorem 2.1).

Everything else — in particular the fast-convergence iteration lemma (Lemma 4.13), the De Giorgi L^∞ tail estimate (Lemma 4.15), the energy-comparison inequality, the truncation iteration (Lemma 4.18) and the gluing — is proved from scratch.

We will also need the elementary pointwise sup-bound on the torsion function. By Talenti's symmetrisation comparison the torsion function of Ω is dominated, after Schwarz symmetrisation, by that of the ball of the same volume; in particular

$$\|u_\Omega\|_{L^\infty(\Omega)} \;\le\; \|u_{B^*}\|_{L^\infty(B^*)} \;=\; \frac{(|\Omega|\,/\omega_n)^{2/n}}{2n} \;=:\; \Lambda_n\,|\Omega|^{2/n}, \qquad \Lambda_n := \frac{\omega_n^{-2/n}}{2n}, \tag{70}$$

where B^* is the ball with $|B^*| = |\Omega|$ and we used $u_{B_r}(x) = (r^2 - |x|^2)/(2n)$. For $|\Omega| = \omega_n$ this reads $\|u_\Omega\|_{L^\infty(\Omega)} \le 1/(2n)$. We record (70) as a consequence of the cited Talenti comparison.

4.2.2. *The fast-convergence iteration lemma.* The De Giorgi scheme reduces an L^∞ bound to the vanishing of a nonlinear recursion. We isolate the underlying numerical lemma and prove it, so that nothing is cited at this step. The mechanism is purely algebraic: a super-quadratic recursion (the increment a_{k+1} depends on a power $a_k^{1+\beta}$ strictly larger than the first power) ultimately beats any *fixed* geometric growth factor b^k, provided the seed a_0 is below an explicit threshold. The threshold is exactly the break-even point of the two competing effects.

Lemma 4.13 (Fast geometric convergence)**.** *Let $(a_k)_{k\ge 0}$ be non-negative reals with*

$$a_{k+1} \;\le\; C\,b^k\,a_k^{1+\beta}, \qquad k \ge 0, \tag{71}$$

where $C > 0$, $b > 1$, $\beta > 0$. If

$$a_0 \;\le\; C^{-1/\beta}\,b^{-1/\beta^2}, \tag{72}$$

then

$$a_k \;\le\; b^{-k/\beta}\,a_0 \qquad \textit{for all } k \ge 0, \tag{73}$$

so $a_k \to 0$.

Proof. Put $r := b^{-1/\beta} \in (0,1)$ and $\theta := C\,a_0^\beta$. The smallness hypothesis (72) is precisely $\theta \le r$. We prove the target $a_k \le r^k a_0$ by induction; this is (73) because $r^k = b^{-k/\beta}$, and $r < 1$ then gives $a_k \to 0$.

Base case $k = 0$: the claim reads $a_0 \le r^0 a_0 = a_0$, an equality.

Inductive step. Assume $a_k \le r^k a_0$. Using (71) and splitting off one factor a_k,

$$a_{k+1} \le C\,b^k\,a_k^{1+\beta} = C\,b^k\,a_k\,a_k^\beta \le C\,b^k\,a_k\,\big(r^k a_0\big)^\beta = \big(C\,a_0^\beta\big)\,(b\,r^\beta)^k\,a_k = \theta\,(b\,r^\beta)^k\,a_k.$$

By the definition of r we have $r^\beta = b^{-1}$, hence $b\,r^\beta = 1$ and the geometric factor collapses:

$$a_{k+1} \le \theta\,a_k \le r\,a_k \le r\cdot r^k a_0 = r^{k+1} a_0,$$

using $\theta \le r$ and the inductive hypothesis $a_k \le r^k a_0$. This closes the induction. □

Remark 4.14. The identity $b\,r^\beta = 1$ with $r = b^{-1/\beta}$ is the algebraic mechanism behind the convergence: the super-quadratic gain $a_k^{1+\beta}$ converts, once a_0 is small, the growing factor b^k into a contraction. The threshold (72) is sharp for this clean conclusion: it is exactly the value of a_0 at which $\theta = r$, i.e. at which the contraction factor in the displayed chain first dips to $r < 1$.

4.2.3. *The De Giorgi L^∞ tail estimate.* We now prove that, for a unit-volume set, the torsion function is quantitatively small far out, with the smallness measured by a *super-linear* power of the tail mass $|\Omega \setminus B_R|$. The super-linearity (exponent $1/n > 0$, or $\frac{1}{2} - \frac{1}{p} > 0$ when $n = 2$) is the whole point: it will let the truncation iteration converge.

Lemma 4.15 (L^∞ tail estimate)**.** *Let $n \geq 3$. There is a dimensional constant $C_\flat = C_\flat(n) > 0$ such that for every open Ω with $|\Omega| = \omega_n$ and every $R \geq 1$,*

$$\|u_\Omega\|_{L^\infty(\Omega\setminus B_{R+1})} \ \leq\ C_\flat\ |\Omega \setminus B_R|^{1/n}\,. \tag{74}$$

For $n = 2$ the same holds with the exponent $1/n$ replaced by $\frac{1}{2} - \frac{1}{p}$ for any fixed $p \in (2, \infty)$ (Remark 4.16); this weaker but still super-linear power is all that the truncation uses.

Proof. If $|\Omega \setminus B_R| = 0$ then $u_\Omega = 0$ a.e. on $\Omega \setminus B_R \supset \Omega \setminus B_{R+1}$ and (74) is trivial, so assume $|\Omega \setminus B_R| > 0$. We carry out a De Giorgi iteration on the annular region $\Omega \setminus B_R$. The plan: build cut-offs that switch on across a shrinking nest of annuli, and levels rising to a target height; show that the volume of the super-level set above the rising level, inside the shrinking annulus, obeys a recursion of the form (71); then choose the target height M so large that the seed is below the threshold (72), forcing the volume to 0 and pinning u_Ω below the height.

Set-up: shrinking radii, cut-offs, and rising levels. For $k \in \mathbb{N}_0$ put

$$R_k := R + 1 - 2^{-k},$$

so that $R_0 = R$, the radii increase to $R_\infty = R + 1$, and $R_k \geq R \geq 1$. Let $\varphi_k \in C^{0,1}(\mathbb{R}^n)$ be the radial cut-off

$$\varphi_k(x) := \phi_k(|x|), \qquad \phi_k = 0 \text{ on } [0, R_{k-1}], \quad \phi_k = 1 \text{ on } [R_k, \infty), \quad \phi_k \text{ affine on } [R_{k-1}, R_k],$$

for $k \geq 1$, and $\varphi_0 \equiv 1$ on $\{|x| \geq R_0\}$, extended by 0 inside (we only use φ_k for $k \geq 1$ in the recursion). Then $0 \leq \varphi_k \leq 1$, $\varphi_k \equiv 0$ on $B_{R_{k-1}}$, $\varphi_k \equiv 1$ on $\mathbb{R}^n \setminus B_{R_k}$, and

$$|\nabla\varphi_k| \leq \frac{1}{R_k - R_{k-1}} = \frac{1}{2^{-(k-1)} - 2^{-k}} = 2^k \leq 2^{k+1}.$$

Choose rising levels

$$s_k := M\ |\Omega \setminus B_R|^{1/n}\ (1 - 2^{-k}), \qquad k \geq 0,$$

where $M = M(n) \geq 1$ is a free constant fixed at the end. Thus $s_0 = 0$, $s_k \uparrow s_\infty := M\,|\Omega \setminus B_R|^{1/n}$, and

$$s_{k+1} - s_k = M\,|\Omega \setminus B_R|^{1/n}\ 2^{-(k+1)}. \tag{75}$$

Caccioppoli inequality from the truncated test function. Fix $k \geq 1$. The function $(u_\Omega - s_k)_+ \in H^1_0(\Omega)$ (it vanishes where $u_\Omega \leq s_k$, in particular near $\partial\Omega$ since $u_\Omega \geq 0$ and $s_k \geq 0$), and φ_k is bounded Lipschitz, so

$$v_k := \varphi_k^2\,(u_\Omega - s_k)_+\ \in\ H^1_0(\Omega)$$

is an admissible test function. Insert it into the weak formulation $\int_\Omega \nabla u_\Omega \cdot \nabla v\,dx = \int_\Omega v\,dx$. Using $\nabla v_k = \varphi_k^2\nabla(u_\Omega - s_k)_+ + 2\varphi_k(u_\Omega - s_k)_+\nabla\varphi_k$ and $\nabla u_\Omega \cdot \nabla(u_\Omega - s_k)_+ = |\nabla(u_\Omega - s_k)_+|^2$ (the gradient of u_Ω equals that of $(u_\Omega - s_k)_+$ on $\{u_\Omega > s_k\}$ and both factors vanish elsewhere), we get

$$\int \varphi_k^2\,|\nabla(u_\Omega - s_k)_+|^2\ dx + 2\int \varphi_k(u_\Omega - s_k)_+\,\nabla\varphi_k \cdot \nabla(u_\Omega - s_k)_+\,dx = \int \varphi_k^2(u_\Omega - s_k)_+\,dx.$$

The middle term is bounded by Young's inequality, $2\,|\varphi_k(u_\Omega - s_k)_+\nabla\varphi_k \cdot \nabla(u_\Omega - s_k)_+| \leq \frac{1}{2}\varphi_k^2\,|\nabla(u_\Omega - s_k)_+|^2 + 2\,|\nabla\varphi_k|^2\,(u_\Omega - s_k)_+^2$, and absorbing the first piece on the left yields the Caccioppoli estimate

$$\int \left|\nabla\big(\varphi_k(u_\Omega - s_k)_+\big)\right|^2\,dx\ \leq\ C_1\!\int |\nabla\varphi_k|^2\,(u_\Omega - s_k)_+^2\,dx\ +\ C_1\!\int \varphi_k^2(u_\Omega - s_k)_+\,dx, \tag{76}$$

with an absolute constant C_1 (one checks $C_1 = 10$ works after expanding $|\nabla(\varphi_k w)|^2 \le 2\varphi_k^2 |\nabla w|^2 + 2|\nabla\varphi_k|^2 w^2$ and combining with the absorbed bound; the precise value is immaterial). Here we wrote $w = (u_\Omega - s_k)_+$.

Sobolev inequality. Set $w_k := \varphi_k(u_\Omega - s_k)_+ \in H_0^1(\mathbb{R}^n)$ and

$$A_k := \{\varphi_k(u_\Omega - s_k)_+ > 0\} = \{u_\Omega > s_k\} \cap \{\varphi_k > 0\} \subset \{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}}).$$

For $n \ge 3$ the Sobolev inequality and Hölder's inequality on the support A_k give

$$\int w_k^2\,dx \le \Big(\int w_k^{2^*}\,dx\Big)^{2/2^*} |A_k|^{1-2/2^*} \le S_n^2\,|A_k|^{2/n}\int |\nabla w_k|^2\,dx, \tag{77}$$

using $1 - 2/2^* = 2/n$. (The $n = 2$ case is handled in Remark 4.16; the resulting estimate has the same shape with $2/n$ replaced by any fixed exponent in $(0,1)$, which only changes the dimensional constants.)

Combining (77) with (76), $|\nabla\varphi_k| \le 2^{k+1}$ (so $|\nabla\varphi_k|^2 \le 4^{k+1}$) and $0 \le \varphi_k \le 1$, we obtain

$$\int w_k^2\,dx \le S_n^2\,C_1\,|A_k|^{2/n}\Big(4^{k+1}\int_{A_k}(u_\Omega - s_k)_+^2\,dx + \int_{A_k}(u_\Omega - s_k)_+\,dx\Big), \tag{78}$$

where for the second term we used $\int \varphi_k^2(u_\Omega - s_k)_+ \le \int_{A_k}(u_\Omega - s_k)_+$. By the global sup-bound (70) at $|\Omega| = \omega_n$ we have $0 \le (u_\Omega - s_k)_+ \le u_\Omega \le 1/(2n) \le 1$ on Ω, hence both $(u_\Omega - s_k)_+^2 \le 1$ and $(u_\Omega - s_k)_+ \le 1$ there, so each integral over A_k is at most $|A_k|$. Therefore

$$\int w_k^2\,dx \le S_n^2\,C_1\,(4^{k+1} + 1)\,|A_k|^{2/n}\,|A_k| \le C_2\,4^k\,|A_k|^{1+2/n}, \tag{79}$$

with $C_2 = C_2(n) := 8\,S_n^2 C_1$ (using $4^{k+1} + 1 \le 8 \cdot 4^k$).

From energy to a measure recursion. Restrict the left-hand side of (79) to the smaller set where the *next* level is exceeded. On $\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_k})$ we have $\varphi_k = 1$ (since this set lies outside B_{R_k}) and $u_\Omega - s_k > s_{k+1} - s_k$, hence $w_k^2 = (u_\Omega - s_k)_+^2 \ge (s_{k+1} - s_k)^2$ there. Therefore

$$(s_{k+1} - s_k)^2\,\big|\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_k})\big| \;\le\; \int_{\mathbb{R}^n} w_k^2\,dx. \tag{80}$$

Introduce the (relative) super-level measures

$$\mu_k := \big|\{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}})\big|, \qquad k \ge 1,$$

which decrease in k (both s_k and R_{k-1} increase). Since $A_k \subset \{u_\Omega > s_k\} \cap (\mathbb{R}^n \setminus B_{R_{k-1}})$ we have $|A_k| \le \mu_k$, and the left-hand set in (80) is exactly $\{u_\Omega > s_{k+1}\} \cap (\mathbb{R}^n \setminus B_{R_{(k+1)-1}})$, of measure μ_{k+1}. Combining (80), (79) and $|A_k| \le \mu_k$,

$$(s_{k+1} - s_k)^2\,\mu_{k+1} \;\le\; C_2\,4^k\,\mu_k^{1+2/n}.$$

Insert the gap (75), $(s_{k+1} - s_k)^2 = M^2\,|\Omega \setminus B_R|^{2/n}\,4^{-(k+1)}$:

$$\mu_{k+1} \;\le\; \frac{C_2\,4^k}{M^2\,|\Omega \setminus B_R|^{2/n}\,4^{-(k+1)}}\,\mu_k^{1+2/n} \;=\; \frac{4\,C_2}{M^2\,|\Omega \setminus B_R|^{2/n}}\,16^k\,\mu_k^{1+2/n}. \tag{81}$$

Normalisation and the iteration lemma. Set $a_k := \mu_k/|\Omega \setminus B_R|$. Since $\mu_k \le \big|\mathbb{R}^n \setminus B_{R_{k-1}} \cap \{u_\Omega > s_k\}\big| \le |\Omega \setminus B_R|$ (because $R_{k-1} \ge R_0 = R$ and $\{u_\Omega > s_k\} \subset \Omega$), we have $0 \le a_k \le 1$; in particular $a_1 \le 1$. Dividing (81) by $|\Omega \setminus B_R|$ and using $\mu_k^{1+2/n} = |\Omega \setminus B_R|^{1+2/n}\,a_k^{1+2/n}$,

$$a_{k+1} \le \frac{4C_2}{M^2\,|\Omega \setminus B_R|^{2/n}}\,16^k\,\frac{|\Omega \setminus B_R|^{1+2/n}}{|\Omega \setminus B_R|}\,a_k^{1+2/n} = \frac{4C_2}{M^2}\,16^k\,a_k^{1+2/n},$$

valid for all $k \geq 1$. To put this in the exact form (71) (indexed from 0) we shift: set $\widetilde{a}_j := a_{j+1}$ for $j \geq 0$. Then for every $j \geq 0$,

$$\widetilde{a}_{j+1} = a_{j+2} \leq \frac{4C_2}{M^2}\, 16^{\,j+1}\, a_{j+1}^{1+2/n} = \underbrace{\frac{64\, C_2}{M^2}}_{=:C}\, 16^{\,j}\, \widetilde{a}_j^{\,1+2/n},$$

which is (71) with

$$C := \frac{64\, C_2}{M^2}, \qquad b := 16, \qquad \beta := \frac{2}{n}.$$

Choose $M = M(n)$ so large that the smallness condition (72) holds at the starting value $\widetilde{a}_0 = a_1 \leq 1$. Since $a_1 \leq 1$ it suffices to require

$$1 \;\leq\; C^{-1/\beta}\, b^{-1/\beta^2} = \left(\tfrac{64 C_2}{M^2}\right)^{-n/2} 16^{-n^2/4}, \quad \text{i.e.} \quad M^2 \;\geq\; 64\, C_2\, 16^{\,n^2/4 \cdot (2/n)} = 64\, C_2\, 16^{\,n/2},$$

which holds for $M := 8\,(C_2)^{1/2} 4^{\,n/2}$ (a dimensional constant, since $C_2 = C_2(n)$). With this M, Lemma 4.13 gives $\widetilde{a}_j \to 0$, i.e. $a_k \to 0$ and hence $\mu_k \to 0$.

Conclusion. Letting $k \to \infty$ in $\mu_k \to 0$ and using $s_k \uparrow s_\infty = M\, |\Omega \setminus B_R|^{1/n}$ and $R_{k-1} \uparrow R+1$, monotone convergence yields

$$\big|\{u_\Omega > s_\infty\} \cap (\mathbb{R}^n \setminus B_{R+1})\big| = \lim_{k\to\infty} \mu_k = 0.$$

Hence $u_\Omega \leq s_\infty = M\,|\Omega \setminus B_R|^{1/n}$ a.e. on $\Omega \setminus B_{R+1}$. Since $u_\Omega \geq 0$, this is (74) with $C_\flat := M = 8\,(C_2)^{1/2} 4^{n/2}$. □

Remark 4.16 (The case $n = 2$)**.** For $n = 2$ the embedding $H_0^1 \hookrightarrow L^{2^*}$ degenerates (formally $2^* = 2n/(n-2) = \infty$), but the Gagliardo–Nirenberg–Sobolev inequality gives, for any fixed $p \in (2, \infty)$, $\|w\|_{L^p(\mathbb{R}^2)} \leq S_{2,p}\, \|\nabla w\|_{L^2(\mathbb{R}^2)}$ for $w \in H_0^1(\mathbb{R}^2)$ with finite-measure support. Repeating the Hölder step with exponent p replaces $2/n = 1$ by $1 - 2/p \in (0, 1)$, so (81) becomes $\mu_{k+1} \leq C\, 16^k\, \mu_k^{1+\beta}$ with $\beta := 1 - 2/p > 0$, and Lemma 4.13 applies. Tracking the level scale: with $s_\infty = M\,|\Omega \setminus B_R|^\alpha$, self-consistency of the recursion forces $\alpha = \frac{p-2}{2p} = \frac{1}{2} - \frac{1}{p}$ (for $n \geq 3$ the analogous bookkeeping gives $\alpha = 1/n$, and the two agree only there). Hence, for $n = 2$, the L^∞ tail estimate holds in the slightly weaker form

$$\|u_\Omega\|_{L^\infty(\Omega \setminus B_{R+1})} \;\leq\; C(p)\; |\Omega \setminus B_R|^{\,\frac{1}{2} - \frac{1}{p}}\,, \qquad \text{any fixed } p \in (2, \infty),$$

in place of the exponent $1/n$ in (74). Fixing once and for all $p = 4$, so that $\frac{1}{2} - \frac{1}{p} = \frac{1}{4} > 0$, this is a genuine super-linear power of the tail mass, which is all the truncation of §4.2.5 uses: the energy-comparison (94) and the tail recursion (96) hold with the exponent $1 + \frac{1}{n}$ replaced by $1 + \kappa$, $\kappa = \frac{1}{4}$, and the geometric iteration (99)–(102) goes through for any $\kappa > 0$ with n-dependent constants. Consequently the global Theorem 4.12 holds for all $n \geq 2$; only the auxiliary exponent in (74) (and the resulting dimensional constants) changes at $n = 2$.

4.2.4. *A suboptimal stability seed.* The truncation iteration needs one quantitative input that the De Giorgi machinery does not by itself supply: a crude (far from sharp) control of the asymmetry by the deficit, used only to make the tail mass small at the first radius and to bound the number of truncation steps. We prove it here from the level-set identity already established, so that the present section introduces no external stability input. The mechanism is: the deficit identity plus the quantitative isoperimetric inequality control a weighted integral of the asymmetries of the superlevel sets; the profile-gap identity shows that this weight has positive mass on the outer collar; and the superlevel transfer shows that on that collar the superlevel sets inherit the asymmetry of Ω itself. The fourth power (in fact a third power) is the price of the crude collar transfer; the sharp second power is the main theorem.

Lemma 4.17 (Suboptimal stability seed)**.** *There is a dimensional constant $C_\dagger = C_\dagger(n) > 0$ such that every open $\Omega \subset \mathbb{R}^n$ of finite measure satisfies*

$$D(\Omega) \ \geq \ \frac{1}{C_\dagger}\,\mathcal{A}(\Omega)^4. \tag{82}$$

Equivalently, $\mathcal{A}(\Omega) \leq \big(C_\dagger D(\Omega)\big)^{1/4}$.

Proof. D and $\mathcal{A}$ are scale invariant, so we may assume $|\Omega| = \omega_n$; then $D(\Omega) = \omega_n^{-(n+2)/n}\delta_T(\Omega)$, $B^* = B_1$. Write $A := \mathcal{A}(\Omega) \in [0,2)$, and recall $E_\rho = \{u > t(\rho)\}$, $|E_\rho| = \omega_n\rho^n$, $a(\rho) = -t'(\rho)$. Three facts proved earlier are used; each holds for any open set of finite measure, since its proof does not use $\Omega \subset B_R$:

(a) the exact deficit identity (Theorem 4.9), $2\delta_T(\Omega) = \int_0^{\|u\|_\infty}(D_H + D_I)/(n^2\omega_n^{2/n}m^{1-2/n})\,dt$, with $D_H, D_I \geq 0$;
(b) the profile-gap identity (62): $0 \leq a(\rho) \leq \rho/n$ and $\int_0^1 \rho^n\big(\frac{\rho}{n} - a(\rho)\big)\,d\rho = \frac{2}{\omega_n}\delta_T(\Omega)$;
(c) the superlevel transfer (Lemma 2.7): $A \leq \mathcal{A}(E_\rho) + 2(1-\rho^n)$ for every $\rho \in (0,1]$.

Step 1 (a level-set lower bound for D_I). Fix a finite-perimeter level ρ and apply the constructive Fraenkel quantitative isoperimetric inequality (Theorem 2.1) to E_ρ; it gives a centre z with $|E_\rho \Delta B_\rho(z)|^2 \leq C_{\mathrm{iso}}(n)\rho^{n+1}(P(E_\rho) - P(B_\rho))$ (this is (8)). Now $\mathcal{A}(E_\rho) \leq |E_\rho\Delta B_\rho(z)|\,/\,|E_\rho|$ (the Fraenkel asymmetry is an infimum over balls of volume $|E_\rho|$, and $B_\rho(z)$ is one such), and $P(E_\rho) - P(B_\rho) = D_I(t(\rho))/(P(E_\rho) + P(B_\rho)) \leq D_I(t(\rho))/P(B_\rho)$. With $|E_\rho| = \omega_n\rho^n$ and $P(B_\rho) = n\omega_n\rho^{n-1}$,

$$\mathcal{A}(E_\rho)^2\,\omega_n^2\rho^{2n} \leq |E_\rho\Delta B_\rho(z)|^2 \leq C_{\mathrm{iso}}(n)\rho^{n+1}\frac{D_I(t(\rho))}{n\omega_n\rho^{n-1}} = \frac{C_{\mathrm{iso}}(n)}{n\omega_n}\rho^2 D_I(t(\rho)),$$

which rearranges to

$$D_I(t(\rho)) \ \geq \ c_1(n)\,\rho^{2n-2}\,\mathcal{A}(E_\rho)^2, \qquad c_1(n) := \frac{n\,\omega_n^3}{C_{\mathrm{iso}}(n)}. \tag{83}$$

Step 2 (restrict to the outer collar $[\rho_A, 1]$). Put $\rho_A := (1 - A/4)^{1/n}$. For $\rho \in [\rho_A, 1]$, $1 - \rho^n \leq 1 - \rho_A^n = A/4$, so (c) gives $\mathcal{A}(E_\rho) \geq A - 2(1-\rho^n) \geq A/2$, and $\rho^n \geq \rho_A^n = 1 - A/4 > 1/2$. In (a) drop $D_H \geq 0$ and change variables $t = t(\rho)$ ($dt = a(\rho)\,d\rho$, $m = \omega_n\rho^n$); keeping only the nonnegative integrand over $[\rho_A, 1]$ and inserting (83),

$$2\delta_T(\Omega) \ \geq \ \int_{\rho_A}^1 \frac{D_I(t(\rho))}{n^2\omega_n^{2/n}(\omega_n\rho^n)^{1-2/n}}\,a(\rho)\,d\rho \ \geq \ c_2(n)\int_{\rho_A}^1 \rho^n\,\mathcal{A}(E_\rho)^2\,a(\rho)\,d\rho,$$

where $c_2(n) = c_1(n)/(n^2\omega_n)$. With $\mathcal{A}(E_\rho) \geq A/2$ on $[\rho_A, 1]$,

$$2\delta_T(\Omega) \ \geq \ \frac{c_2(n)\,A^2}{4}\int_{\rho_A}^1 \rho^n a(\rho)\,d\rho. \tag{84}$$

Step 3 (the profile gap bounds the collar mass below). By (b) and $\frac{\rho}{n} - a \geq 0$,

$$\int_{\rho_A}^1 \rho^n a\,d\rho = \int_{\rho_A}^1 \frac{\rho^{n+1}}{n}\,d\rho - \int_{\rho_A}^1 \rho^n\Big(\frac{\rho}{n} - a\Big)d\rho \ \geq \ \frac{1-\rho_A^{n+2}}{n(n+2)} - \frac{2}{\omega_n}\delta_T(\Omega),$$

the inequality because the subtracted integral is at most the full integral $\int_0^1 \rho^n(\frac{\rho}{n}-a) = \frac{2}{\omega_n}\delta_T(\Omega)$. Since $\rho_A \le 1$, $\rho_A^{n+2} \le \rho_A^n = 1 - A/4$, so $1-\rho_A^{n+2} \ge A/4$ and

$$\int_{\rho_A}^1 \rho^n a\,d\rho \;\ge\; \frac{A}{4n(n+2)} - \frac{2}{\omega_n}\delta_T(\Omega). \tag{85}$$

Step 4 (conclusion). Insert (85) into (84):

$$2\delta_T(\Omega) \;\ge\; \frac{c_2(n)A^2}{4}\Big(\frac{A}{4n(n+2)} - \frac{2}{\omega_n}\delta_T(\Omega)\Big) = c_3(n)A^3 - c_4(n)A^2\delta_T(\Omega),$$

$c_3(n) = c_2(n)/(16n(n+2))$, $c_4(n) = c_2(n)/(2\omega_n)$. Therefore $\delta_T(\Omega)(2+c_4(n)A^2) \ge c_3(n)A^3$, and since $A<2$, $\delta_T(\Omega) \ge c_3(n)A^3/(2+4c_4(n)) =: c_5(n)A^3$. Finally $A<2$ gives $A^4 \le 2A^3$, hence $\delta_T(\Omega) \ge \frac{1}{2}c_5(n)A^4$; multiplying by $\omega_n^{-(n+2)/n}$ yields (82) with $C_\dagger(n) = 2\omega_n^{(n+2)/n}/c_5(n)$. □

4.2.5. *The truncation construction.* We now build the bounded surrogate. The estimate driving the construction is an *energy-comparison inequality*: cutting Ω off at radius $\sim k$ costs only a higher-order amount of energy, controlled by the tail mass through Lemma 4.15. The logic of the proof is a feedback loop: the energy comparison (Step 1) becomes, via Saint–Venant, a recursion bounding the tail mass two radii out by the tail mass here plus the deficit (Step 2); a stopping index K marks where the tail first drops below the deficit scale (Step 3); the recursion is shown to force K to be dimensionally bounded (the heart of the matter); finally we cut at $K+3$, rescale to unit volume, and verify the four advertised properties (Steps 5–7).

Lemma 4.18 (Bounded truncation)**.** *There exist dimensional constants $C_\sharp = C_\sharp(n) > 0$, $\delta = \delta(n) \in (0,1]$ and $d = d(n) \ge 1$ such that for every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \le \delta$ there is an open set $\widetilde{\Omega}$ with*

$$\big|\widetilde{\Omega}\big| \;=\; \omega_n, \tag{86}$$

$$\operatorname{diam}(\widetilde{\Omega}) \;\le\; d, \tag{87}$$

$$D(\widetilde{\Omega}) \;\le\; C_\sharp\, D(\Omega), \tag{88}$$

$$\mathcal{A}(\Omega) \;\le\; \mathcal{A}(\widetilde{\Omega}) + C_\sharp\, D(\Omega). \tag{89}$$

Proof. Normalise so that the ball realising the asymmetry of Ω is B_1 (translate Ω; both $\mathcal{A}$ and D are translation invariant), i.e. $\mathcal{A}(\Omega) = |\Omega \,\triangle\, B_1|/\omega_n$. We will repeatedly use the tail-mass notation

$$b_k := \frac{|\Omega\setminus B_k|}{\omega_n}, \qquad k \ge 1, \tag{90}$$

which is non-increasing in k and satisfies $0 \le b_k \le 1$.

Step 0: the tail mass is small. Since B_1 realises the asymmetry, $|\Omega\setminus B_1| \le |\Omega\,\triangle\, B_1| = \omega_n\mathcal{A}(\Omega)$. Combined with the suboptimal seed Lemma 4.17,

$$b_1 \;\le\; \mathcal{A}(\Omega) \;\le\; \big(C_\dagger\, D(\Omega)\big)^{1/4}. \tag{91}$$

In particular, since $b_k \le b_1$ for all $k \ge 1$, every tail mass is $\le (C_\dagger D(\Omega))^{1/4}$, which can be made smaller than any prescribed dimensional threshold by shrinking $\delta(n)$ (recall $D(\Omega) \le \delta(n)$). This is the only use of Lemma 4.17 in the construction.

Step 1: the energy-comparison inequality. For $k \ge 1$ let φ_k be the Lipschitz cut-off

$$\varphi_k(x) := \min\big\{1, (k+2-|x|)_+\big\},$$

so $\varphi_k \equiv 1$ on B_{k+1}, $\varphi_k \equiv 0$ outside B_{k+2}, $0 \le \varphi_k \le 1$ and $|\nabla\varphi_k| \le 1$ everywhere (it is 1 only on the annulus $B_{k+2} \setminus B_{k+1}$). Then $u_k := \varphi_k\, u_\Omega \in H^1_0(\Omega \cap B_{k+2})$ is admissible for the variational problem defining $E(\Omega \cap B_{k+2})$, so $E(\Omega \cap B_{k+2}) \le J_{\Omega\cap B_{k+2}}(u_k) = \frac{1}{2}\int |\nabla u_k|^2 - \int u_k$. We use the pointwise algebraic identity

$$|\nabla(\varphi_k u_\Omega)|^2 = \nabla u_\Omega \cdot \nabla(u_\Omega \varphi_k^2) + |\nabla\varphi_k|^2\, u_\Omega^2,$$

which is verified by expanding both sides ($\nabla(u_\Omega\varphi_k^2) = \varphi_k^2 \nabla u_\Omega + 2\varphi_k u_\Omega \nabla\varphi_k$, so the right-hand side equals $\varphi_k^2\, |\nabla u_\Omega|^2 + 2\varphi_k u_\Omega \nabla u_\Omega \cdot \nabla\varphi_k + |\nabla\varphi_k|^2\, u_\Omega^2 = |\varphi_k \nabla u_\Omega + u_\Omega \nabla\varphi_k|^2$). Integrating and applying the weak equation $\int \nabla u_\Omega \cdot \nabla(u_\Omega\varphi_k^2) = \int u_\Omega \varphi_k^2$ with the admissible test function $u_\Omega\varphi_k^2 \in H^1_0(\Omega)$, we get

$$J_{\Omega\cap B_{k+2}}(u_k) = \tfrac{1}{2}\int u_\Omega \varphi_k^2 + \tfrac{1}{2}\int |\nabla\varphi_k|^2\, u_\Omega^2 - \int \varphi_k u_\Omega.$$

Now use $\frac{1}{2}\varphi_k^2 - \varphi_k = -\frac{1}{2}(1 - (1-\varphi_k)^2) = -\frac{1}{2} + \frac{1}{2}(1-\varphi_k)^2$ to write $\frac{1}{2}\int u_\Omega\varphi_k^2 - \int \varphi_k u_\Omega = -\frac{1}{2}\int u_\Omega + \frac{1}{2}\int (1-\varphi_k)^2 u_\Omega = E(\Omega) + \frac{1}{2}\int (1-\varphi_k)^2 u_\Omega$, where we used $E(\Omega) = -\frac{1}{2}\int u_\Omega$ from (67). Therefore

$$E(\Omega \cap B_{k+2}) \le J_{\Omega\cap B_{k+2}}(u_k) = E(\Omega) + \tfrac{1}{2}\int (1-\varphi_k)^2 u_\Omega + \tfrac{1}{2}\int |\nabla\varphi_k|^2\, u_\Omega^2. \tag{92}$$

Both correction integrals are supported in $\mathbb{R}^n \setminus B_{k+1}$: indeed $1 - \varphi_k = 0$ on B_{k+1} and $\nabla\varphi_k = 0$ on $B_{k+1} \cup (\mathbb{R}^n \setminus B_{k+2})$. Hence, with $0 \le 1 - \varphi_k \le 1$ and $|\nabla\varphi_k| \le 1$,

$$E(\Omega \cap B_{k+2}) - E(\Omega) \;\le\; \tfrac{1}{2}\, |\Omega \setminus B_{k+1}| \Big(\sup_{\Omega\setminus B_{k+1}} u_\Omega + \sup_{\Omega\setminus B_{k+1}} u_\Omega^2 \Big). \tag{93}$$

By Lemma 4.15 applied with $R = k \ge 1$ (so $R + 1 = k + 1$ and the tail region is $\Omega \setminus B_{k+1}$), $\sup_{\Omega\setminus B_{k+1}} u_\Omega \le C_\flat\, |\Omega \setminus B_k|^{1/n} = C_\flat\, \omega_n^{1/n} b_k^{1/n}$, and likewise the square is bounded by $C_\flat^2 \omega_n^{2/n} b_k^{2/n}$. Also $|\Omega \setminus B_{k+1}| = \omega_n b_{k+1} \le \omega_n b_k$. Plugging in, and using $b_k \le 1$ so that $b_k^{2/n} \le b_k^{1/n}$,

$$E(\Omega \cap B_{k+2}) \;\le\; E(\Omega) + C\, b_{k+1}\big(b_k^{1/n} + b_k^{2/n}\big) \;\le\; E(\Omega) + C\, b_k^{1+1/n}, \tag{94}$$

for a dimensional constant $C = C(n)$ (we absorbed $\frac{1}{2}\omega_n(C_\flat\omega_n^{1/n} + C_\flat^2\omega_n^{2/n})$ and $b_{k+1} \le b_k$ into C). This is the energy-comparison inequality.

Step 2: turning energy comparison into a tail recursion. By the Saint–Venant inequality (69) applied to the set $\Omega \cap B_{k+2}$, whose volume is $|\Omega \cap B_{k+2}| = \omega_n(1 - b_{k+2})$,

$$E(B_1)\, |B_1|^{-(n+2)/n} \le E(\Omega \cap B_{k+2})\; |\Omega \cap B_{k+2}|^{-(n+2)/n},$$

that is

$$E(B_1)\, \omega_n^{-(n+2)/n} \big(\omega_n(1 - b_{k+2})\big)^{(n+2)/n} \le E(\Omega \cap B_{k+2}),$$

i.e.

$$E(B_1)\, (1 - b_{k+2})^{(n+2)/n} \le E(\Omega \cap B_{k+2}).$$

Combine with (94) and the definition $E(\Omega) = E(B_1) + \omega_n^{(n+2)/n} D(\Omega)$ (which is (68) at $|\Omega| = \omega_n$):

$$E(B_1)(1 - b_{k+2})^{(n+2)/n} \le E(\Omega) + C\, b_k^{1+1/n} = E(B_1) + \omega_n^{(n+2)/n} D(\Omega) + C\, b_k^{1+1/n}.$$

Rearranging, and recalling $E(B_1) < 0$, divide by $-E(B_1) > 0$:

$$1 - (1 - b_{k+2})^{(n+2)/n} \le \frac{\omega_n^{(n+2)/n} D(\Omega) + C\, b_k^{1+1/n}}{-E(B_1)} =: \widehat{C}_1\big(D(\Omega) + b_k^{1+1/n}\big),$$

with $\widehat{C}_1 = \widehat{C}_1(n) > 0$ (here $\widehat{C}_1 = \max\{\omega_n^{(n+2)/n}, C\}/(-E(B_1))$). To extract b_{k+2} we use the elementary lower bound

$$1-(1-t)^p \ \geq \ t \qquad \text{for all } t\in[0,1],\ p\geq 1, \tag{95}$$

which holds because $r\mapsto(1-t)^r$ is non-increasing on $[0,\infty)$, so $(1-t)^p\leq(1-t)^1=1-t$ for $p=(n+2)/n\geq 1$. Applying (95) with $t=b_{k+2}\in[0,1]$ and $p=(n+2)/n$,

$$b_{k+2} \ \leq \ 1-(1-b_{k+2})^{(n+2)/n} \ \leq \ \widehat{C}\,\big(D(\Omega)+b_k^{1+1/n}\big), \qquad \widehat{C}:=\widehat{C}_1. \tag{96}$$

Step 3: the geometric iteration to a bounded stopping index. Define the stopping index

$$K:=\max\big\{k\geq 1:\ b_k\geq 2\widehat{C}\,D(\Omega)\big\}, \tag{97}$$

with the convention $K:=0$ if $b_1<2\widehat{C}D(\Omega)$ (then the tail is already controlled at the first radius and Step 5 applies with $K=0$). The set in (97) is finite because $b_k\to 0$ as $k\to\infty$ (the tail mass of a finite-measure set vanishes), so K is well defined. We prove

$$K \ \leq \ K(n) \tag{98}$$

for a dimensional $K(n)$, provided $\delta(n)$ is small enough. Assume $K\geq 3$ (otherwise (98) is trivial).

Self-improving contraction. For every k with $k+2\leq K$ we have, by definition of K, that $b_{k+2}\geq 2\widehat{C}D(\Omega)$, i.e. $\widehat{C}D(\Omega)\leq\frac{1}{2}b_{k+2}$. Feeding this into (96),

$$b_{k+2}\leq\widehat{C}D(\Omega)+\widehat{C}b_k^{1+1/n}\leq\tfrac{1}{2}b_{k+2}+\widehat{C}b_k^{1+1/n},$$

and absorbing $\frac{1}{2}b_{k+2}$ on the left,

$$b_{k+2} \ \leq \ 2\widehat{C}\,b_k^{1+1/n} \qquad \text{whenever } k+2\leq K. \tag{99}$$

Smallness from the seed. By Step 0, $b_1\leq(C_\dagger D(\Omega))^{1/4}\leq(C_\dagger\delta(n))^{1/4}$. Choose $\delta(n)$ small enough that

$$b_1 \ \leq \ (C_\dagger\,\delta(n))^{1/4} \ \leq \ (2\widehat{C})^{-2n} \ =: \ \eta_n\in(0,1). \tag{100}$$

Then $2\widehat{C}\,b_k^{1/(2n)}\leq 2\widehat{C}\,\eta_n^{1/(2n)}=1$ for all $k\geq 1$ (as $b_k\leq b_1\leq\eta_n$), so (99) self-improves to

$$b_{k+2} \ = \ \big(2\widehat{C}\,b_k^{1/(2n)}\big)\,b_k^{1+1/(2n)} \ \leq \ b_k^{1+1/(2n)} \qquad \text{whenever } k+2\leq K. \tag{101}$$

Iterating along odd indices. Apply (101) repeatedly, starting from $k=1$, as long as the index stays $\leq K$. Writing the odd indices $1,3,5,\dots$ and setting $\gamma:=1+\frac{1}{2n}>1$, induction gives, for every $j\geq 0$ with $2j+1\leq K$,

$$b_{2j+1} \ \leq \ b_1^{\gamma^{\,j}}. \tag{102}$$

Indeed (102) holds at $j=0$, and if it holds at j with $2(j+1)+1=2j+3\leq K$, then (101) (at $k=2j+1$, legitimate since $2j+3\leq K$) gives $b_{2j+3}\leq b_{2j+1}^{\gamma}\leq\big(b_1^{\gamma^j}\big)^{\gamma}=b_1^{\gamma^{j+1}}$.

Bounding K. Let $j_K:=\lfloor(K-1)/2\rfloor$, the largest j with $2j+1\leq K$. By the maximality of K and (102),

$$2\widehat{C}\,D(\Omega) \ \leq \ b_{2j_K+1} \ \leq \ b_1^{\gamma^{\,j_K}} \ \leq \ \big(C_\dagger D(\Omega)\big)^{\gamma^{\,j_K}/4}, \tag{103}$$

using $b_1\leq(C_\dagger D(\Omega))^{1/4}$. Take logarithms (recall $D(\Omega)\leq\delta(n)<1$, so $\log D(\Omega)<0$); writing $L:=-\log D(\Omega)>0$, (103) reads

$$\log(2\widehat{C})-L \ \leq \ \frac{\gamma^{\,j_K}}{4}\big(\log C_\dagger-L\big).$$

For $\delta(n)$ small, $L = -\log D(\Omega) \geq -\log \delta(n)$ is large; in particular we may take $\delta(n)$ small enough that $L \geq 2\max\{\left|\log(2\widehat{C})\right|, |\log C_\dagger|\}$, whence $\log(2\widehat{C}) - L \geq -\frac{3}{2}L$ and $\log C_\dagger - L \leq -\frac{1}{2}L$. The displayed inequality then forces $-\frac{3}{2}L \leq -\frac{\gamma^{j_K}}{8}L$, i.e. $\gamma^{j_K} \leq 12$. Since $\gamma = 1 + \frac{1}{2n} > 1$ this bounds $j_K \leq \log 12/\log\gamma =: j_*(n)$, hence $K \leq 2j_K + 2 \leq 2j_*(n) + 2 =: K(n)$, a dimensional constant. This proves (98).

Step 4: tail at the stopping radius. By the maximality (97), $b_{K+1} < 2\widehat{C}D(\Omega)$; we record this for the next step.

Step 5: the truncated set and its volume. By the definition (97) of K and $K \leq K(n)$, $b_{K+1} < 2\widehat{C}D(\Omega)$, hence

$$|\Omega \setminus B_{K+1}| = \omega_n b_{K+1} < 2\widehat{C}\,\omega_n\,D(\Omega). \tag{104}$$

In fact we cut one radius further out to have room for the cut-off: by monotonicity $b_{K+3} \leq b_{K+1} < 2\widehat{C}D(\Omega)$, so

$$|\Omega \cap B_{K+3}| = \omega_n(1 - b_{K+3}) \geq \omega_n(1 - 2\widehat{C}D(\Omega)), \qquad |\Omega \setminus B_{K+3}| < 2\widehat{C}\,\omega_n D(\Omega). \tag{105}$$

Define

$$r := \Big(\frac{|\Omega \cap B_{K+3}|}{\omega_n}\Big)^{1/n} = (1 - b_{K+3})^{1/n} \in \big[(1 - 2\widehat{C}D(\Omega))^{1/n},\, 1\big], \qquad \widetilde{\Omega} := r^{-1}\,(\Omega \cap B_{K+3}).$$

Then $\left|\widetilde{\Omega}\right| = r^{-n}\,|\Omega \cap B_{K+3}| = \omega_n$, which is (86). Since $\Omega \cap B_{K+3} \subset B_{K+3}$ and $r \geq (1 - 2\widehat{C}\delta(n))^{1/n} \geq \frac{1}{2}$ for $\delta(n)$ small,

$$\operatorname{diam}(\widetilde{\Omega}) = r^{-1}\operatorname{diam}(\Omega \cap B_{K+3}) \leq r^{-1} \cdot 2(K+3) \leq 4\big(K(n) + 3\big) =: d(n),$$

giving (87). We record for later the linearised bound

$$0 \leq 1 - r^n = b_{K+3} < 2\widehat{C}D(\Omega), \qquad 0 \leq 1 - r \leq 1 - r^n < 2\widehat{C}D(\Omega), \tag{106}$$

the middle inequality because $r \leq 1$ implies $1 - r \leq 1 - r^n$.

Step 6: deficit of the surrogate. Apply the energy comparison (94) at $k = K+1$ (so the cut radius is $k + 2 = K + 3$):

$$E(\Omega \cap B_{K+3}) \leq E(\Omega) + C\,b_{K+1}^{1+1/n} \leq E(\Omega) + C\,(2\widehat{C}D(\Omega))^{1+1/n} \leq E(\Omega) + C'\,D(\Omega),$$

where in the last step we used $D(\Omega) \leq \delta(n) \leq 1$ so that $D(\Omega)^{1+1/n} \leq D(\Omega)$, and absorbed constants into $C' = C'(n)$. Since E scales as $E(t\Omega') = t^{n+2}E(\Omega')$ and $\widetilde{\Omega} = r^{-1}(\Omega \cap B_{K+3})$,

$$E(\widetilde{\Omega}) = r^{-(n+2)}E(\Omega \cap B_{K+3}) \leq r^{-(n+2)}\big(E(\Omega) + C'D(\Omega)\big).$$

Because $\left|\widetilde{\Omega}\right| = \omega_n$, $D(\widetilde{\Omega}) = \omega_n^{-(n+2)/n}(E(\widetilde{\Omega}) - E(B_1))$. Using $E(\Omega) = E(B_1) + \omega_n^{(n+2)/n}D(\Omega)$ and $r \leq 1$ (so $r^{-(n+2)} \geq 1$), with $E(B_1) < 0$,

$$\begin{aligned} E(\widetilde{\Omega}) - E(B_1) &\leq r^{-(n+2)}\big(E(B_1) + \omega_n^{(n+2)/n}D(\Omega) + C'D(\Omega)\big) - E(B_1) \\ &= \big(r^{-(n+2)} - 1\big)E(B_1) + r^{-(n+2)}\big(\omega_n^{(n+2)/n} + C'\big)D(\Omega). \end{aligned}$$

For the first term, $r^{-(n+2)} - 1 \geq 0$ and $E(B_1) < 0$, so $(r^{-(n+2)} - 1)E(B_1) \leq 0$ and we may bound it by $(r^{-(n+2)} - 1)|E(B_1)|$. Quantitatively, by (106), $r^n \geq 1 - 2\widehat{C}D(\Omega)$, so $r^{-(n+2)} = r^{-n} \cdot r^{-2} \leq (1 - 2\widehat{C}D(\Omega))^{-(n+2)/n} \leq 1 + C''D(\Omega)$ for $\delta(n)$ small (Bernoulli/Taylor bound, since $2\widehat{C}D(\Omega) \leq 2\widehat{C}\delta(n) \leq \frac{1}{2}$). Hence $r^{-(n+2)} - 1 \leq C''D(\Omega)$ and $r^{-(n+2)} \leq 1 + C''D(\Omega) \leq 2$, giving

$$E(\widetilde{\Omega}) - E(B_1) \leq C''D(\Omega) \cdot |E(B_1)| + 2\big(\omega_n^{(n+2)/n} + C'\big)D(\Omega) \leq C_2'\,D(\Omega),$$

with $C_2' = C_2'(n)$. Dividing by $\omega_n^{(n+2)/n}$,

$$D(\widetilde{\Omega}) = \omega_n^{-(n+2)/n}\big(E(\widetilde{\Omega}) - E(B_1)\big) \le \omega_n^{-(n+2)/n} C_2'\, D(\Omega) =: C_3\, D(\Omega), \tag{107}$$

which is (88) with $C_3 = C_3(n)$.

Step 7: asymmetry loss. Let $B_1(x_0)$ realise the asymmetry of $\widetilde{\Omega}$, so $\mathcal{A}(\widetilde{\Omega}) = \left|\widetilde{\Omega} \triangle B_1(x_0)\right| / \omega_n$. We compare the asymmetry of Ω to that of $\widetilde{\Omega}$ through the chain of inclusions relating Ω, $\Omega \cap B_{K+3}$ and $\widetilde{\Omega} = r^{-1}(\Omega \cap B_{K+3})$. For any test ball $B_1(y)$,

$$|\Omega \triangle B_1(y)| \le |\Omega \triangle (\Omega \cap B_{K+3})| + |(\Omega \cap B_{K+3}) \triangle B_1(y)|.$$

The first term is $|\Omega \setminus B_{K+3}| < 2\widehat{C}\omega_n D(\Omega)$ by (105). For the second, scale by r: writing $E_r := \Omega \cap B_{K+3} = r\widetilde{\Omega}$ and $B_1(y) = r\, B_{1/r}(y/r)$,

$$\left|(r\widetilde{\Omega}) \triangle (r\, B_{1/r}(y/r))\right| = r^n \left|\widetilde{\Omega} \triangle B_{1/r}(y/r)\right| \le r^n \Big(\left|\widetilde{\Omega} \triangle B_1(y/r)\right| + \left|B_1(y/r) \triangle B_{1/r}(y/r)\right|\Big).$$

The last term is the volume difference of concentric balls of radii 1 and $1/r \ge 1$: $\left|B_1 \triangle B_{1/r}\right| = \omega_n(r^{-n} - 1)$. Hence

$$r^n \left|B_1(y/r) \triangle B_{1/r}(y/r)\right| = r^n\, \omega_n(r^{-n} - 1) = \omega_n(1 - r^n) < 2\widehat{C}\, \omega_n D(\Omega),$$

using (106). Choosing $y := rx_0$ so that $y/r = x_0$ is the optimal centre for $\widetilde{\Omega}$, the middle term becomes $r^n \left|\widetilde{\Omega} \triangle B_1(x_0)\right| = r^n \omega_n \mathcal{A}(\widetilde{\Omega}) \le \omega_n \mathcal{A}(\widetilde{\Omega})$ (as $r \le 1$). Collecting,

$$|\Omega \triangle B_1(rx_0)| \le 2\widehat{C}\omega_n D(\Omega) + \omega_n \mathcal{A}(\widetilde{\Omega}) + 2\widehat{C}\omega_n D(\Omega),$$

and dividing by ω_n and taking the infimum over translates (which only decreases the left side),

$$\mathcal{A}(\Omega) \le \frac{|\Omega \triangle B_1(rx_0)|}{\omega_n} \le \mathcal{A}(\widetilde{\Omega}) + 4\widehat{C}\, D(\Omega). \tag{108}$$

This is (89) with the constant $4\widehat{C}$. Taking $C_\sharp := \max\{C_3, 4\widehat{C}\} = C_\sharp(n)$ proves (88)–(89) simultaneously, and we already have (86)–(87). This completes the proof. $\square$

Remark 4.19 (Translation into a fixed ball)**.** The diameter bound (87) is translation invariant, so after a translation we may assume $\widetilde{\Omega} \subset B_{d(n)}$. Both $\mathcal{A}$ and D are translation invariant, so (86)–(89) are preserved.

4.2.6. *The far branch.* When the deficit is not small the inequality is trivial: the asymmetry is bounded ($\mathcal{A} < 2$) while the deficit is bounded below, so a quadratic bound holds with a crude constant. No truncation is needed.

Lemma 4.20 (Large-deficit quadratic bound)**.** *For every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \ge \delta(n)/4$,*

$$E(\Omega) - E(B_1) \;\ge\; \omega_n^{(n+2)/n}\, \frac{\delta(n)}{16}\, \mathcal{A}(\Omega)^2. \tag{109}$$

Proof. Since $|\Omega| = \omega_n$ gives $\mathcal{A}(\Omega) < 2$, we have $\mathcal{A}(\Omega)^2/4 < 1$. Hence

$$D(\Omega) \ge \frac{\delta(n)}{4} = \frac{\delta(n)}{4} \cdot 1 \ge \frac{\delta(n)}{4} \cdot \frac{\mathcal{A}(\Omega)^2}{4} = \frac{\delta(n)}{16}\mathcal{A}(\Omega)^2.$$

Multiplying by $\omega_n^{(n+2)/n}$ and using $E(\Omega) - E(B_1) = \omega_n^{(n+2)/n} D(\Omega)$ gives (109). $\square$

4.2.7. *The near branch.* Fix the dimensional working radius

$$R_*(n) := \max\{d(n), 2\}, \tag{110}$$

so $R_* \geq d(n)$ (the surrogates of Lemma 4.18 lie in B_{R_*} after translation, by Remark 4.19) and $R_* \geq 2$ (the hypothesis of Theorem 4.11).

Lemma 4.21 (Small-deficit transfer)**.** *Set*

$$\widetilde{c}_{\mathrm{SV}}(n) := \omega_n^{-(n+2)/n}\, c_{\mathrm{SV}}\big(n, R_*(n)\big) > 0. \tag{111}$$

For every open Ω *with* $|\Omega| = \omega_n$ *and* $D(\Omega) \leq \delta(n)$*,*

$$\mathcal{A}(\Omega)^2 \;\leq\; \Big(\frac{4\,C_\sharp(n)}{\widetilde{c}_{\mathrm{SV}}(n)} + 4\,C_\sharp(n)^2\Big)\, D(\Omega). \tag{112}$$

Proof. Apply Lemma 4.18 to get $\widetilde{\Omega}$ satisfying (86)–(89); by Remark 4.19 assume $\widetilde{\Omega} \subset B_{d(n)} \subset B_{R_*(n)}$. Theorem 4.11 at $R = R_*(n)$ gives $E(\widetilde{\Omega}) - E(B_1) \geq c_{\mathrm{SV}}(n, R_*)\mathcal{A}(\widetilde{\Omega})^2$; dividing by $\omega_n^{(n+2)/n}$ and using $\big|\widetilde{\Omega}\big| = \omega_n$,

$$D(\widetilde{\Omega}) \;\geq\; \widetilde{c}_{\mathrm{SV}}(n)\, \mathcal{A}(\widetilde{\Omega})^2. \tag{113}$$

Combining (113) with (88),

$$\mathcal{A}(\widetilde{\Omega})^2 \leq \frac{D(\widetilde{\Omega})}{\widetilde{c}_{\mathrm{SV}}} \leq \frac{C_\sharp}{\widetilde{c}_{\mathrm{SV}}}\, D(\Omega).$$

Now use (89), the elementary $(a+b)^2 \leq 2a^2 + 2b^2$, and $D(\Omega) \leq \delta(n) \leq 1$ (so $D(\Omega)^2 \leq D(\Omega)$):

$$\begin{aligned}\mathcal{A}(\Omega)^2 &\leq \big(\mathcal{A}(\widetilde{\Omega}) + C_\sharp D(\Omega)\big)^2 \leq 2\mathcal{A}(\widetilde{\Omega})^2 + 2C_\sharp^2 D(\Omega)^2\\ &\leq \frac{2C_\sharp}{\widetilde{c}_{\mathrm{SV}}} D(\Omega) + 2C_\sharp^2 D(\Omega) \leq \Big(\frac{4C_\sharp}{\widetilde{c}_{\mathrm{SV}}} + 4C_\sharp^2\Big) D(\Omega),\end{aligned}$$

which is (112). □

Corollary 4.22 (Small-deficit quadratic bound)**.** *With*

$$c_{\mathrm{near}}(n) := \frac{\omega_n^{(n+2)/n}}{4C_\sharp(n)\big(1/\widetilde{c}_{\mathrm{SV}}(n) + C_\sharp(n)\big)} > 0, \tag{114}$$

every open Ω *with* $|\Omega| = \omega_n$ *and* $D(\Omega) \leq \delta(n)$ *satisfies* $E(\Omega) - E(B_1) \geq c_{\mathrm{near}}(n)\, \mathcal{A}(\Omega)^2$*.*

Proof. Rearrange (112) to $D(\Omega) \geq \mathcal{A}(\Omega)^2 / \big(4C_\sharp/\widetilde{c}_{\mathrm{SV}} + 4C_\sharp^2\big)$ and multiply by $\omega_n^{(n+2)/n}$, using $E(\Omega) - E(B_1) = \omega_n^{(n+2)/n} D(\Omega)$. □

4.2.8. *Proof of the global theorem.* We now glue the far and near branches, then remove the volume normalisation.

Proof of Theorem 4.12. Set

$$c_{\mathrm{SV}}^{\mathrm{glob}}(n) := \min\Big\{\, c_{\mathrm{near}}(n),\ \omega_n^{(n+2)/n}\, \tfrac{\delta(n)}{16} \,\Big\} > 0. \tag{115}$$

Let Ω be open with $|\Omega| = \omega_n$.

Case A: $D(\Omega) \geq \delta(n)/4$. Lemma 4.20 gives $E(\Omega) - E(B_1) \geq \omega_n^{(n+2)/n}(\delta(n)/16)\mathcal{A}(\Omega)^2 \geq c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$.

Case B: $D(\Omega) \leq \delta(n)$ (which includes $D(\Omega) \leq \delta(n)/4$). Corollary 4.22 gives $E(\Omega) - E(B_1) \geq c_{\mathrm{near}}(n)\mathcal{A}(\Omega)^2 \geq c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$.

The two cases overlap on $\delta(n)/4 \leq D(\Omega) \leq \delta(n)$ and together cover $[0, \infty)$; in either case (64) holds with $c_{\mathrm{SV}}^{\mathrm{glob}}(n)$ as in (115), a dimensional constant because $\delta(n)$, $c_{\mathrm{near}}(n)$ depend only on n (recall $R_*(n)$ and hence $c_{\mathrm{SV}}(n, R_*(n))$ are dimensional).

General volume. For arbitrary finite-measure Ω put $\Omega_* := (\omega_n/|\Omega|)^{1/n}\Omega$, so $|\Omega_*| = \omega_n$, $\mathcal{A}(\Omega_*) = \mathcal{A}(\Omega)$ and $D(\Omega_*) = D(\Omega)$ by scale invariance of $\mathcal{A}$ and D. Applying (64) to Ω_* reads $D(\Omega) \geq \omega_n^{-(n+2)/n} c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$. Multiplying by $|\Omega|^{(n+2)/n}$ and using $E(\Omega)\,|\Omega|^{-(n+2)/n} - E(B^*)\,|B^*|^{-(n+2)/n} = D(\Omega)$ with $|B^*| = |\Omega|$, then $E = -T/2$, gives

$$T(B^*) - T(\Omega) = 2\big(E(\Omega) - E(B^*)\big) \geq 2\, c_{\mathrm{SV}}^{\mathrm{glob}}(n)\Big(\tfrac{|\Omega|}{\omega_n}\Big)^{(n+2)/n}\mathcal{A}(\Omega)^2,$$

which is (65). □

4.3. **Affine first variation of the moving-ball distance.** We now turn to the proof of the bounded Saint–Venant inequality (Theorem 4.11), the input used in §4.2. The technical engine is a finite-perimeter one-sided derivative for the moving-ball distance $q(\rho)$: pushing the superlevel E_ρ forward by a homothety plus an affine translation produces a competitor for $q(\rho+h)$, whose symmetric-difference cost is bounded by a single boundary integral involving the velocity V_ρ, the homothetic field $H_{z,\rho}$, and the translation parameter a.

Lemma 4.23 (Affine shell estimate)**.** *Let ρ be shell-admissible. Fix $z, a \in \mathbb{R}^n$, and define*

$$T_h(x) := z + ha + \left(1 + \frac{h}{\rho}\right)(x - z) = x + h\left(\frac{x-z}{\rho} + a\right).$$

Then

$$\limsup_{h\downarrow 0} \frac{|E_{\rho+h}\Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho} |V_\rho - H_{z,\rho} - a\cdot\nu_\rho|\, d\mathcal{H}^{n-1}. \tag{116}$$

Proof. Let $E = E_\rho$, $W(x) = (x-z)/\rho + a$, and $w = w(\rho)$. Since $\Omega \subset B_R$, W is bounded on Ω. The finite Radon measure

$$\left(\frac{w}{|\nabla u|} + |W| + \left|\frac{w}{|\nabla u|} - W\cdot\nu_\rho\right|\right)\mathcal{H}^{n-1}\llcorner\partial^* E$$

is finite. Fix $\varepsilon > 0$. By inner regularity and $\partial^* E \subset \Omega$ up to $\mathcal{H}^{n-1}$-null sets, choose a compact $K \subset \partial^* E \cap \Omega$ such that this measure of $\partial^* E \setminus K$ is at most ε. Then choose $U \in \mathcal{U}$ with $K \subset U \Subset \Omega$.

We first estimate the symmetric difference in U. On a neighbourhood of $\overline{U}$, interior elliptic regularity gives $u \in C^1$, and uniformly for $x \in U$,

$$T_h^{-1}(x) = x - hW(x) + O(h^2), \qquad u(T_h^{-1}x) = u(x) - h\nabla u(x)\cdot W(x) + o(h).$$

Also $t(\rho+h) = t(\rho) - hw + o(h)$. Set

$$F(x) := w + \nabla u(x)\cdot W(x).$$

For $x \in U$,

$$x \in E_{\rho+h} \iff u(x) > t(\rho) - hw + o(h),$$

while

$$x \in T_h(E) \iff u(x) > t(\rho) + h\nabla u(x)\cdot W(x) + o(h).$$

Thus the part of the symmetric difference in U is contained, up to a uniform $o(h)$ outward error, in the two-sided slab whose endpoints are $\min(-w, \nabla u\cdot W)$ and $\max(-w, \nabla u\cdot W)$. Precisely, fix $\delta > 0$. Choose rational c, λ, z_0, a_0 with $c \in [-1, 0]$, $1 \leq \lambda \leq \rho_*^{-1}$, and

$$|c + w| + \sup_{\overline{U}} |\nabla u\cdot(\lambda(x - z_0) + a_0) - \nabla u\cdot W| \leq \delta.$$

Then choose a rational buffer $r > 2\delta$. For all sufficiently small $h > 0$, the local symmetric difference is contained in the registered slab with endpoints

$$\min(c, \nabla u \cdot (\lambda(x - z_0) + a_0)) - r, \qquad \max(c, \nabla u \cdot (\lambda(x - z_0) + a_0)) + r.$$

Lemma 2.8, followed by $\delta, r \downarrow 0$, gives

$$\limsup_{h \downarrow 0} \frac{|(E_{\rho+h} \Delta T_h(E)) \cap U|}{h} \leq \int_{\partial^* E \cap U} \frac{|F|}{|\nabla u|} \, d\mathcal{H}^{n-1}.$$

On the complement use

$$E_{\rho+h} \Delta T_h(E) \subset (E_{\rho+h} \Delta E) \cup (E \Delta T_h(E)).$$

For the level-set part, the exact volume derivative and the registered localized one-sided coarea formula on U give the complement estimate. Indeed,

$$|E_{\rho+h} \setminus E| = \omega_n\big((\rho + h)^n - \rho^n\big) = h\, P(B_\rho) + o(h) = h\, w \int_{\partial^* E} \frac{1}{|\nabla u|} \, d\mathcal{H}^{n-1} + o(h),$$

where the last equality is the shell-admissible flux identity. On U, the same rational-buffer use of Lemma 2.8, now with one endpoint 0, gives

$$\lim_{h \downarrow 0} \frac{|(E_{\rho+h} \setminus E) \cap U|}{h} = w \int_{\partial^* E \cap U} \frac{1}{|\nabla u|} \, d\mathcal{H}^{n-1}.$$

Subtracting the local shell from the exact total shell yields

$$\limsup_{h \downarrow 0} \frac{|(E_{\rho+h} \Delta E) \setminus U|}{h} \leq w \int_{\partial^* E \setminus U} \frac{1}{|\nabla u|} \, d\mathcal{H}^{n-1} \leq w \int_{\partial^* E \setminus K} \frac{1}{|\nabla u|} \, d\mathcal{H}^{n-1}.$$

For the affine deformation part, choose $\eta \in C_c^1(\mathbb{R}^n)$ with $\eta = 0$ on a neighbourhood $N_K \Subset U$ of K, $\eta = 1$ on a fixed neighbourhood of $\overline{B_{R+1}} \setminus U$, and $0 \leq \eta \leq 1$. Since $E \subset B_R$ and W is bounded, $T_h(E) \subset B_{R+1}$ for all small $h > 0$; hence $(E \Delta T_h(E)) \setminus U \subset \overline{B_{R+1}} \setminus U \subset \{\eta = 1\}$. As $\eta = 0$ near K and $0 \leq \eta \leq 1$, Lemma 2.9 gives

$$\limsup_{h \downarrow 0} \frac{|(E \Delta T_h(E)) \setminus U|}{h} \leq \int_{\partial^* E} \eta\, |W| \, d\mathcal{H}^{n-1} \leq \int_{\partial^* E \setminus K} |W| \, d\mathcal{H}^{n-1}.$$

The two tail terms are bounded by 2ε. Therefore

$$\limsup_{h \downarrow 0} \frac{|E_{\rho+h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho \cap U} \frac{|F|}{|\nabla u|} \, d\mathcal{H}^{n-1} + 2\varepsilon \leq \int_{\partial^* E_\rho} \frac{|F|}{|\nabla u|} \, d\mathcal{H}^{n-1} + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$, and using

$$\frac{F}{|\nabla u|} = \frac{w}{|\nabla u|} + \frac{\nabla u \cdot W}{|\nabla u|} = V_\rho - W \cdot \nu_\rho = V_\rho - H_{z,\rho} - a \cdot \nu_\rho$$

on $\partial^* E_\rho$, proves the estimate. □

Theorem 4.24 (One-sided first variation of q)**.** *For every $\rho \in (0, 1]$, define*

$$q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

At every shell-admissible $\rho \in (0, 1)$,

$$D^+ q(\rho) := \limsup_{h \downarrow 0} \frac{q(\rho + h) - q(\rho)}{h} \leq \frac{n}{\rho} q(\rho) + \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1}, \tag{117}$$

where z_ρ is any minimiser of $z \mapsto |E_\rho \Delta B_\rho(z)|$.

Proof. The minimiser exists. Indeed, $z \mapsto |E_\rho \Delta B_\rho(z)|$ is continuous by translation continuity in L^1, and if $|z| > R + \rho$, then $B_\rho(z) \cap E_\rho = \emptyset$, so the value is $2|E_\rho|$. Thus the infimum is attained in a closed ball.

Fix a minimiser z_ρ and $a \in \mathbb{R}^n$. For the affine map in Lemma 4.23,

$$T_h(B_\rho(z_\rho)) = B_{\rho+h}(z_\rho + ha).$$

Therefore

$$\begin{aligned} q(\rho + h) &\le |E_{\rho+h} \Delta T_h(B_\rho(z_\rho))| \\ &\le |E_{\rho+h} \Delta T_h(E_\rho)| + |T_h(E_\rho) \Delta T_h(B_\rho(z_\rho))|. \end{aligned}$$

Since T_h is injective affine with constant Jacobian $(1 + h/\rho)^n$,

$$|T_h(E_\rho) \Delta T_h(B_\rho(z_\rho))| = \left(1 + \frac{h}{\rho}\right)^n q(\rho).$$

Subtract $q(\rho)$, divide by h, take the limsup as $h \downarrow 0$, and use Lemma 4.23. Then take the infimum over a. □

Lemma 4.25 (Unconditional Lipschitz bound for q)**.** *For* $0 < \rho \le \sigma \le 1$,

$$|q(\sigma) - q(\rho)| \le 2\omega_n(\sigma^n - \rho^n) \le 2n\omega_n|\sigma - \rho|.$$

Consequently q is absolutely continuous and differentiable a.e.

Proof. Use the same centre for the two radii and the nestedness $E_\rho \subset E_\sigma$, $B_\rho(z) \subset B_\sigma(z)$:

$$q(\sigma) \le q(\rho) + |E_\sigma \setminus E_\rho| + |B_\sigma \setminus B_\rho| = q(\rho) + 2\omega_n(\sigma^n - \rho^n).$$

Interchange ρ and σ to get the reverse inequality. □

4.4. **The moving-ball endpoint trace.** The Lipschitz bound $|q(\rho) - q(\sigma)| \le 2n\omega_n|\rho - \sigma|$ just obtained (Lemma 4.25) lets us trade pointwise information about q at the bad-density boundary ρ_δ for a Lebesgue-measure average over a fixed-width slab. The endpoint trace below converts the budget bound (59) into a quadratic estimate on $q(\rho_\delta)^2$, and is the geometric content that closes Theorem 4.29.

Lemma 4.26 (Moving-ball endpoint trace)**.** *Let*

$$L_0 := \frac{1 - \rho_*}{8}, \qquad J := [\rho_\delta - L_0, \rho_\delta].$$

Assume $\delta_T(\Omega)$ is small enough that $\rho_\delta \ge (1 + \rho_*)/2$. Then $J \subset [\rho_*, \rho_\delta]$ and

$$q(\rho_\delta)^2 \le C(n, R, \rho_*)\, \delta_T(\Omega).$$

Proof. The inclusion $J \subset [\rho_*, \rho_\delta]$ follows from $\rho_\delta - \rho_* \ge (1 - \rho_*)/2$. Let

$$G_I := \{\rho \in J : D_I(t(\rho)) \le \eta_I\}, \qquad G = G_I \cap G_\tau.$$

On G, Theorem 2.1 gives the pointwise volume bound

$$q(\rho)^2 = \inf_z |E_\rho \Delta B_\rho(z)|^2 \le C(n, \rho_*) D_I(t(\rho)), \qquad d\rho \le \tau_0^{-1} d\mu(\rho),$$

so, by the budget (59),

$$\int_G q(\rho)^2\, d\rho \le \tau_0^{-1} \int_G q(\rho)^2\, d\mu \le C\, \delta_T(\Omega).$$

On $J \setminus G$, the measure is $O(\delta_T)$ by the proof of Proposition 4.28, while $q \le 2\omega_n$. Hence

$$\int_J q(\rho)^2 \, d\rho \le C \, \delta_T(\Omega).$$

There is therefore $\rho_0 \in J$ with

$$q(\rho_0)^2 \le C L_0^{-1} \delta_T(\Omega).$$

Since q is absolutely continuous and $\rho_0 \le \rho_\delta$,

$$q(\rho_\delta) \le q(\rho_0) + \int_{\rho_0}^{\rho_\delta} q'_+(\rho) \, d\rho.$$

By Cauchy–Schwarz and Proposition 4.28,

$$q(\rho_\delta) \le C \delta_T(\Omega)^{1/2} + L_0^{1/2} \left(\int_J (q'_+)^2 \, d\rho \right)^{1/2} \le C \delta_T(\Omega)^{1/2}.$$

□

4.5. **The positive kinetic estimate.** The affine shell estimate of §4.3 provides an upper bound on the one-sided derivative of q in terms of the level-set defects $D_H + D_I$ and an arbitrary choice of centre and translation. The positive kinetic estimate below complements it: at a distance-minimising centre it produces a *lower* bound on the homothetic-velocity defect of order $q(\rho)$ itself, which is the mechanism that converts the level-set budget (59) into a quantitative estimate on q.

Lemma 4.27 (Transfer to a distance-minimising centre)**.** *After reducing $\eta_I = \eta_I(n, R, \rho_*)$ if necessary, the following holds. Let $\rho \in G_I$ be a level where Theorem 2.5 holds, and let $z_\rho^{\min}$ minimise $z \mapsto |E_\rho \Delta B_\rho(z)|$. Then*

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho^{\min}, \rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \le C(n, R, \rho_*) D_I(t(\rho)). \tag{118}$$

Proof. Let z_ρ^{sc} be the centre from Theorem 2.5. Minimality gives

$$|E_\rho \Delta B_\rho(z_\rho^{\min})| \le |E_\rho \Delta B_\rho(z_\rho^{\mathrm{sc}})| \le C D_I(t(\rho))^{1/2}.$$

Thus

$$|B_\rho(z_\rho^{\min}) \Delta B_\rho(z_\rho^{\mathrm{sc}})| \le C D_I(t(\rho))^{1/2}.$$

For equal balls with centre distance d,

$$|B_\rho(z_1) \Delta B_\rho(z_2)| \ge c(n) \rho^{n-1} \min\{d, \rho\}.$$

Indeed, in coordinates $x = z_1 + s\,e + y$ with $e = (z_2 - z_1)/d$ and $y \perp e$, one has $B_\rho(z_1) \setminus B_\rho(z_2) = \{s^2 + |y|^2 < \rho^2, \ s \le d/2\}$; integrating the $(n-1)$-dimensional cross-sections over $s \in [0, \frac{1}{2} \min\{d, \rho\}]$, where $\rho^2 - s^2 \ge \frac{3}{4}\rho^2$, gives $|B_\rho(z_1) \setminus B_\rho(z_2)| \ge \frac{1}{2}\omega_{n-1}(3/4)^{(n-1)/2}\rho^{n-1} \min\{d, \rho\}$, and the symmetric difference is twice this. Choose η_I so small that the preceding upper bound forces the linear branch $d < \rho$. Since $\rho \ge \rho_*$,

$$|z_\rho^{\min} - z_\rho^{\mathrm{sc}}| \le C(n, \rho_*) D_I(t(\rho))^{1/2}.$$

The homothetic trace is affine in the centre:

$$H_{z_\rho^{\min}, \rho} - H_{z_\rho^{\mathrm{sc}}, \rho} = -\frac{(z_\rho^{\min} - z_\rho^{\mathrm{sc}}) \cdot \nu_\rho}{\rho}.$$

On G_I, the perimeter is uniformly bounded because $D_I(t(\rho)) \le \eta_I$ and $\rho \in [\rho_*, 1]$. Hence

$$\int_{\partial^* E_\rho} |H_{z_\rho^{\min},\rho} - H_{z_\rho^{\mathrm{sc}},\rho}| \, d\mathcal{H}^{n-1} \le C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Combining this with (16) via the triangle inequality,

$$\int_{\partial^* E_\rho} |H_{z_\rho^{\min},\rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \le \int_{\partial^* E_\rho} |H_{z_\rho^{\min},\rho} - H_{z_\rho^{\mathrm{sc}},\rho}| \, d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} |H_{z_\rho^{\mathrm{sc}},\rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \le C(n, R, \rho_*) D_I(t(\rho))^{1/2},$$

and squaring gives (118). □

Proposition 4.28 (Positive kinetic bound)**.** *Let $J \subset [\rho_*, \rho_\delta]$ be an interval. Then*

$$\int_J (q'_+(\rho))^2 \, d\rho \le C(n, R, \rho_*) \, \delta_T(\Omega),$$

where $q'_+(\rho) = \max\{q'(\rho), 0\}$ at a.e. differentiability points.

Proof. Let

$$G := J \cap G_I \cap G_\tau, \qquad \mathcal{B} := J \setminus G.$$

Since q is Lipschitz, q' exists a.e.; shell-admissible radii have full measure in G_τ. At points where both properties hold, $D^+ q = q'$. The right side of Theorem 4.24 is nonnegative, so the theorem also bounds q'_+. On G, outside this null set, Theorem 4.24, Theorem 2.5, Lemma 4.27, and Lemma 2.6 give

$$(q'_+)^2 \le C(n, R, \rho_*) \big(D_H(t(\rho)) + D_I(t(\rho)) \big).$$

Indeed $q^2 \le C D_I$ on G_I by (16), and

$$\inf_a \int |V_\rho - H_{z_\rho^{\min},\rho} - a \cdot \nu_\rho| \le \int |V_\rho - \bar{V}_\rho| + \int |H_{z_\rho^{\min},\rho} - \bar{V}_\rho|.$$

On G_τ, $d\rho \le \tau_0^{-1} d\mu$. Hence the budget (59) yields

$$\int_G (q'_+)^2 \, d\rho \le C \, \delta_T(\Omega).$$

It remains to pay $\mathcal{B}$. By Lemma 4.25, $q'_+ \le 2n\omega_n$ a.e. The bad-density part has $|J \cap B_\tau| \le C \, \delta_T(\Omega)$. For the bad-isoperimetric part on good density,

$$|J \cap G_\tau \cap G_I^c| \le \tau_0^{-1} \mu(G_I^c) \le \frac{C}{\eta_I} \int_J D_I(t(\rho)) \, d\mu(\rho) \le C \, \delta_T(\Omega).$$

Therefore $|\mathcal{B}| \le C \, \delta_T(\Omega)$, and

$$\int_{\mathcal{B}} (q'_+)^2 \, d\rho \le C \, \delta_T(\Omega).$$

□

4.6. **Closure of the bounded estimate and the Kohler–Jobin transfer.** We now combine the pieces. The endpoint trace (Lemma 4.26) bounds $q(\rho_\delta)^2$ by $\delta_T(\Omega)$; the superlevel asymmetry transfer (Lemma 2.7) passes from $q(\rho_\delta)$ back to $\mathcal{A}(\Omega)$; and the bound on ρ_δ from Proposition 4.10 guarantees that the chosen radius is admissible whenever the deficit is small enough. This yields the bounded Saint–Venant stability inequality, and the proof of Theorem 1.1 is then completed by gluing the global Saint–Venant inequality from §4.2 to the Kohler–Jobin transfer of §2.4.

Theorem 4.29 (Bounded sharp Saint–Venant stability)**.** *There are computable constants $C = C(n, R, \rho_*)$ and $\delta_0 = \delta_0(n, R, \rho_*) > 0$, depending only on the level-set identity constants, the Fusco–Julin constant $C_{\mathrm{FJ}}(n)$, and the explicit constants above, such that every bounded open*

$\Omega \subset B_R$, $|\Omega| = \omega_n$, *satisfies*

$$\mathcal{A}(\Omega)^2 \le C\,\delta_T(\Omega).$$

Proof. Let κ be the constant fixed in Proposition 4.10. First assume $\delta_T(\Omega) \le \delta_0$, with δ_0 chosen so that $\rho_\delta = 1 - \kappa\sqrt{\delta_T(\Omega)} \ge (1+\rho_*)/2$, $\delta_T(\Omega) \le 1$, and the smallness threshold η_I used in Theorem 2.5 and Lemma 4.27 holds. Lemma 4.26 gives

$$q(\rho_\delta)^2 \le C\delta_T(\Omega).$$

Since $|E_{\rho_\delta}| = \omega_n \rho_\delta^n$ and $\rho_\delta \ge \rho_*$,

$$\mathcal{A}(E_{\rho_\delta})^2 \le \frac{q(\rho_\delta)^2}{\omega_n^2 \rho_\delta^{2n}} \le C(n,\rho_*)\delta_T(\Omega).$$

Moreover $E_{\rho_\delta} \subset \Omega$, and by the definition of ρ_δ,

$$|\Omega \setminus E_{\rho_\delta}| = \omega_n(1-\rho_\delta^n) \le n\omega_n(1-\rho_\delta) = n\omega_n\kappa\,\delta_T(\Omega)^{1/2}.$$

Lemma 2.7 gives

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\rho_\delta}) + C(n,\rho_*)\delta_T(\Omega)^{1/2}.$$

Squaring proves the estimate in the small-deficit regime.

If $\delta_T(\Omega) > \delta_0$, then $\mathcal{A}(\Omega) < 2$, so

$$\mathcal{A}(\Omega)^2 \le \frac{4}{\delta_0}\delta_T(\Omega).$$

Enlarging C completes the proof. □

4.6.1. *From Saint–Venant to Faber–Krahn.* We now use Theorem 2.10 just proved: it lets us bound $\lambda_1(\Omega)$ from below by a function of the torsion $T(\Omega)$ alone. The strategy of Part 2 is to (i) turn the multiplicative Kohler–Jobin inequality into an additive lower bound for $\lambda_1(\Omega) - L$ in terms of the torsion deficit $\sigma = T_0 - T(\Omega)$, (ii) bound this from below linearly in σ via a one-variable convexity estimate, and (iii) feed in the global Saint–Venant stability bound $\sigma \gtrsim \mathcal{A}^2$ to close the loop. A large-deficit branch handles the regime where the linearisation is wasteful.

Throughout, $L := \lambda_1(B^*)$, $T_0 := T(B^*)$, $\beta = (n+2)/2$, $p := 1/\beta = 2/(n+2) \in (0,1]$.

Lemma 4.30 (Pointwise Kohler–Jobin transfer)**.** *For every open* $\Omega \subset \mathbb{R}^n$ *with* $|\Omega| = |B^*|$,

$$\lambda_1(\Omega) \;\ge\; L\left(\frac{T_0}{T(\Omega)}\right)^{1/\beta} = L\left(\frac{T_0}{T(\Omega)}\right)^{2/(n+2)}. \tag{119}$$

Proof. Theorem 2.10 with the ball $B = B^*$ (which has $|B^*| = |\Omega|$, $T(B^*) = T_0$, $\lambda_1(B^*) = L$) reads $T(\Omega)\,\lambda_1(\Omega)^\beta \ge T(B^*)\,\lambda_1(B^*)^\beta = T_0 L^\beta$. Divide by $T(\Omega) > 0$: $\lambda_1(\Omega)^\beta \ge L^\beta\, T_0/T(\Omega)$. Raise both nonnegative sides to the power $1/\beta > 0$ (strictly increasing): $\lambda_1(\Omega) \ge L\,(T_0/T(\Omega))^{1/\beta}$. Finally $1/\beta = 1/((n+2)/2) = 2/(n+2)$, giving the displayed exponent. □

By Saint–Venant (28) at fixed volume $|\Omega| = |B^*|$, the ball maximises torsion, so $T(\Omega) \le T(B^*) = T_0$ and the torsion deficit $\sigma := \sigma(\Omega) := T_0 - T(\Omega) \ge 0$ is well defined. Put $s := \sigma/T_0 \in [0,1)$ (it is < 1 because $T(\Omega) > 0$ for Ω with positive measure and finite λ_1). Then $T(\Omega) = T_0 - \sigma = T_0(1-s)$, so $T_0/T(\Omega) = 1/(1-s) = (1-s)^{-1}$ and $(T_0/T(\Omega))^p = (1-s)^{-p}$ with $p = 2/(n+2)$. Subtracting L from both sides of (119),

$$\lambda_1(\Omega) - L \;\ge\; L\big[(1-s)^{-p} - 1\big], \qquad p = \tfrac{2}{n+2}. \tag{120}$$

Lemma 4.31 (An elementary one-variable inequality)**.** *For* $p \in (0,1]$ *and* $s \in [0,1)$, $(1-s)^{-p} - 1 \ge p\,s$.

Proof. Set $f(s) := (1-s)^{-p} - 1$ on $[0,1)$. Then $f(0) = (1-0)^{-p} - 1 = 1 - 1 = 0$. By the chain rule, $f'(s) = (-p)(1-s)^{-p-1} \cdot (-1) = p\,(1-s)^{-(p+1)}$. For $s \in [0,1)$ we have $0 < 1-s \le 1$, so $(1-s)^{-(p+1)} \ge 1^{-(p+1)} = 1$ (a base in $(0,1]$ raised to a positive power $p+1>0$ is at most 1, hence its reciprocal is at least 1). Therefore $f'(s) \ge p$ on $[0,1)$. By the fundamental theorem of calculus, $f(s) = f(0) + \int_0^s f'(t)\,dt = \int_0^s f'(t)\,dt \ge \int_0^s p\,dt = p\,s$. This is the claim. (The inequality is the convexity-type linear lower bound: f is convex with slope at least p throughout, so it dominates its tangent line ps at the origin.) □

Lemma 4.32 (Small-deficit linearisation). *If $|\Omega| = |B^*|$ and $\sigma \le T_0/2$, then $\lambda_1(\Omega) - L \ge \dfrac{2L}{(n+2)T_0}\sigma$.*

Proof. The hypothesis $\sigma \le T_0/2$ means $s = \sigma/T_0 \le 1/2 < 1$, so $s \in [0,1)$ and Lemma 4.31 applies with $p = 2/(n+2) \in (0,1]$ (indeed $p \le 1$ since $n \ge 2$ gives $n+2 \ge 4 > 2$). Chaining (120) with Lemma 4.31,

$$\lambda_1(\Omega) - L \;\ge\; L\big[(1-s)^{-p} - 1\big] \;\ge\; L\,p\,s = L \cdot \frac{2}{n+2} \cdot \frac{\sigma}{T_0} = \frac{2L}{(n+2)\,T_0}\,\sigma,$$

where the first inequality is (120), the second is Lemma 4.31, and the substitutions are $p = 2/(n+2)$ and $s = \sigma/T_0$. □

Lemma 4.33 (Large-deficit branch). *If $|\Omega| = |B^*|$ and $\sigma > T_0/2$, then $\lambda_1(\Omega) - L \ge L(2^{2/(n+2)} - 1)$, hence, since $\mathcal{A}(\Omega) < 2$,*

$$\lambda_1(\Omega) - L \;\ge\; \frac{L(2^{2/(n+2)} - 1)}{4}\,\mathcal{A}(\Omega)^2. \tag{121}$$

Proof. The hypothesis $\sigma > T_0/2$ means $T(\Omega) = T_0 - \sigma < T_0 - T_0/2 = T_0/2$, hence $T_0/T(\Omega) > 2$. Since $x \mapsto x^p$ with $p = 2/(n+2) > 0$ is strictly increasing, $(T_0/T(\Omega))^p > 2^p = 2^{2/(n+2)}$, and (119) gives

$$\lambda_1(\Omega) \;\ge\; L\,(T_0/T(\Omega))^{2/(n+2)} \;>\; L\,2^{2/(n+2)},$$

so $\lambda_1(\Omega) - L \ge L\,(2^{2/(n+2)} - 1)$ (the strict inequality is weakened to $\ge$, which is all we need). To convert this constant lower bound into the stated $\mathcal{A}^2$ form, recall that at fixed volume the Fraenkel asymmetry obeys $\mathcal{A}(\Omega) \in [0,2)$, so $\mathcal{A}(\Omega)^2 \le 4$, i.e. $\mathcal{A}(\Omega)^2/4 \le 1$. Multiplying the constant bound by $\mathcal{A}(\Omega)^2/4 \le 1$ (which only decreases the right-hand side, preserving the inequality):

$$\lambda_1(\Omega) - L \;\ge\; L(2^{2/(n+2)} - 1) \;\ge\; L(2^{2/(n+2)} - 1) \cdot \frac{\mathcal{A}(\Omega)^2}{4} = \frac{L(2^{2/(n+2)} - 1)}{4}\,\mathcal{A}(\Omega)^2,$$

the second inequality because $2^{2/(n+2)} - 1 > 0$ and $\mathcal{A}^2/4 \le 1$. This is (121). □

Theorem 4.34 (Sharp Faber–Krahn via Kohler–Jobin). *By the global sharp Saint–Venant stability estimate of Theorem 4.12, with $c_{\mathrm{SV}}^{\mathrm{glob}}(n) > 0$,*

$$T(B^*) - T(\Omega) = 2\big(E(\Omega) - E(B^*)\big) \;\ge\; 2\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)\Big(\frac{|\Omega|}{\omega_n}\Big)^{(n+2)/n}\mathcal{A}(\Omega)^2 \tag{122}$$

for all open bounded $\Omega \subset \mathbb{R}^n$, $0 < |\Omega| < \infty$. Then

$$\delta_{\mathrm{FK}}(\Omega) = \Big(\frac{|\Omega|}{\omega_n}\Big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \;\ge\; c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2, \tag{123}$$

with the explicit constant

$$c_{\mathrm{FK}}(n) = \min\left\{ \frac{4\,L_n\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)\,T_n},\quad \frac{L_n\big(2^{2/(n+2)}-1\big)}{4} \right\}, \tag{124}$$

equivalently $\mathcal{A}(\Omega)^2 \le C_{\mathrm{FK}}(n)\,\delta_{\mathrm{FK}}(\Omega)$ *with* $C_{\mathrm{FK}}(n) = 1/c_{\mathrm{FK}}(n)$.

Proof. Reduction by scaling. The deficit δ_{FK} and the asymmetry $\mathcal{A}$ are scale invariant: $\mathcal{A}(t\Omega) = \mathcal{A}(\Omega)$ since both $|\cdot \triangle \cdot|$ and $|B^*|$ scale by t^n and cancel; and $\delta_{\mathrm{FK}}(t\Omega) = (t^n\,|\Omega|\,/\omega_n)^{2/n}(t^{-2}\lambda_1(\Omega) - t^{-2}\lambda_1(B^*)) = t^2(|\Omega|\,/\omega_n)^{2/n}\cdot t^{-2}(\lambda_1(\Omega)-\lambda_1(B^*)) = \delta_{\mathrm{FK}}(\Omega)$ by the dilation laws of §2.4.1. The hypothesis (122) is likewise scale invariant: both sides carry the factor $(|\Omega|\,/\omega_n)^{(n+2)/n}\cdot$(scale-free) and T scales by t^{n+2} consistently. Hence we may rescale so that $|\Omega| = \omega_n$, whence $B^* = B_1$, $L = \lambda_1(B_1) = L_n$, $T_0 = T(B_1) = T_n$, and the volume prefactor $(|\Omega|\,/\omega_n)^{2/n} = 1$, so that $\delta_{\mathrm{FK}}(\Omega) = \lambda_1(\Omega) - L_n$ in this normalisation. Under this normalisation, (122) becomes $\sigma = T_n - T(\Omega) \ge 2c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$ (the prefactor $(|\Omega|\,/\omega_n)^{(n+2)/n} = 1$). Split on the torsion deficit $\sigma = T_n - T(\Omega)$.

Case 1 ($\sigma \le T_n/2$). Lemma 4.32 applies (its hypothesis $\sigma \le T_0/2 = T_n/2$ holds), giving $\lambda_1(\Omega) - L_n \ge \frac{2L_n}{(n+2)T_n}\,\sigma$. Now insert the Saint–Venant lower bound $\sigma \ge 2c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$:

$$\lambda_1(\Omega) - L_n \;\ge\; \frac{2L_n}{(n+2)T_n}\,\sigma \;\ge\; \frac{2L_n}{(n+2)T_n}\cdot 2c_{\mathrm{SV}}^{\mathrm{glob}}(n)\,\mathcal{A}(\Omega)^2 = \frac{4L_n\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)T_n}\,\mathcal{A}(\Omega)^2.$$

The first inequality is Lemma 4.32; the second multiplies the positive coefficient $\frac{2L_n}{(n+2)T_n}$ by the lower bound for σ (preserving direction since the coefficient is positive); the final equality collects the constants. This matches the first entry of the minimum in (124).

Case 2 ($\sigma > T_n/2$). Lemma 4.33 applies (its hypothesis $\sigma > T_0/2 = T_n/2$ holds), giving directly $\lambda_1(\Omega) - L_n \ge \frac{L_n(2^{2/(n+2)}-1)}{4}\mathcal{A}(\Omega)^2$, the second entry of the minimum in (124). (Note this branch does *not* use the Saint–Venant input: a large torsion deficit already forces a large spectral gap through Kohler–Jobin alone.)

Conclusion. In either case $\delta_{\mathrm{FK}}(\Omega) = \lambda_1(\Omega) - L_n \ge c\,\mathcal{A}(\Omega)^2$ with c the constant of the relevant case; taking the smaller of the two constants, which is the worst case across the dichotomy, gives the uniform bound $\delta_{\mathrm{FK}}(\Omega) \ge c_{\mathrm{FK}}(n)\mathcal{A}(\Omega)^2$ with $c_{\mathrm{FK}}(n)$ the minimum (124). Undoing the scaling preserves the inequality, every term being scale invariant as shown. The equivalent reciprocal form $\mathcal{A}(\Omega)^2 \le C_{\mathrm{FK}}(n)\delta_{\mathrm{FK}}(\Omega)$ with $C_{\mathrm{FK}} = 1/c_{\mathrm{FK}}$ follows by dividing by the positive constant $c_{\mathrm{FK}}(n)$. ☐

Proof of Theorem 1.1. For an open set of finite measure with $\lambda_1(\Omega) = +\infty$ the inequality is trivial, since its left side $\delta_{\mathrm{FK}}(\Omega)$ is then $+\infty$ (the prefactor is finite and positive, multiplying $\lambda_1(\Omega) - \lambda_1(B^*) = +\infty$) while the right side $c_{\mathrm{FK}}(n)\mathcal{A}(\Omega)^2 \le 4c_{\mathrm{FK}}(n)$ is finite (using $\mathcal{A}(\Omega)^2 \le 4$). Otherwise $\lambda_1(\Omega) < \infty$, and Theorem 4.34 applies: its Saint–Venant hypothesis (122) is exactly the global estimate (65) of Theorem 4.12, so (123) holds with $c_{\mathrm{FK}}(n)$ as in (124). This is precisely the assertion of Theorem 1.1. ☐

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